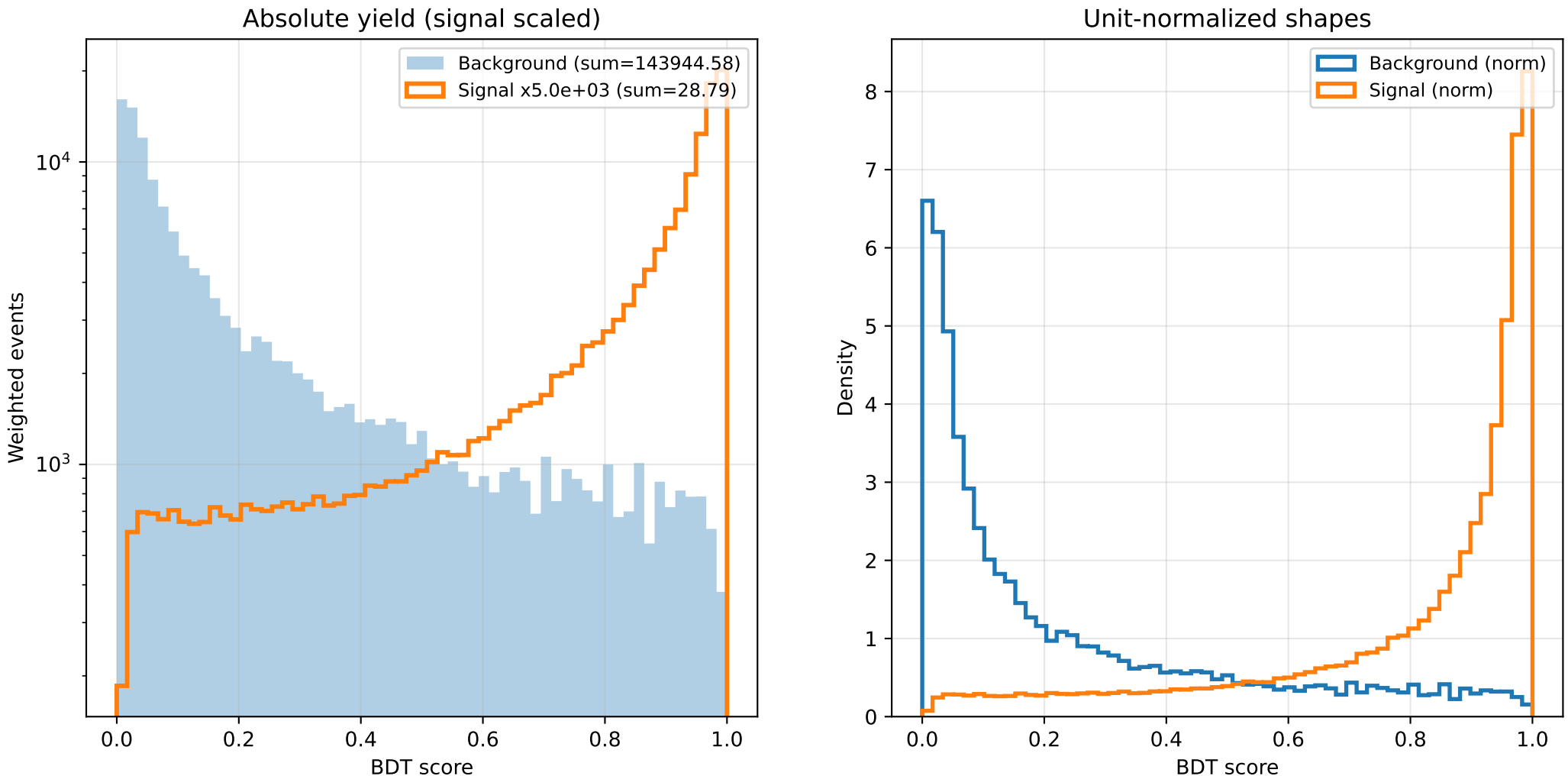
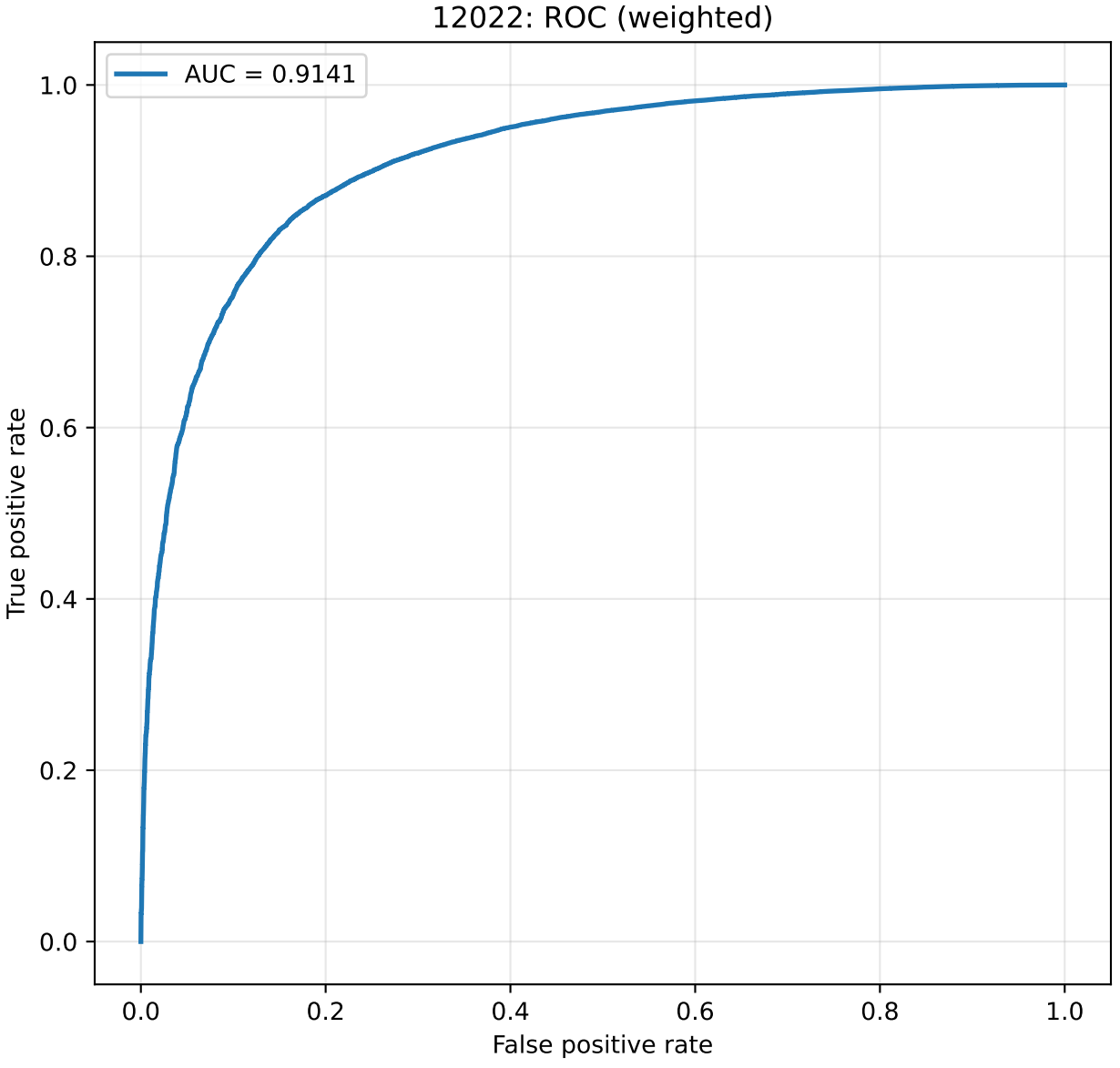
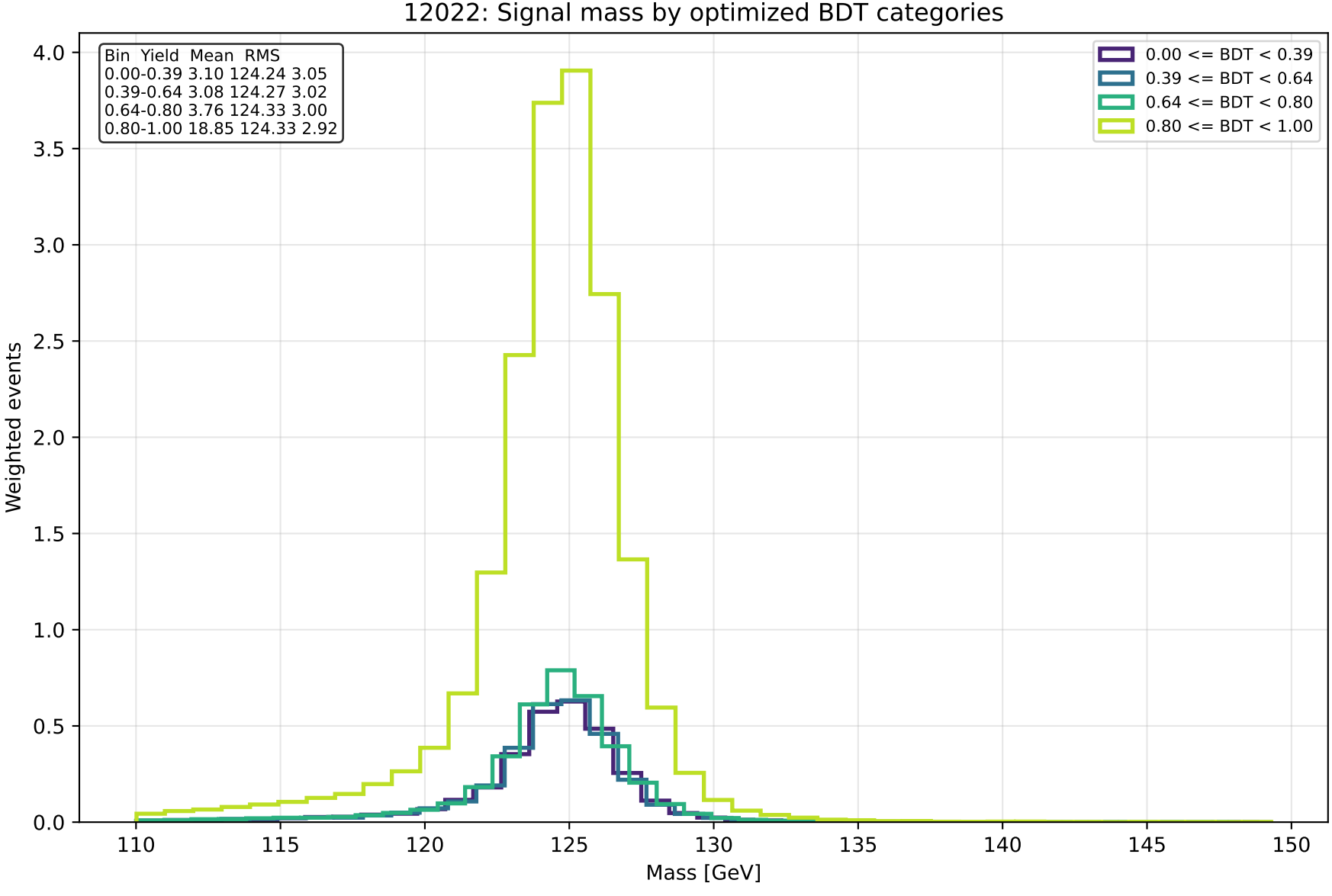
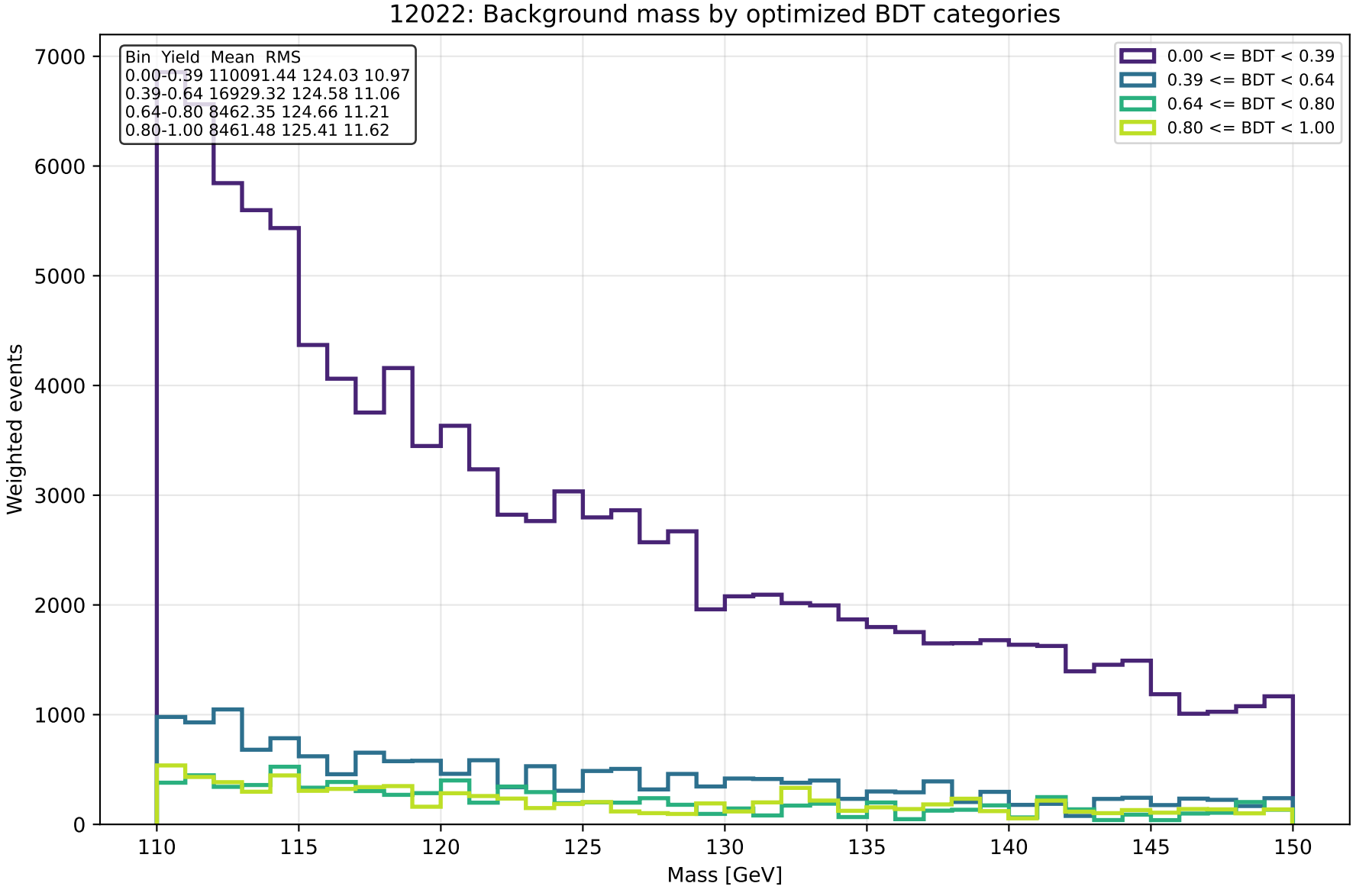

12022: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=143944.58)
Signal x5.0e+03 (sum=28.79)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
1
2
3
4
5
6
7
8
BDT score

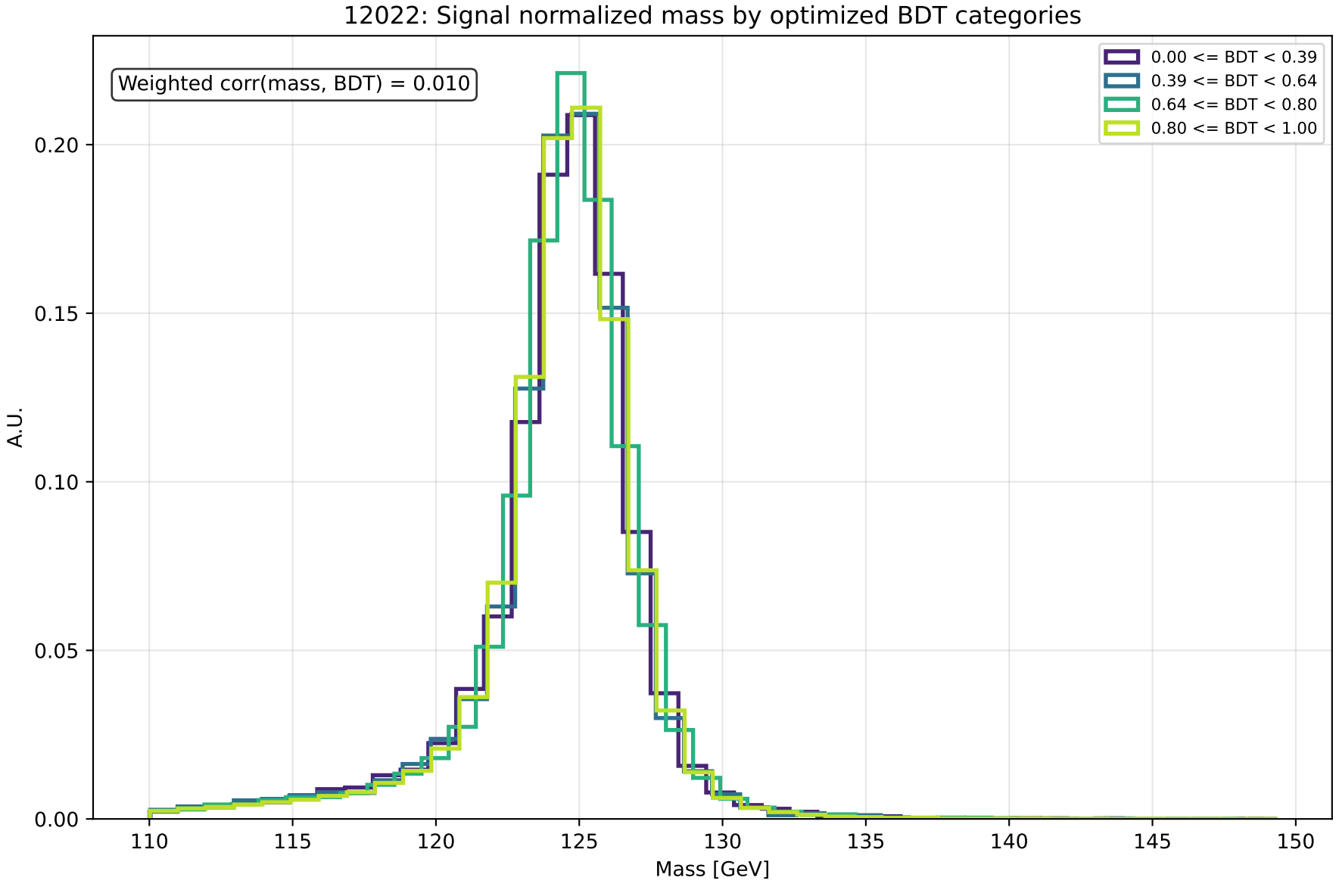

12022: ROC (weighted)
AUC = 0.9141
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

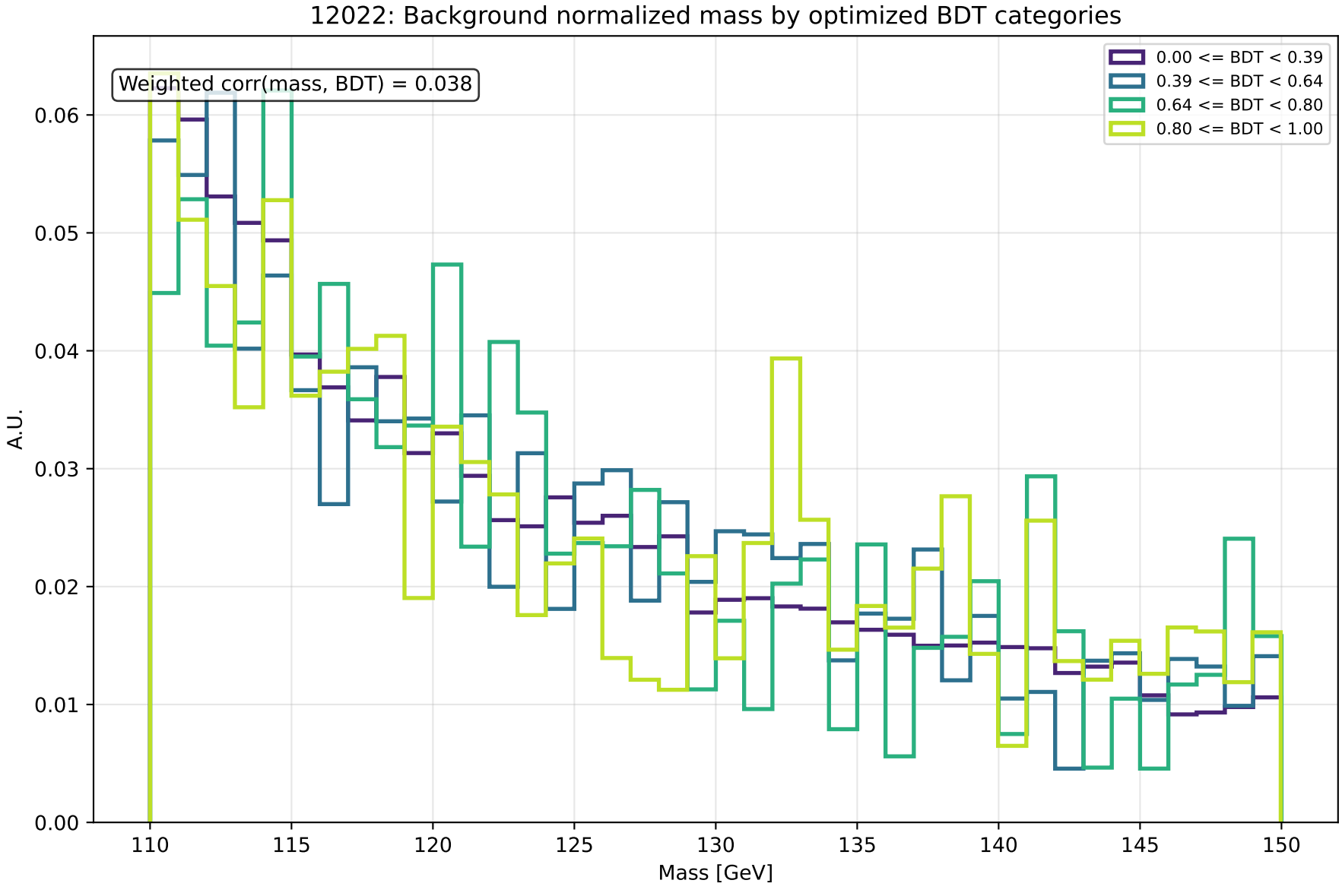
12022: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=0.000, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.386	3.097	110091.437	2.81e-05	0.009
2	0.386-0.641	3.083	16929.322	0.000182	0.024
3	0.641-0.804	3.765	8462.348	0.000445	0.041
4	0.804-1.000	18.848	8461.477	0.00223	0.205


12022: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.39 3.10 124.24 3.05
0.39-0.64 3.08 124.27 3.02
0.64-0.80 3.76 124.33 3.00
0.80-1.00 18.85 124.33 2.92
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
110
115
120
125
130
135
140
145
150


12022: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.39 110091.44 124.03 10.97
0.39-0.64 16929.32 124.58 11.06
0.64-0.80 8462.35 124.66 11.21
0.80-1.00 8461.48 125.41 11.62
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0
1000
2000
3000
4000
5000
6000
7000
110
115
120
125
130
135
140
145
150


12022: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
110
115
120
125
130
135
140
145
150
Mass [GeV]

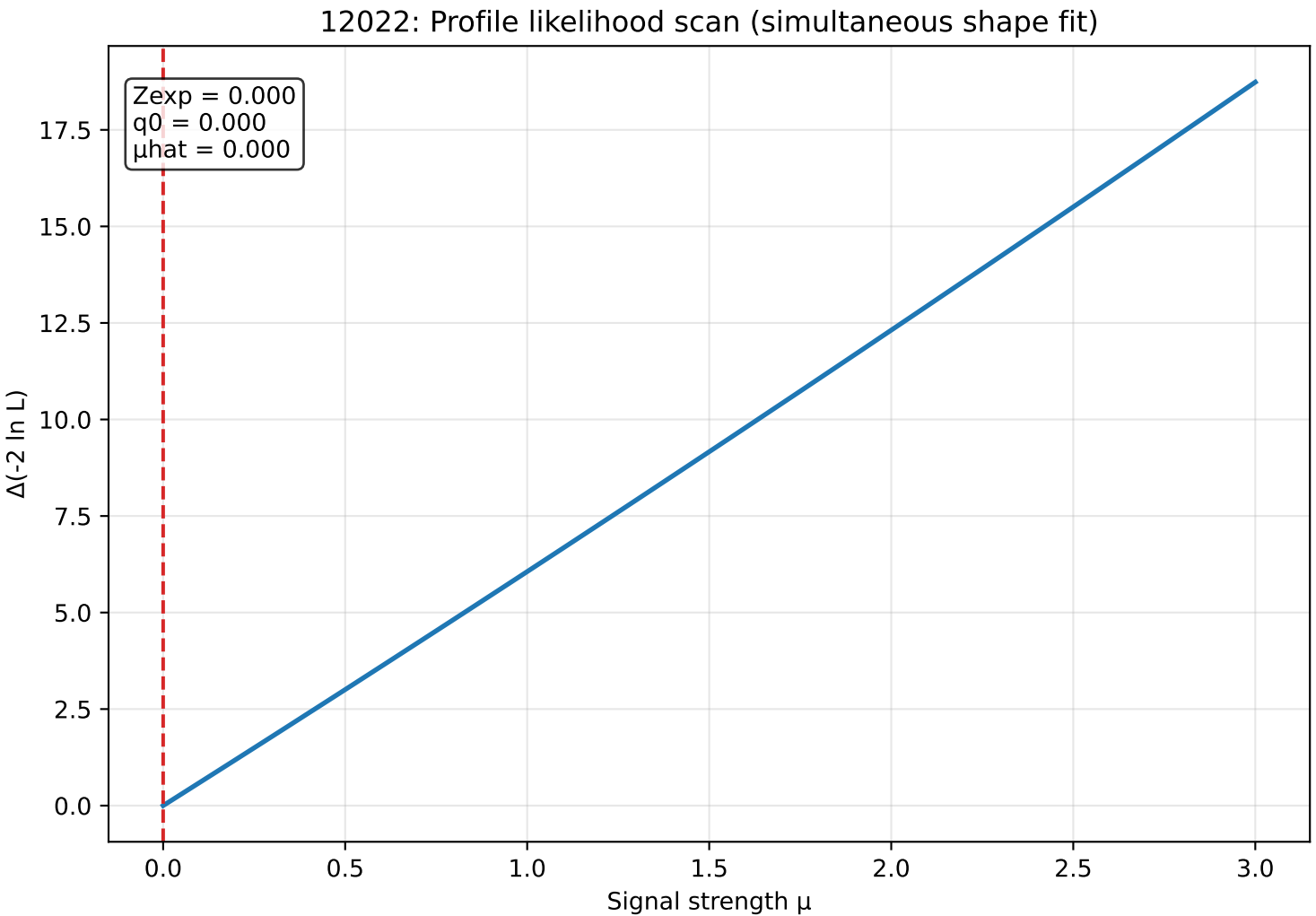

12022: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.038
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

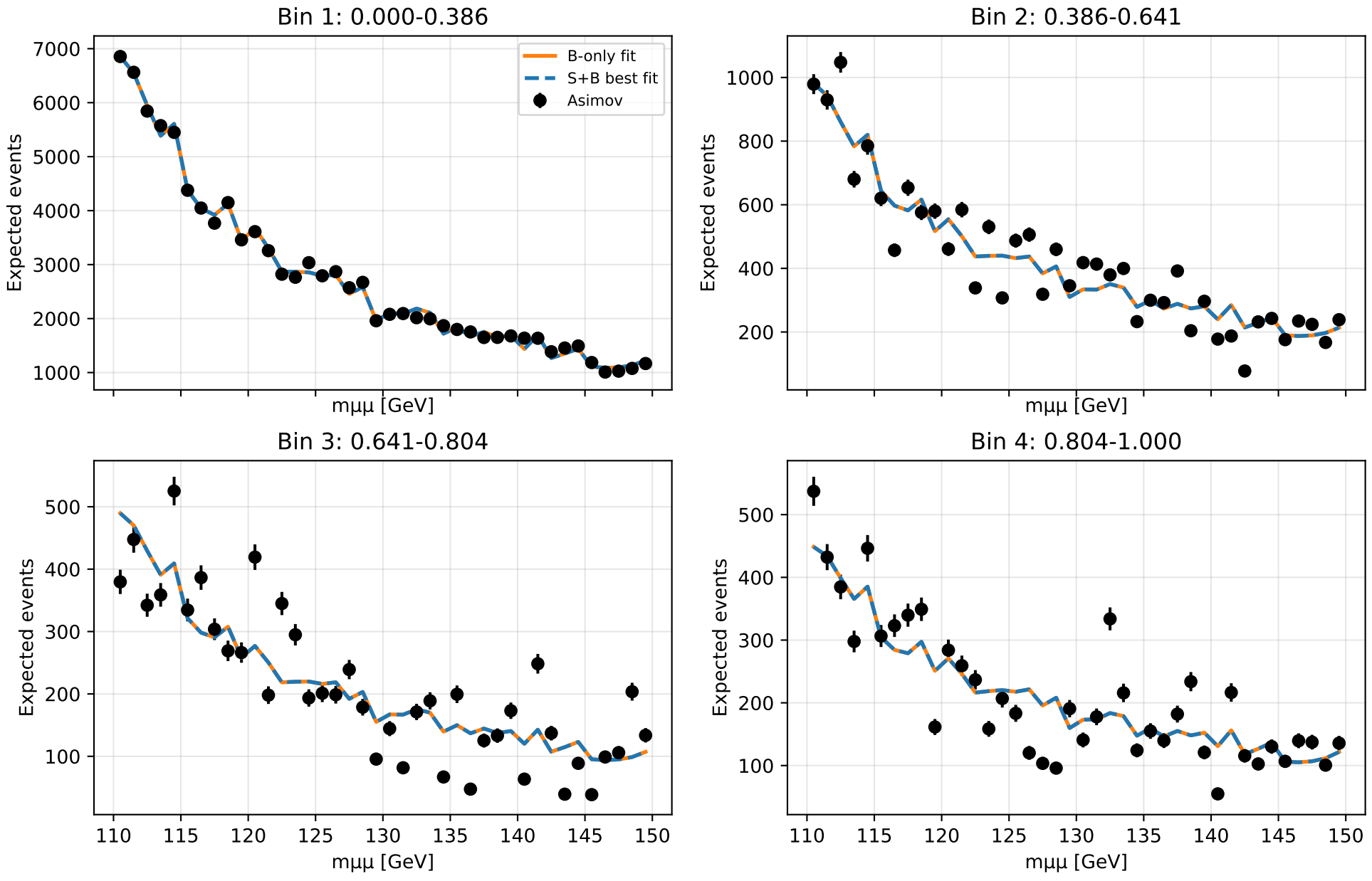
12022: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	0.0000
q0	0.0000
mu_hat	0.0000
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

12022: Shape-likelihood bin yields (Asimov)

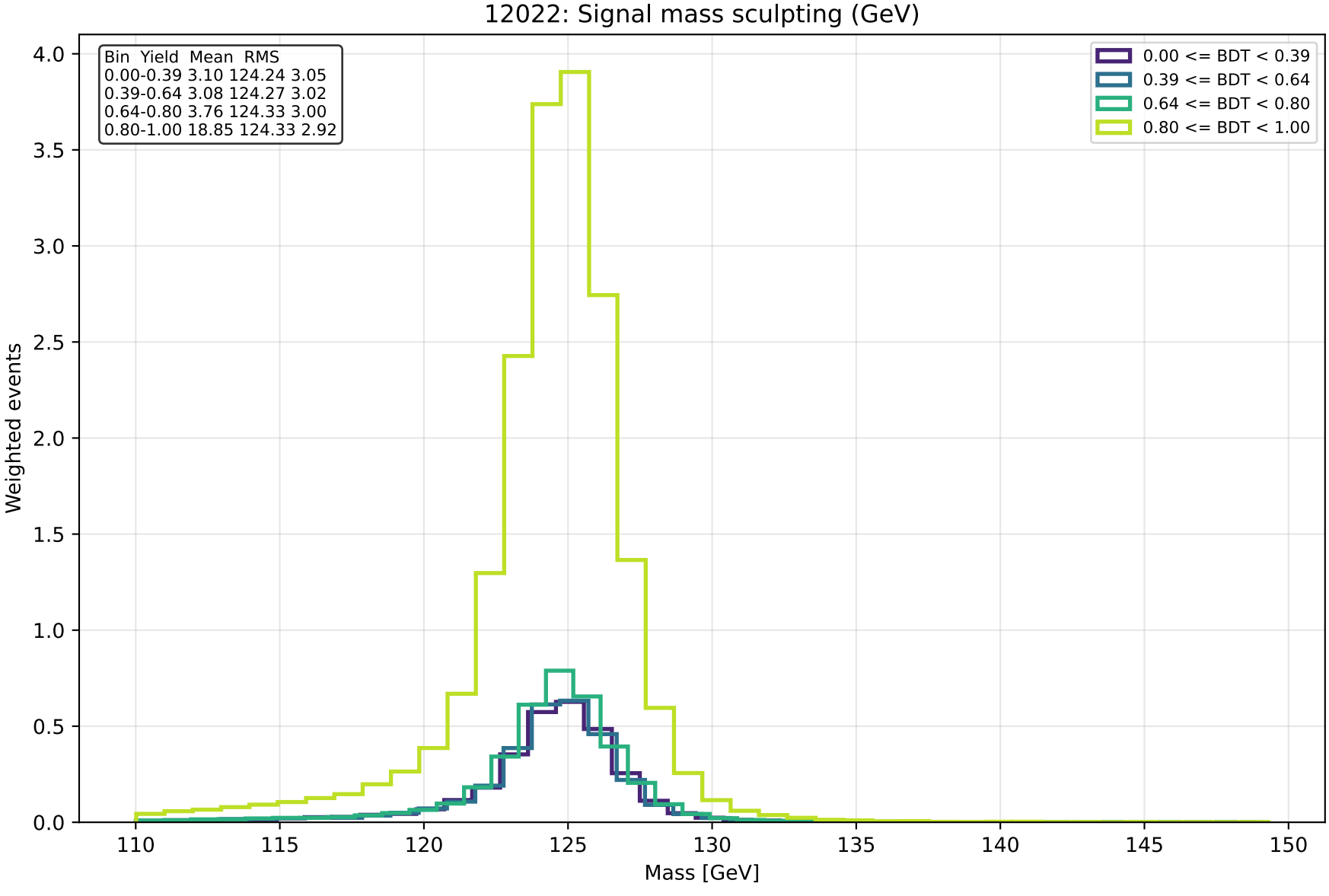
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.386	3.097	110091.437	110091.437	110091.437
2	0.386-0.641	3.083	16929.322	16929.322	16929.322
3	0.641-0.804	3.765	8462.348	8462.348	8462.348
4	0.804-1.000	18.848	8461.477	8461.477	8461.477

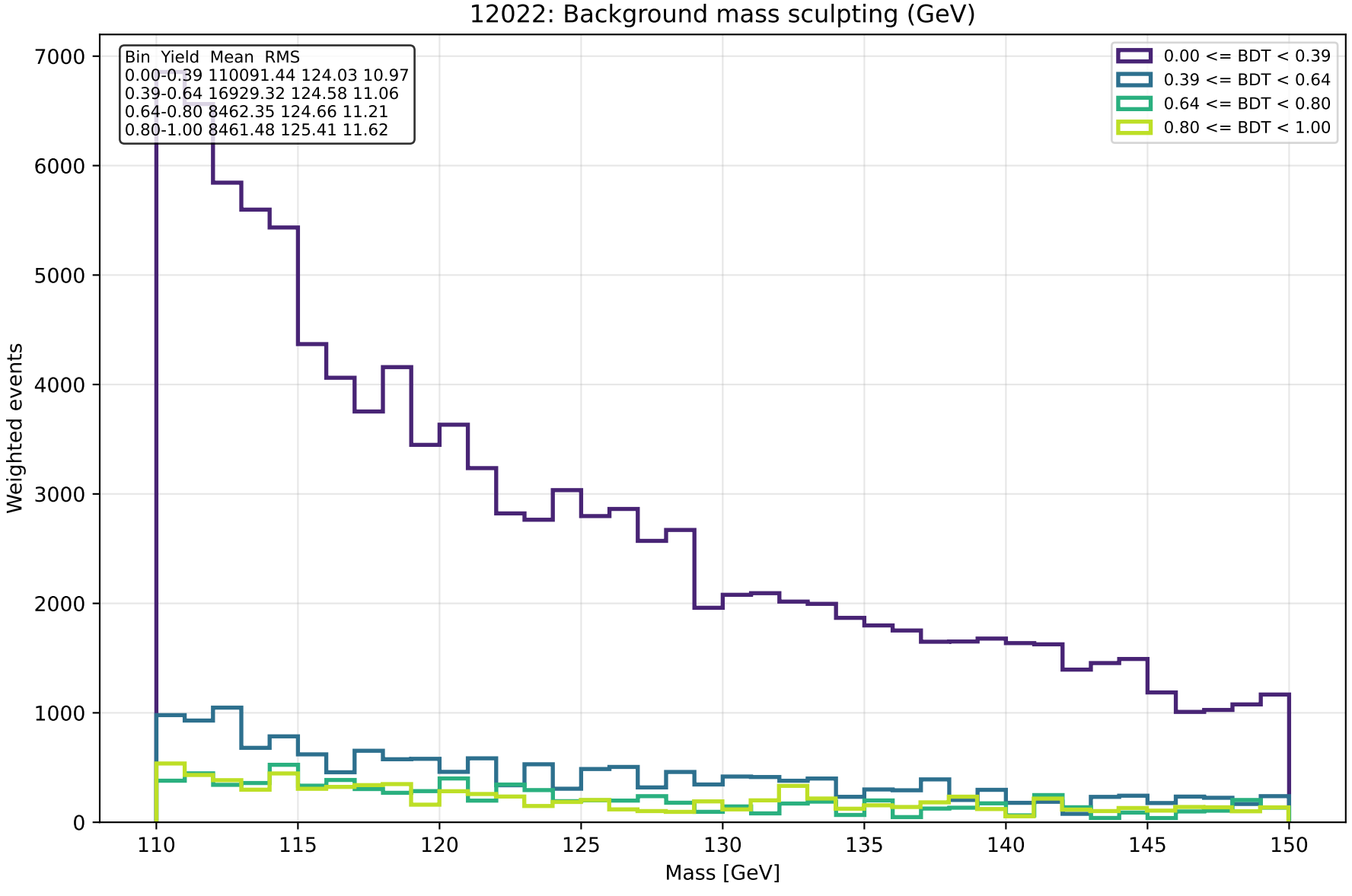

12022: Profile likelihood scan (simultaneous shape fit)
Zexp = 0.000
q0 = 0.000
μhat = 0.000
Δ(-2 ln L)
Signal strength μ
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0

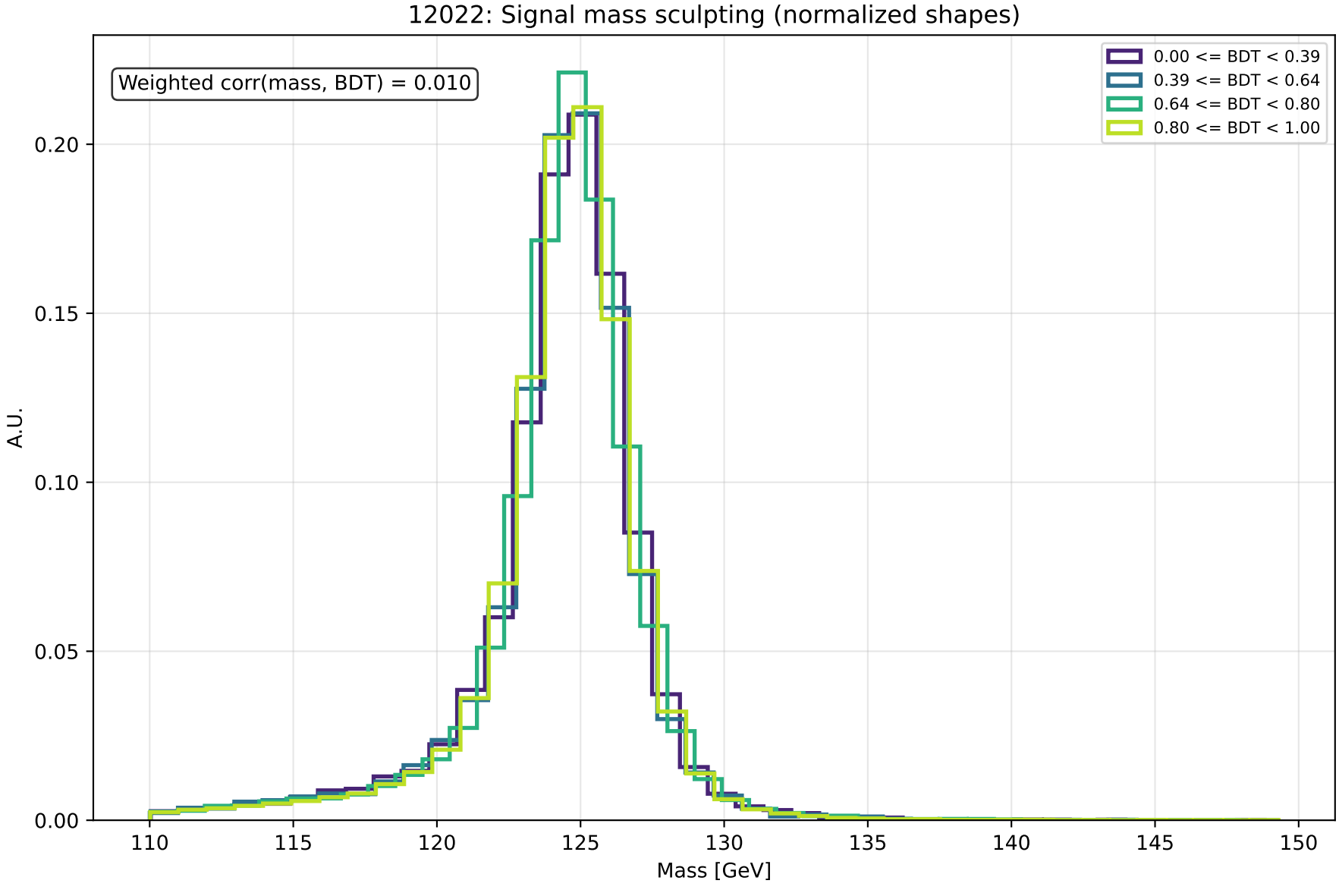

12022: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.386
Bin 2: 0.386-0.641
Bin 3: 0.641-0.804
Bin 4: 0.804-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

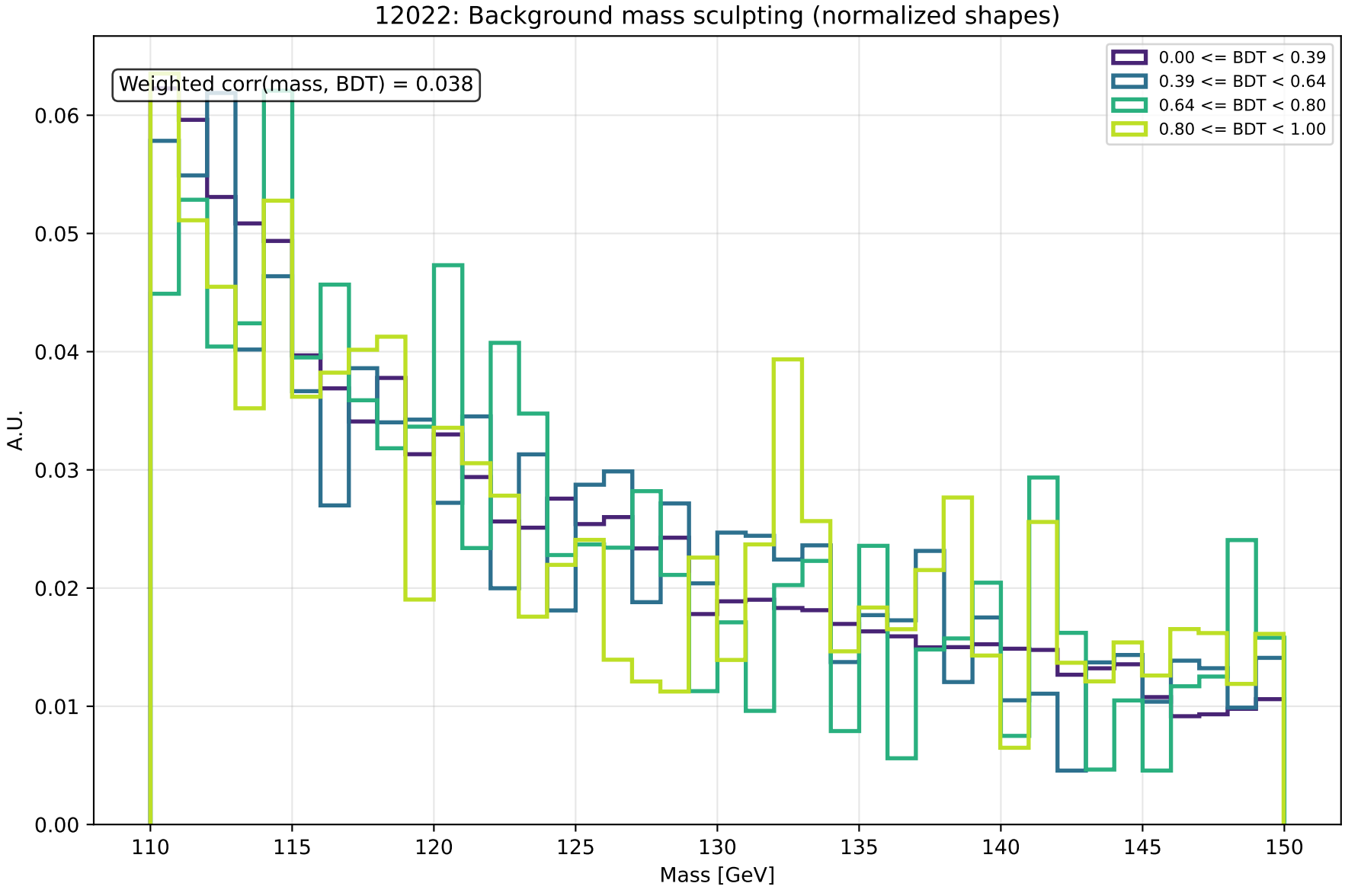
12022: Expected yield per optimized BDT bin (110-150 GeV)
S total=28.79, B total=143944.58

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.386	3.097	10.8%	100.0%	110091.437	0.0002
0.386-0.641	3.083	10.7%	89.2%	16929.322	0.000759
0.641-0.804	3.765	13.1%	78.5%	8462.348	0.00134
0.804-1.000	18.848	65.5%	65.5%	8461.477	0.00223

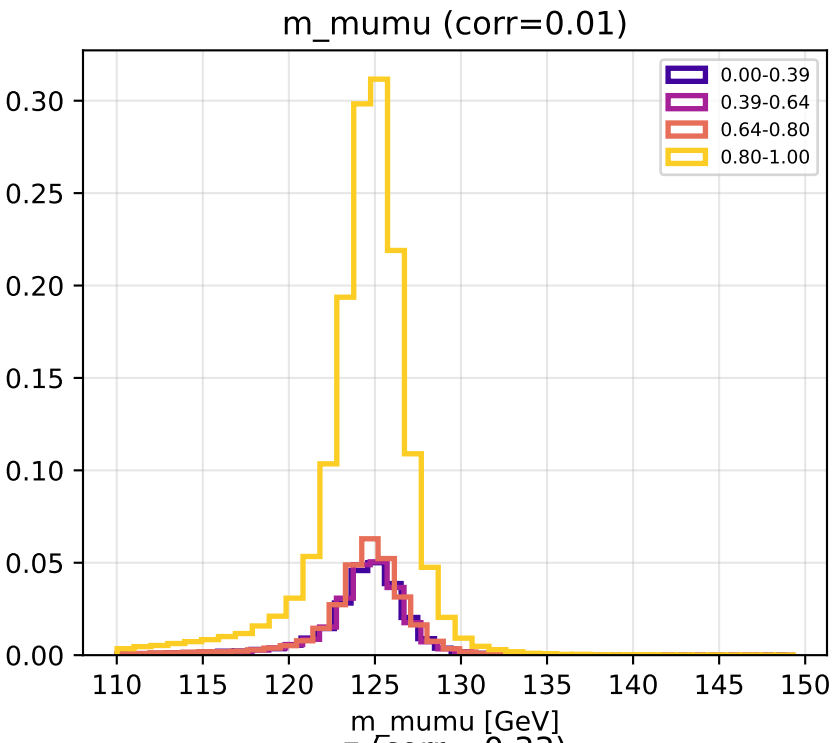

12022: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.39 3.10 124.24 3.05
0.39-0.64 3.08 124.27 3.02
0.64-0.80 3.76 124.33 3.00
0.80-1.00 18.85 124.33 2.92
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
110
115
120
125
130
135
140
145
150

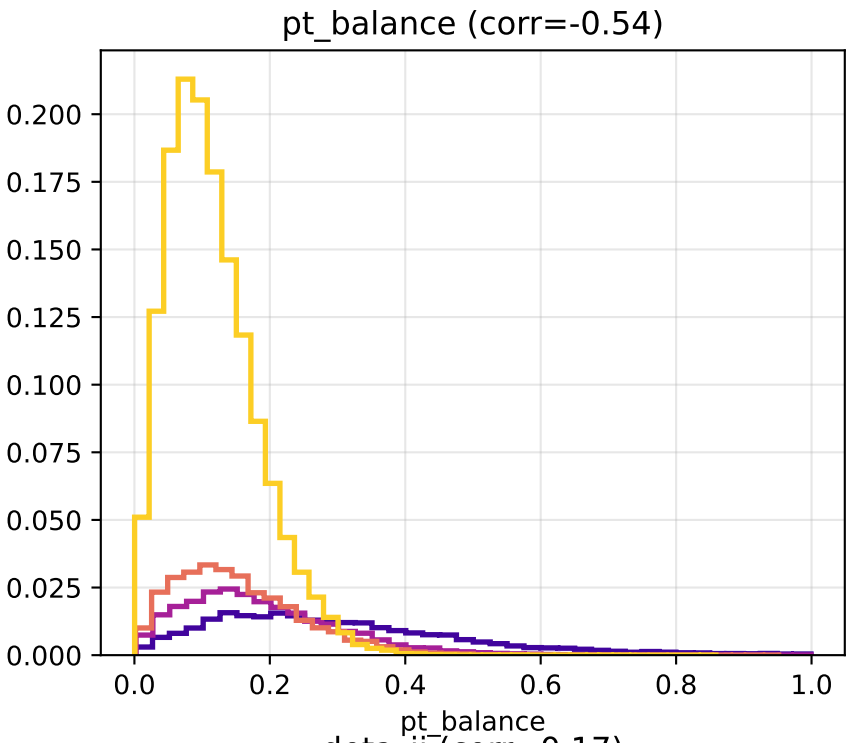

12022: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.39 110091.44 124.03 10.97
0.39-0.64 16929.32 124.58 11.06
0.64-0.80 8462.35 124.66 11.21
0.80-1.00 8461.48 125.41 11.62
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0
1000
2000
3000
4000
5000
6000
7000
110
115
120
125
130
135
140
145
150

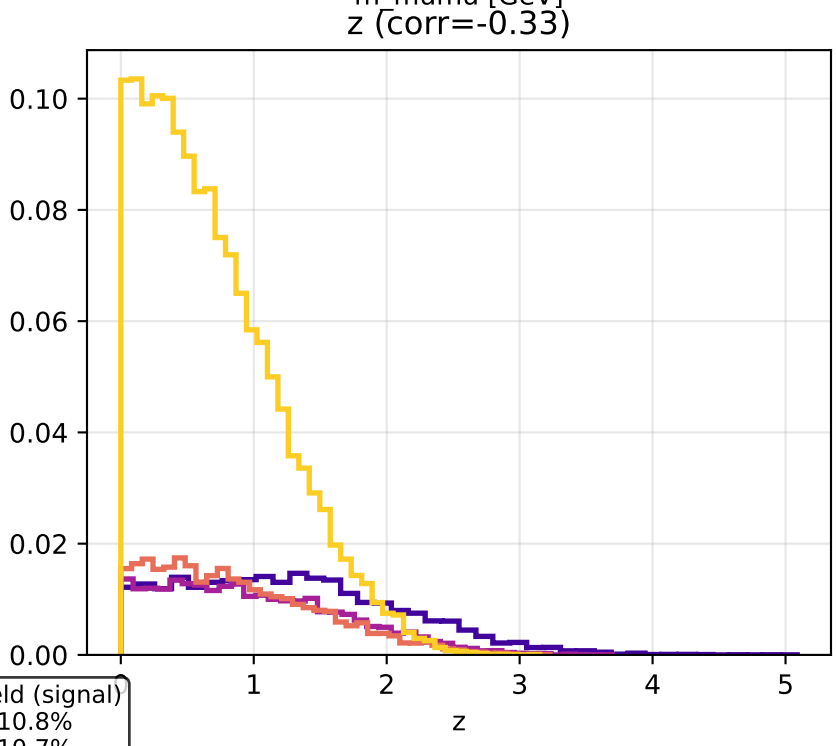

12022: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
110
115
120
125
130
135
140
145
150
Mass [GeV]

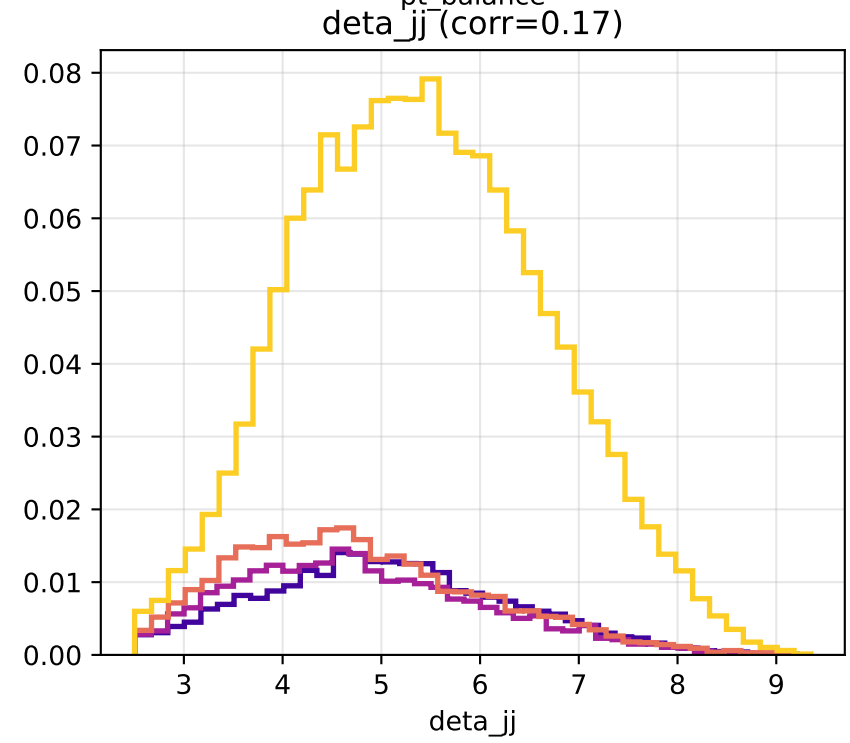

12022: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.038
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

12022: Signal distributions by BDT bin

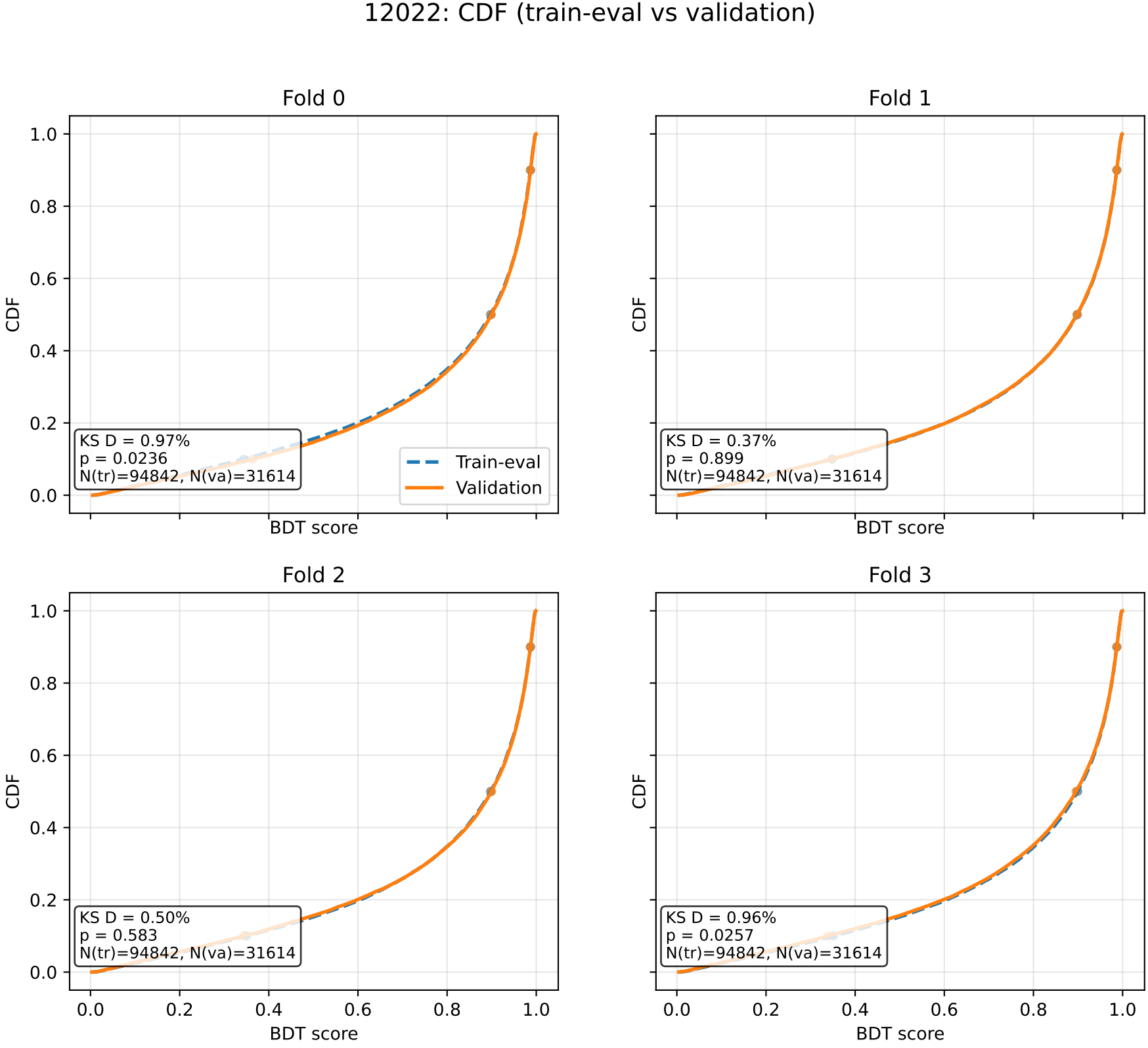








BDT bin yield (signal)
0.00-0.39: 10.8%
0.39-0.64: 10.7%
0.64-0.80: 13.1%
0.80-1.00: 65.5%

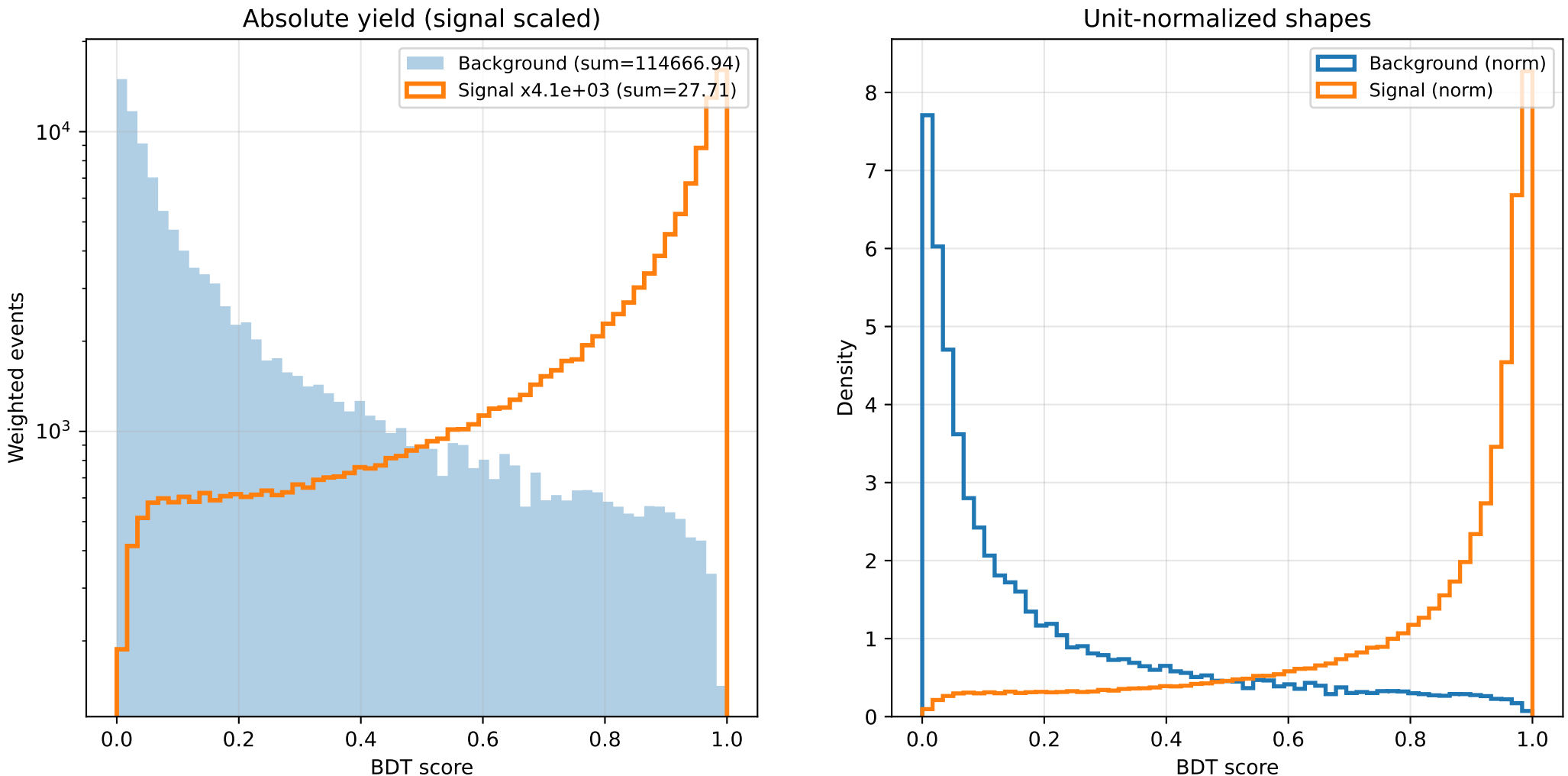

12022: CDF (train-eval vs validation)
Fold 0
KS D = 0.97%
p = 0.0236
N(tr)=94842, N(va)=31614
Train-eval
Validation
Fold 1
KS D = 0.37%
p = 0.899
N(tr)=94842, N(va)=31614
Fold 2
KS D = 0.50%
p = 0.583
N(tr)=94842, N(va)=31614
Fold 3
KS D = 0.96%
p = 0.0257
N(tr)=94842, N(va)=31614
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

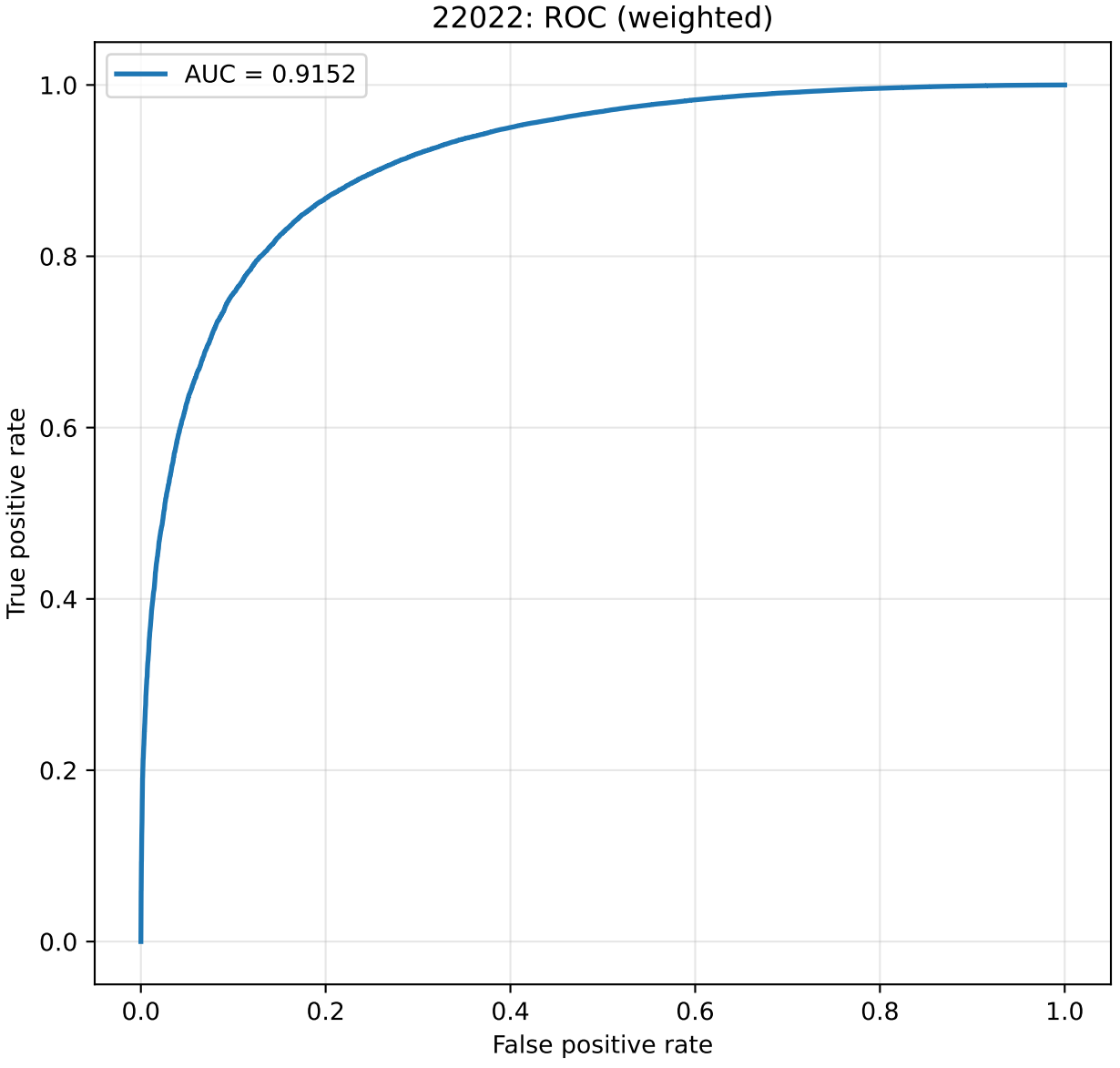
12022: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.97%	31614	0.001	0.20%	FALSE
1	0.37%	31614	0.001	0.20%	FALSE
2	0.50%	31614	0.001	0.20%	FALSE
3	0.96%	31614	0.001	0.20%	FALSE

12022: Top-10 feature importance (gain)
12022_featureimportance_DY.txt

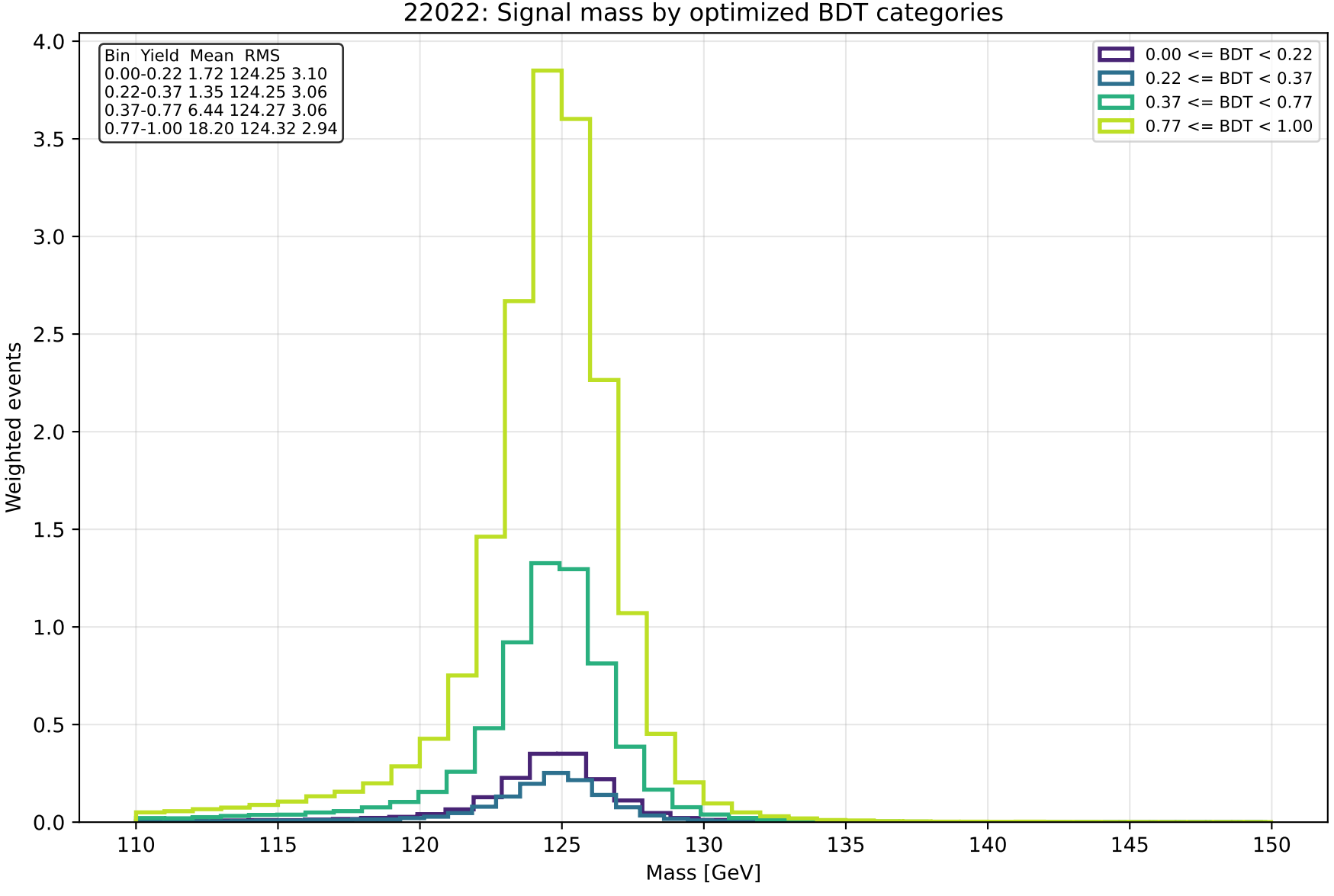
Feature	Gain %
pt_balance	11.97%
z	11.41%
deta_jj	10.34%
njets	9.72%
pt_j2	6.57%
dphi_mumu_jj	4.80%
jet1_eta	4.77%
costheta_CS	4.17%
pt_j1	4.06%
mu1_eta	4.01%

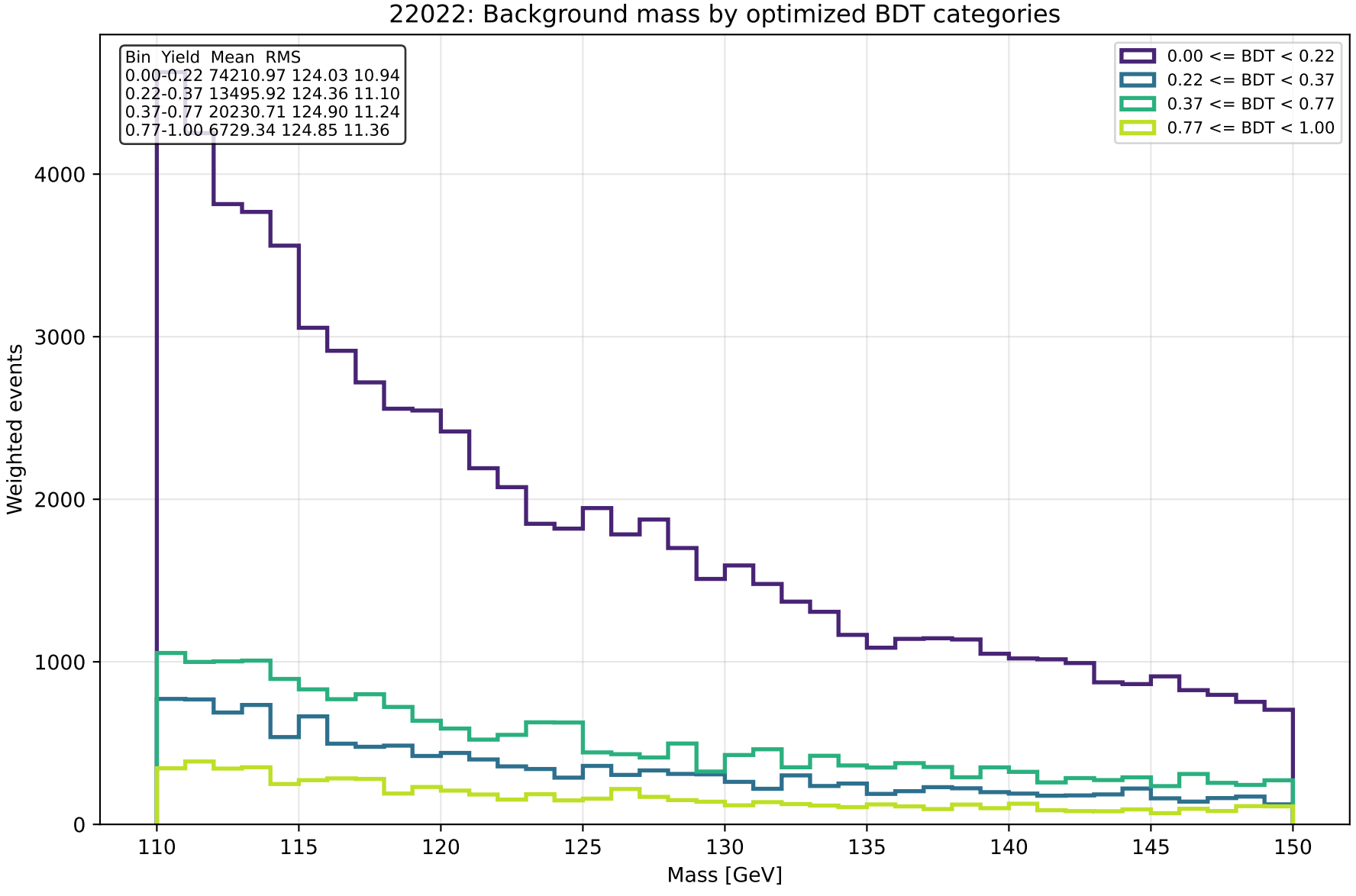

22022: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=114666.94)
Signal x4.1e+03 (sum=27.71)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
1
2
3
4
5
6
7
8
BDT score

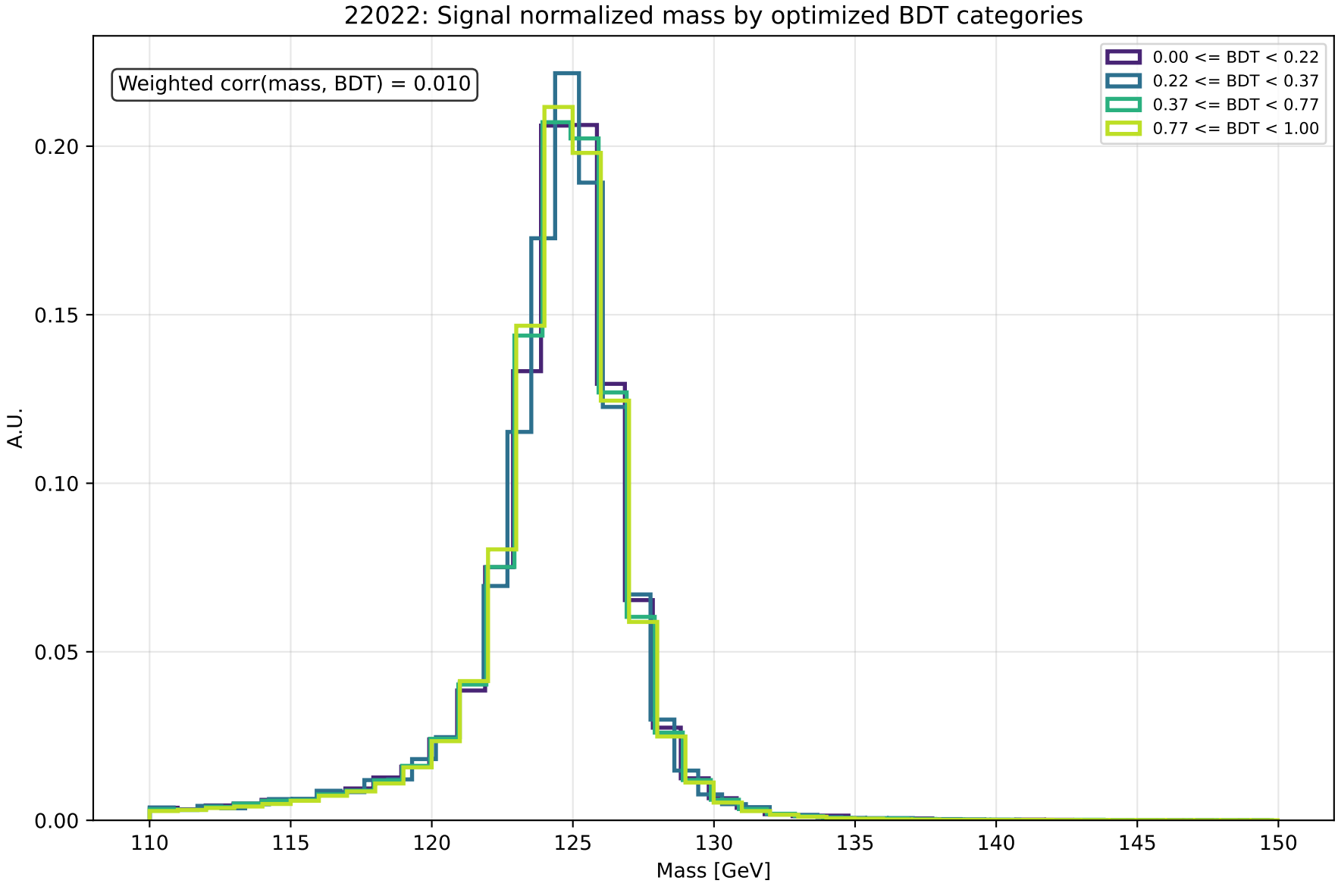

22022: ROC (weighted)
AUC = 0.9152
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

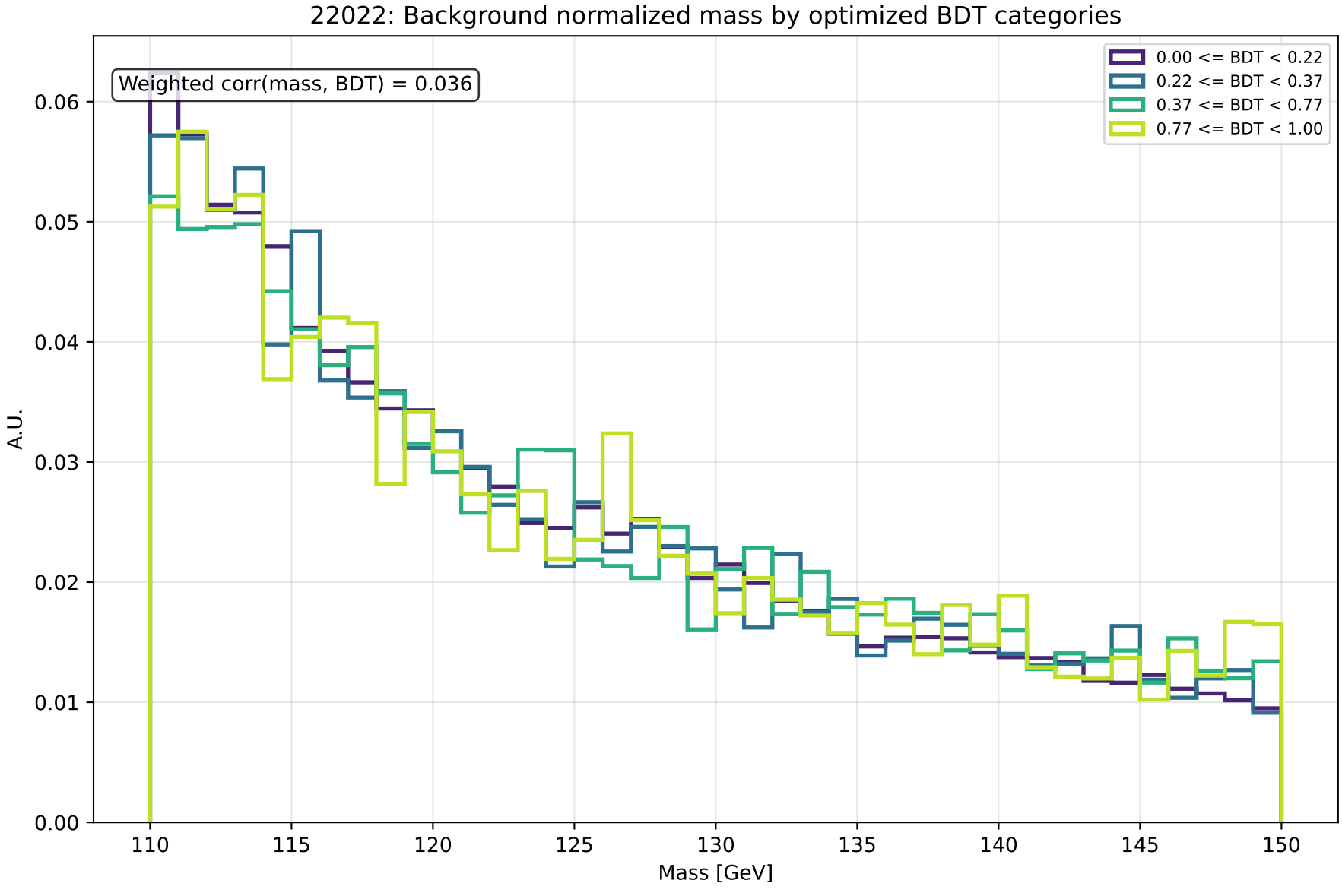
22022: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=0.444, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.220	1.716	74210.971	2.31e-05	0.006
2	0.220-0.365	1.346	13495.923	9.97e-05	0.012
3	0.365-0.770	6.441	20230.707	0.000318	0.045
4	0.770-1.000	18.203	6729.336	0.0027	0.222


22022: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.22 1.72 124.25 3.10
0.22-0.37 1.35 124.25 3.06
0.37-0.77 6.44 124.27 3.06
0.77-1.00 18.20 124.32 2.94
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
110
115
120
125
130
135
140
145
150


22022: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.22 74210.97 124.03 10.94
0.22-0.37 13495.92 124.36 11.10
0.37-0.77 20230.71 124.90 11.24
0.77-1.00 6729.34 124.85 11.36
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
0
1000
2000
3000
4000
110
115
120
125
130
135
140
145
150
Mass [GeV]


22022: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
110
115
120
125
130
135
140
145
150
Mass [GeV]

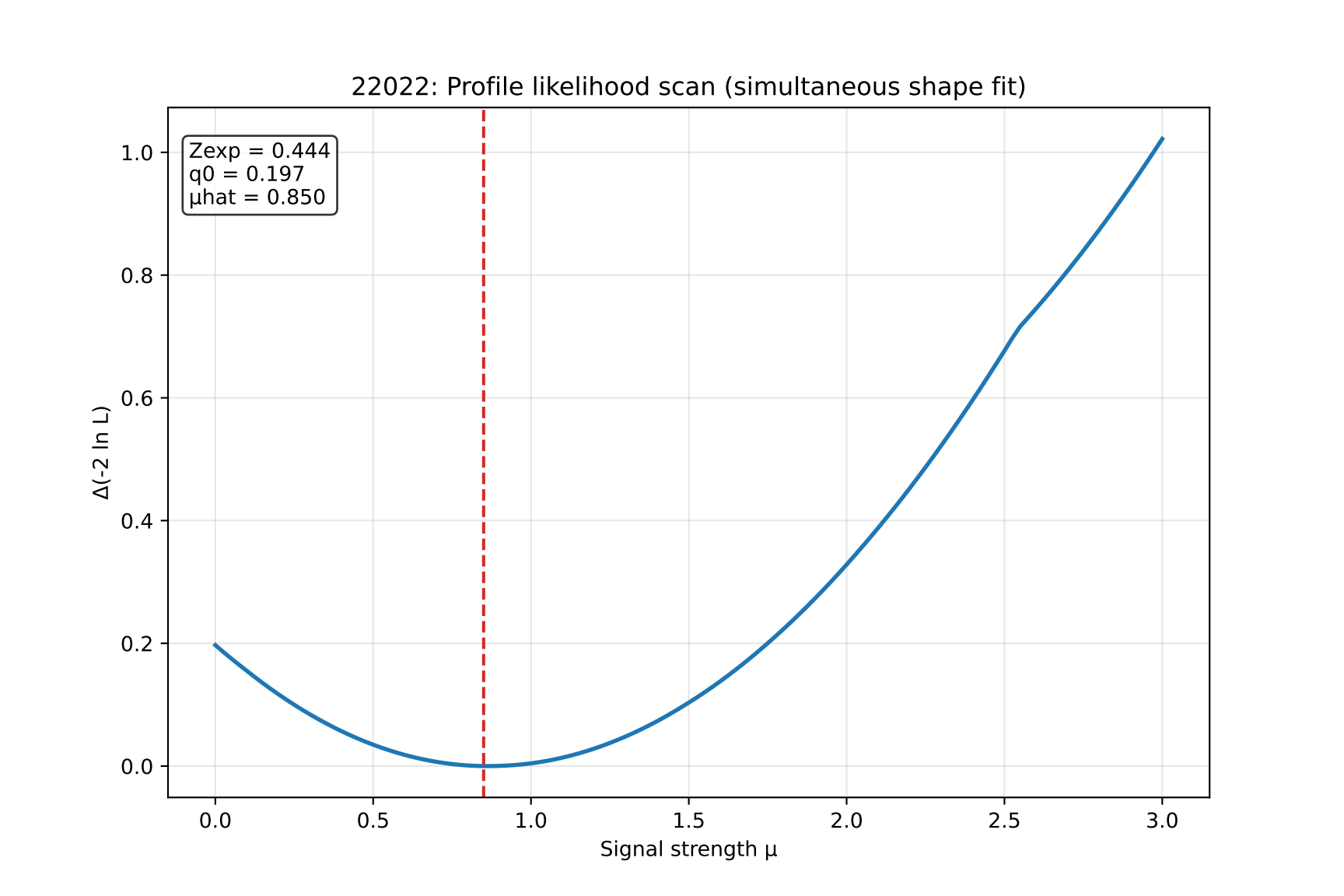

22022: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.036
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

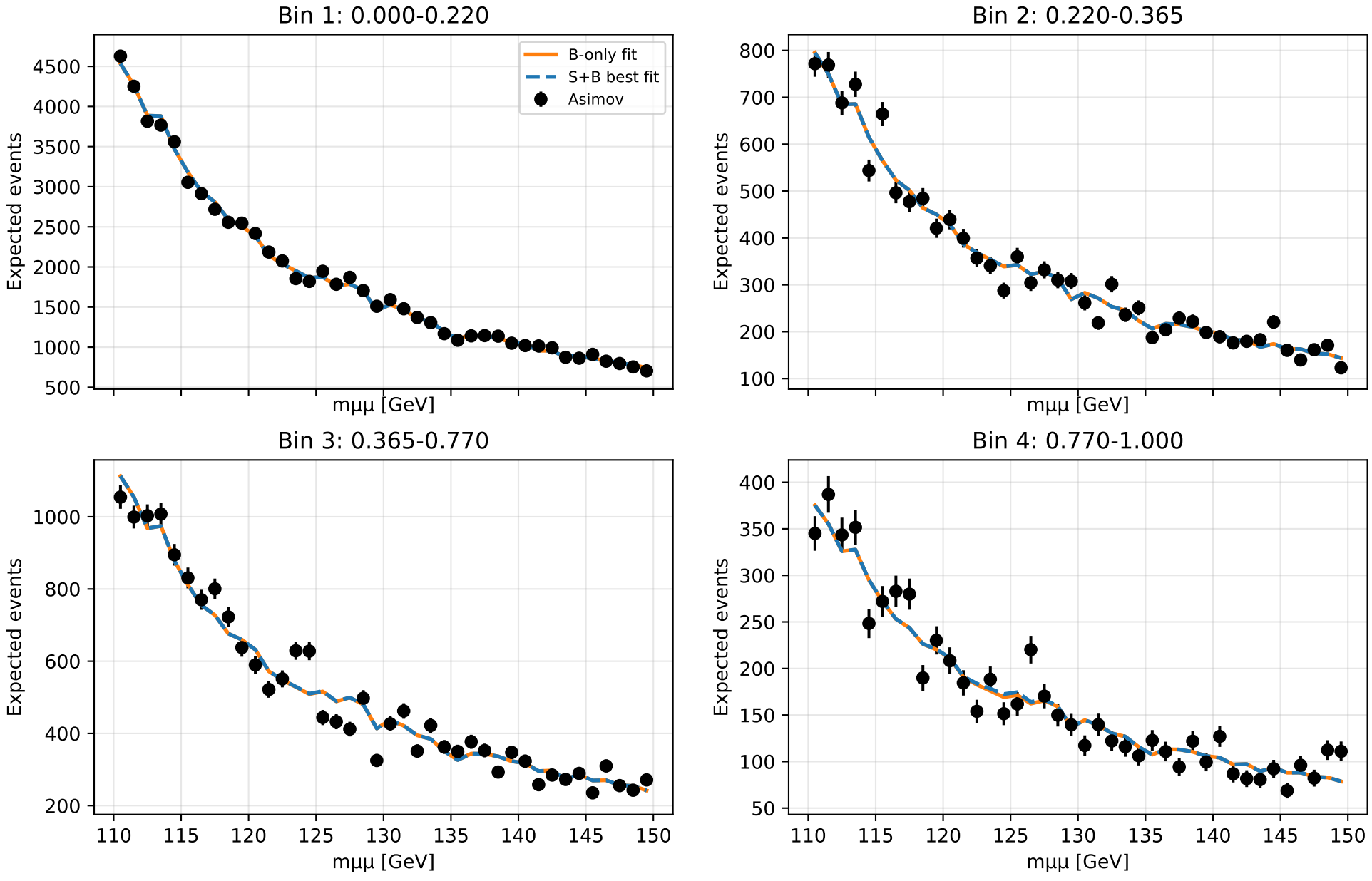
22022: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	0.4439
q0	0.1971
mu_hat	0.8500
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

22022: Shape-likelihood bin yields (Asimov)

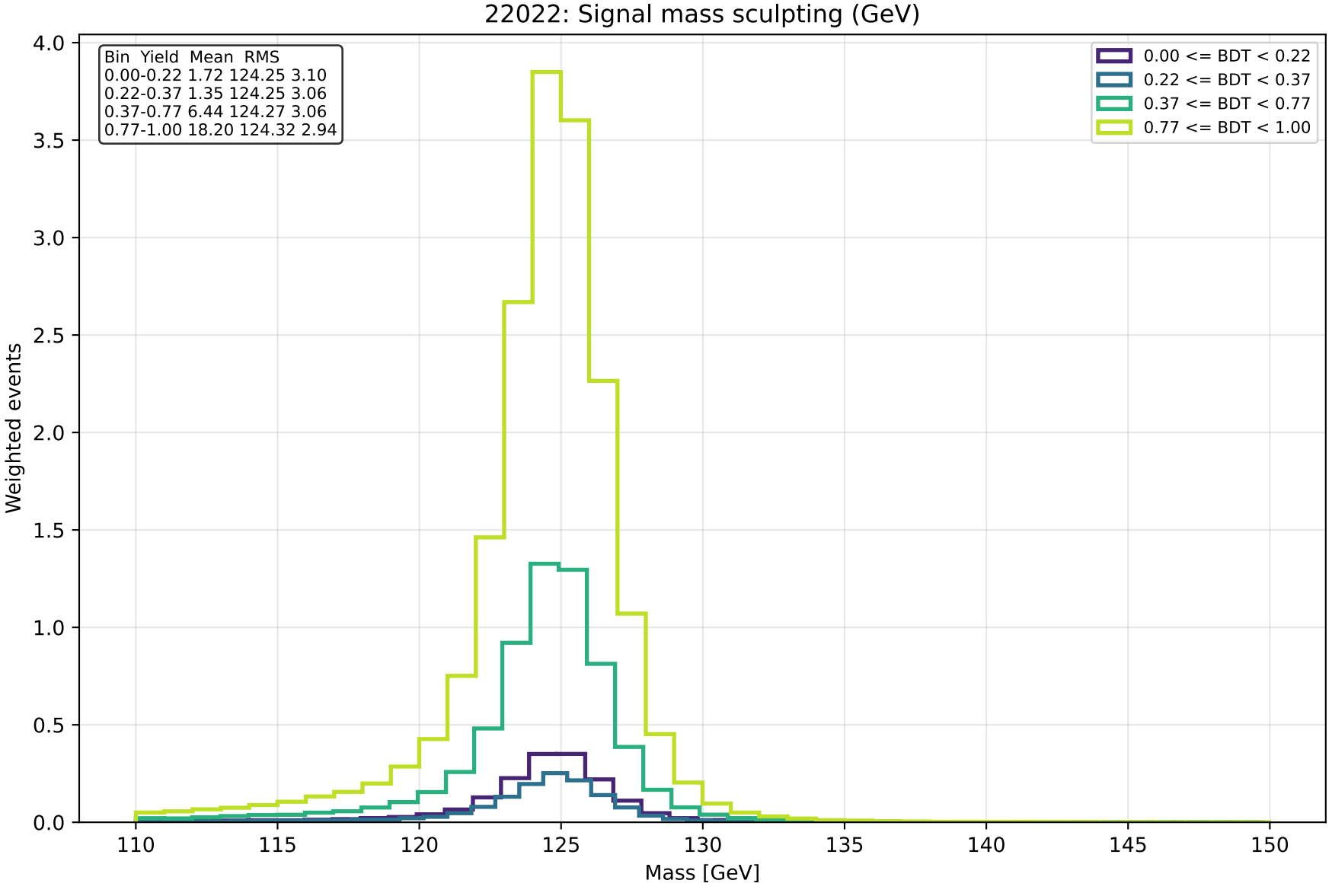
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.220	1.716	74210.971	74210.971	74212.430
2	0.220-0.365	1.346	13495.923	13495.923	13497.067
3	0.365-0.770	6.441	20230.707	20230.707	20236.183
4	0.770-1.000	18.203	6729.336	6729.336	6744.808

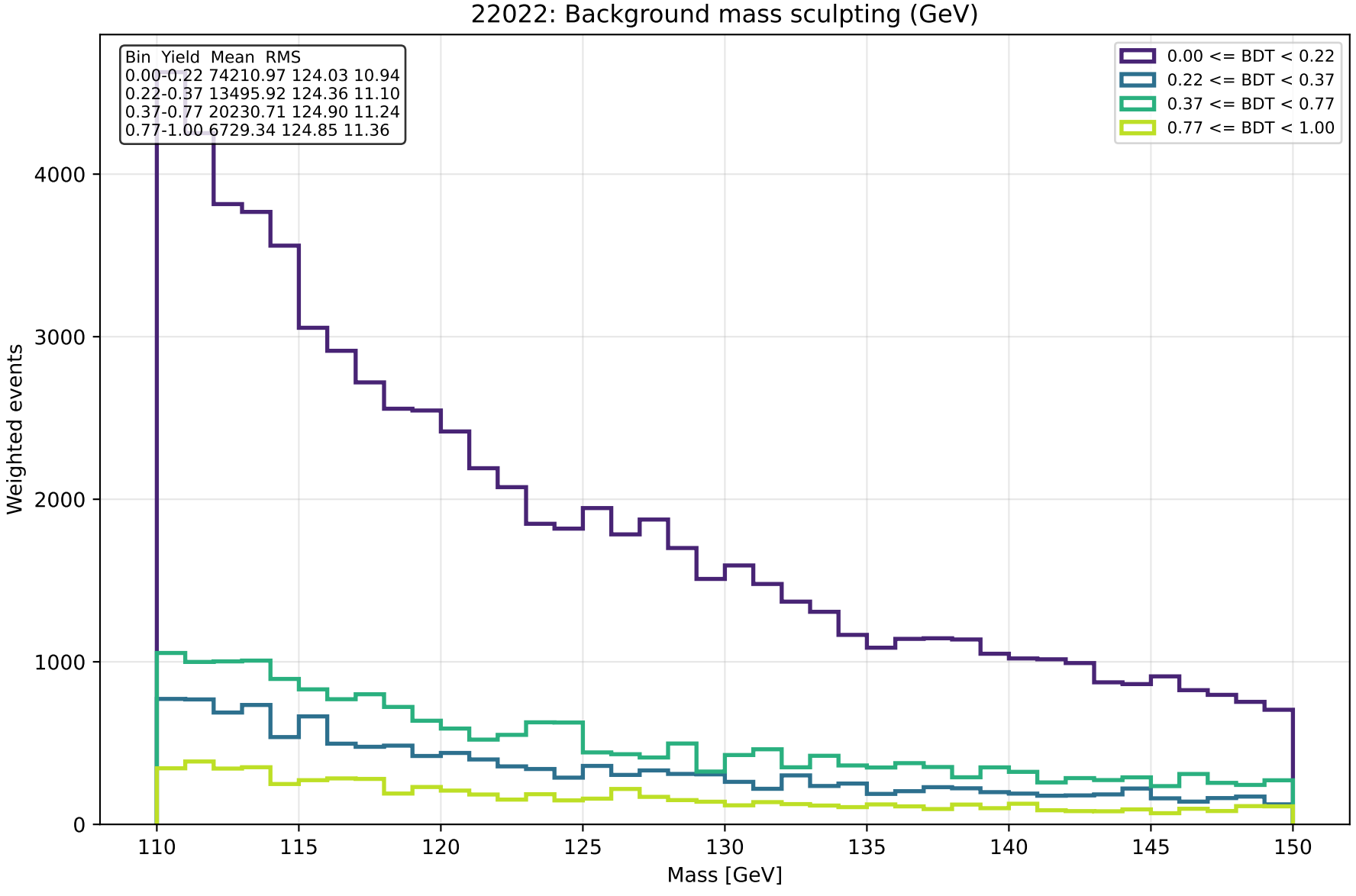

22022: Profile likelihood scan (simultaneous shape fit)
Zexp = 0.444
q0 = 0.197
μhat = 0.850
Δ(-2 ln L)
Signal strength μ
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0

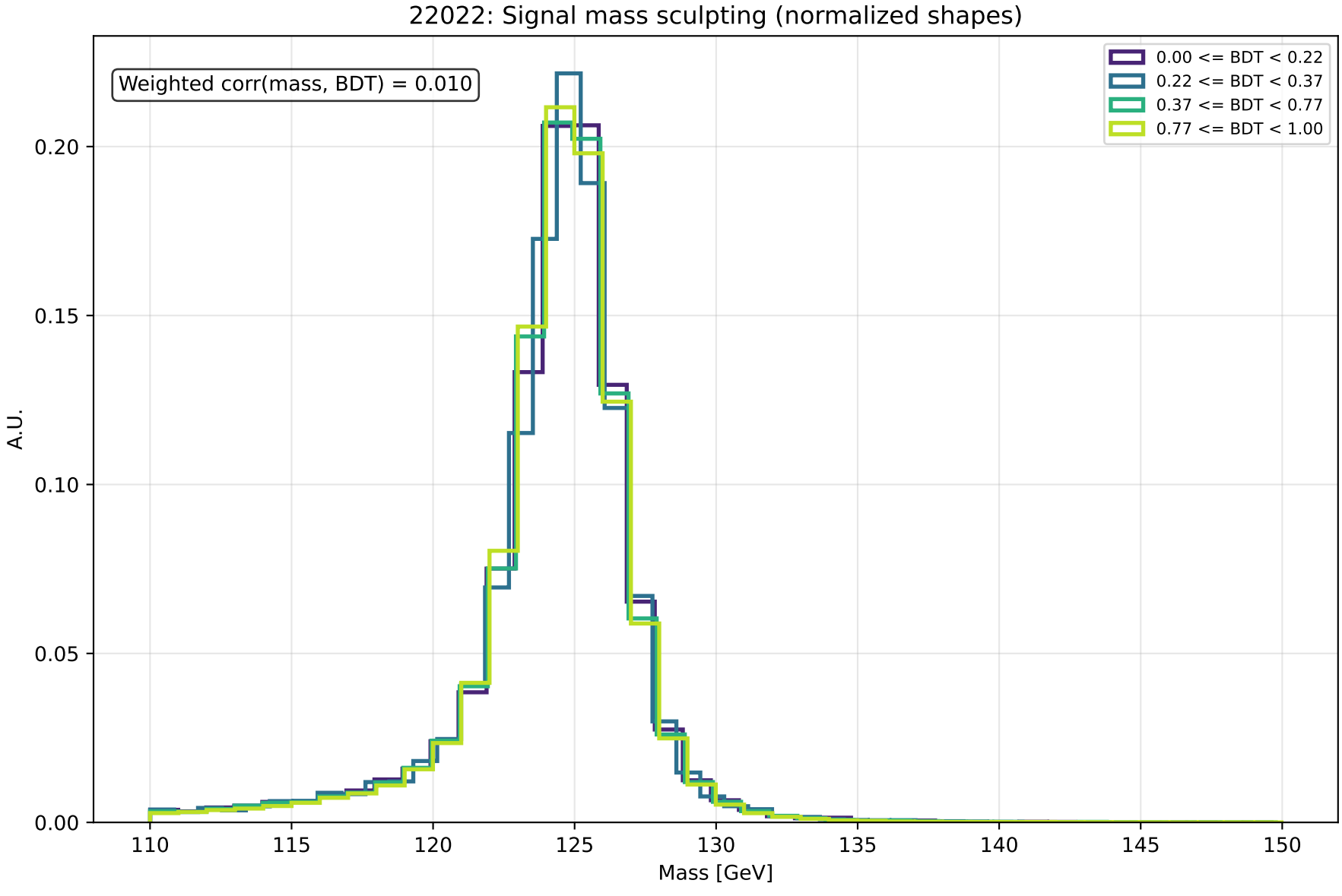

22022: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.220
Bin 2: 0.220-0.365
Bin 3: 0.365-0.770
Bin 4: 0.770-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

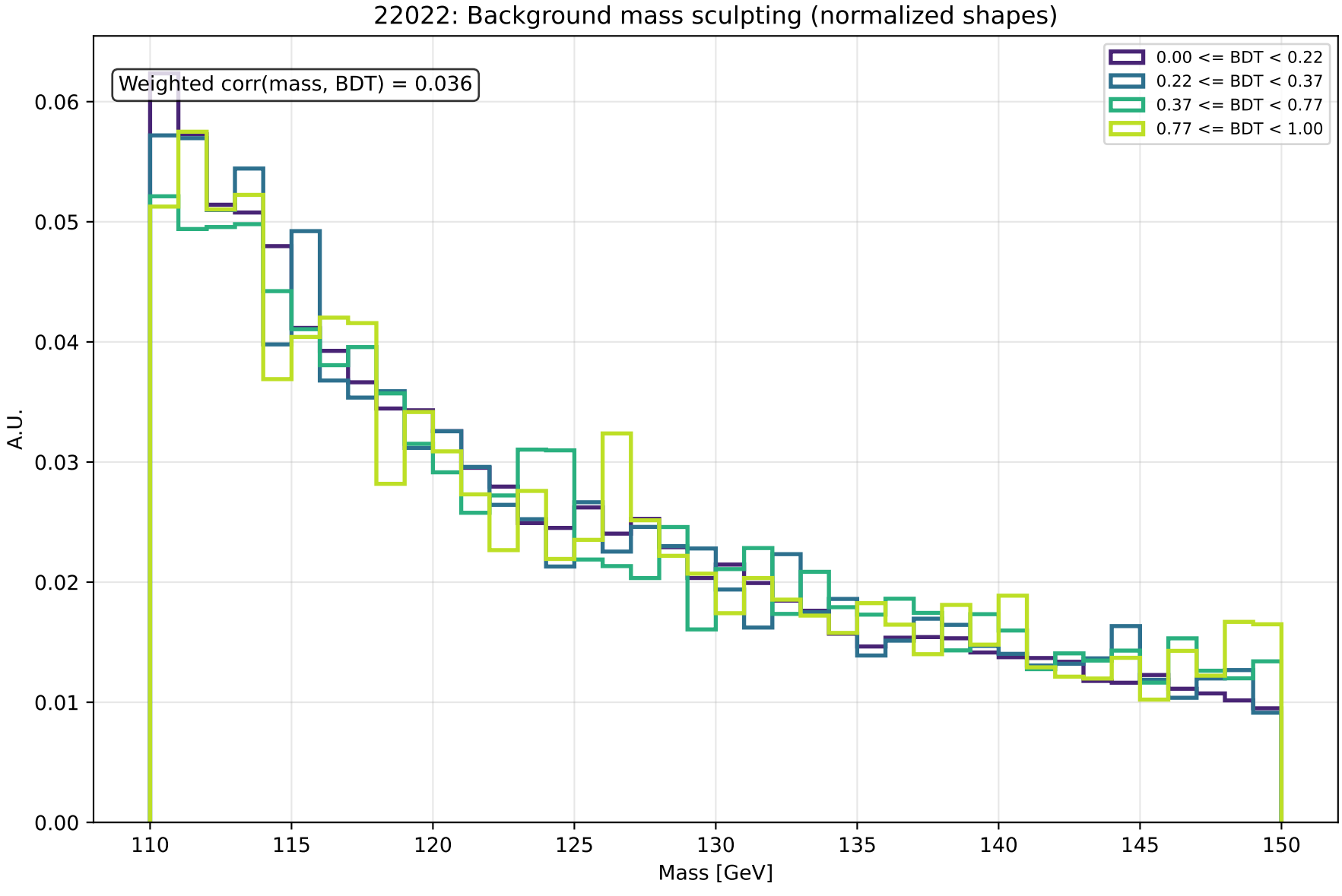
22022: Expected yield per optimized BDT bin (110-150 GeV)
S total=27.71, B total=114666.94

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.220	1.716	6.2%	100.0%	74210.971	0.000242
0.220-0.365	1.346	4.9%	93.8%	13495.923	0.000642
0.365-0.770	6.441	23.2%	88.9%	20230.707	0.000914
0.770-1.000	18.203	65.7%	65.7%	6729.336	0.0027

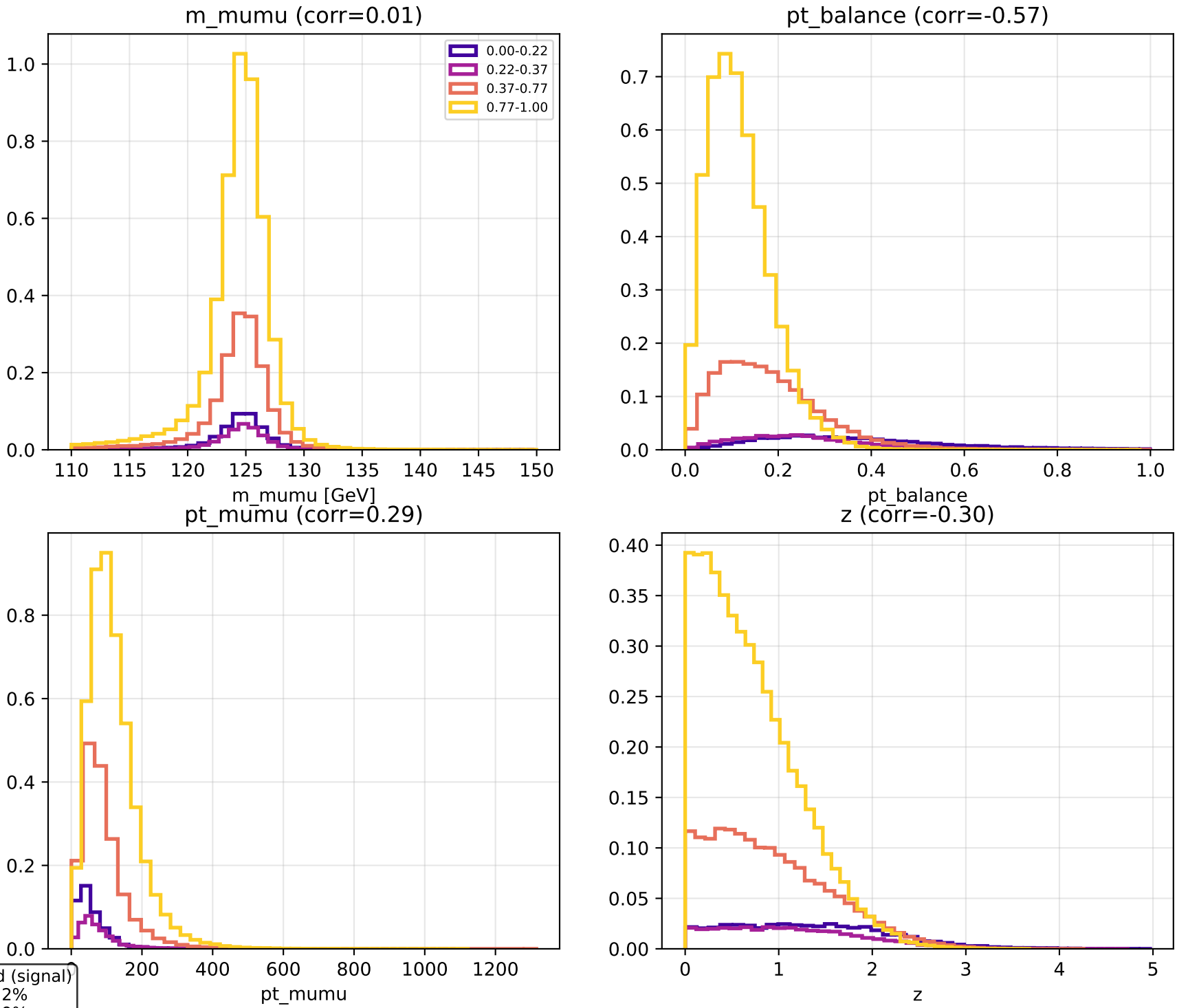

22022: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.22 1.72 124.25 3.10
0.22-0.37 1.35 124.25 3.06
0.37-0.77 6.44 124.27 3.06
0.77-1.00 18.20 124.32 2.94
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
110
115
120
125
130
135
140
145
150


22022: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.22 74210.97 124.03 10.94
0.22-0.37 13495.92 124.36 11.10
0.37-0.77 20230.71 124.90 11.24
0.77-1.00 6729.34 124.85 11.36
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
Weighted events
0
1000
2000
3000
4000
110
115
120
125
130
135
140
145
150
Mass [GeV]

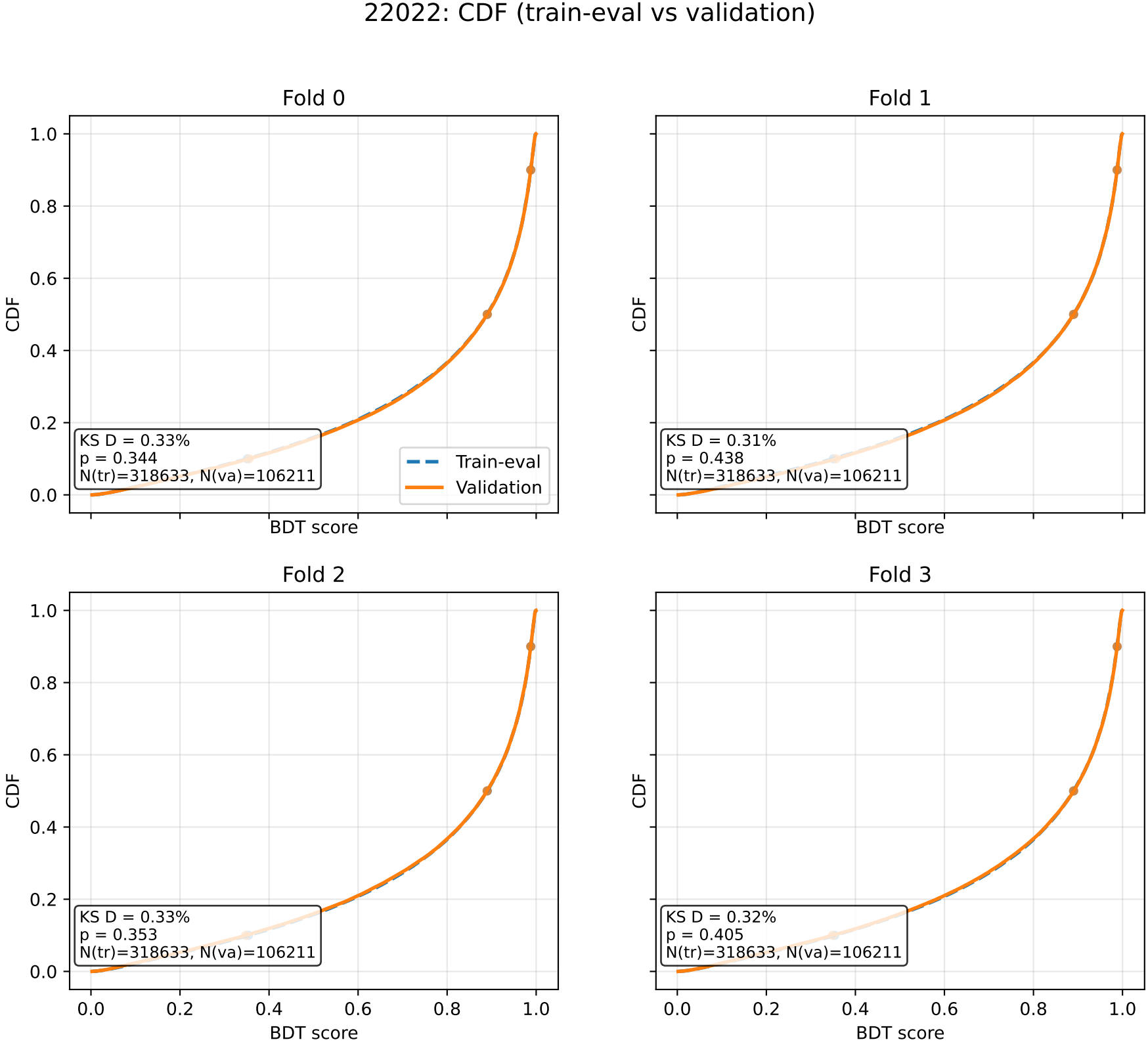

22022: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
110
115
120
125
130
135
140
145
150
Mass [GeV]


22022: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.036
0.00 <= BDT < 0.22
0.22 <= BDT < 0.37
0.37 <= BDT < 0.77
0.77 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

22022: Signal distributions by BDT bin



BDT bin yield (signal)
0.00-0.22: 6.2%
0.22-0.37: 4.9%
0.37-0.77: 23.2%
0.77-1.00: 65.7%

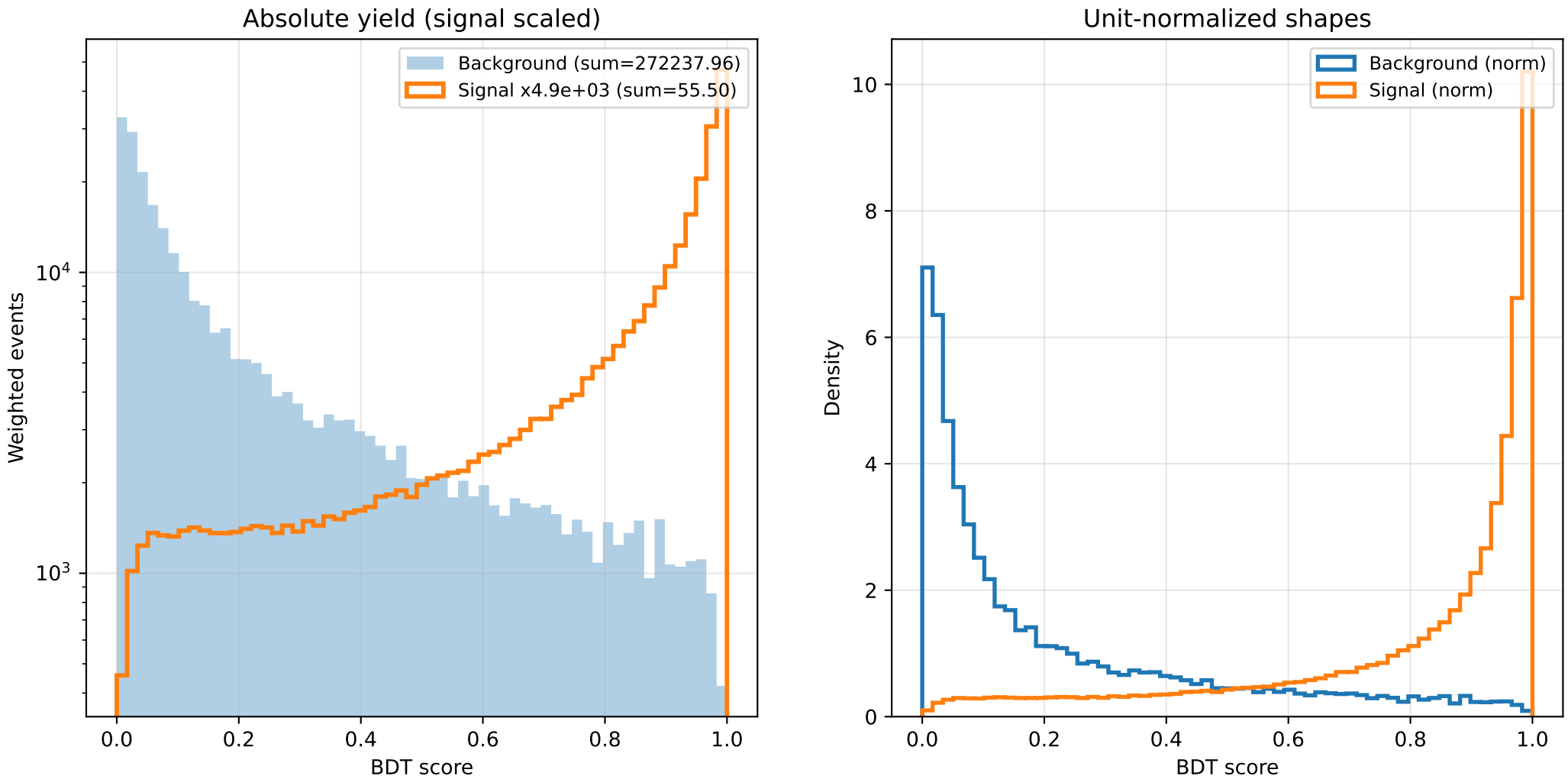

22022: CDF (train-eval vs validation)
Fold 0
KS D = 0.33%
p = 0.344
N(tr)=318633, N(va)=106211
Train-eval
Validation
Fold 1
KS D = 0.31%
p = 0.438
N(tr)=318633, N(va)=106211
Fold 2
KS D = 0.33%
p = 0.353
N(tr)=318633, N(va)=106211
Fold 3
KS D = 0.32%
p = 0.405
N(tr)=318633, N(va)=106211
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

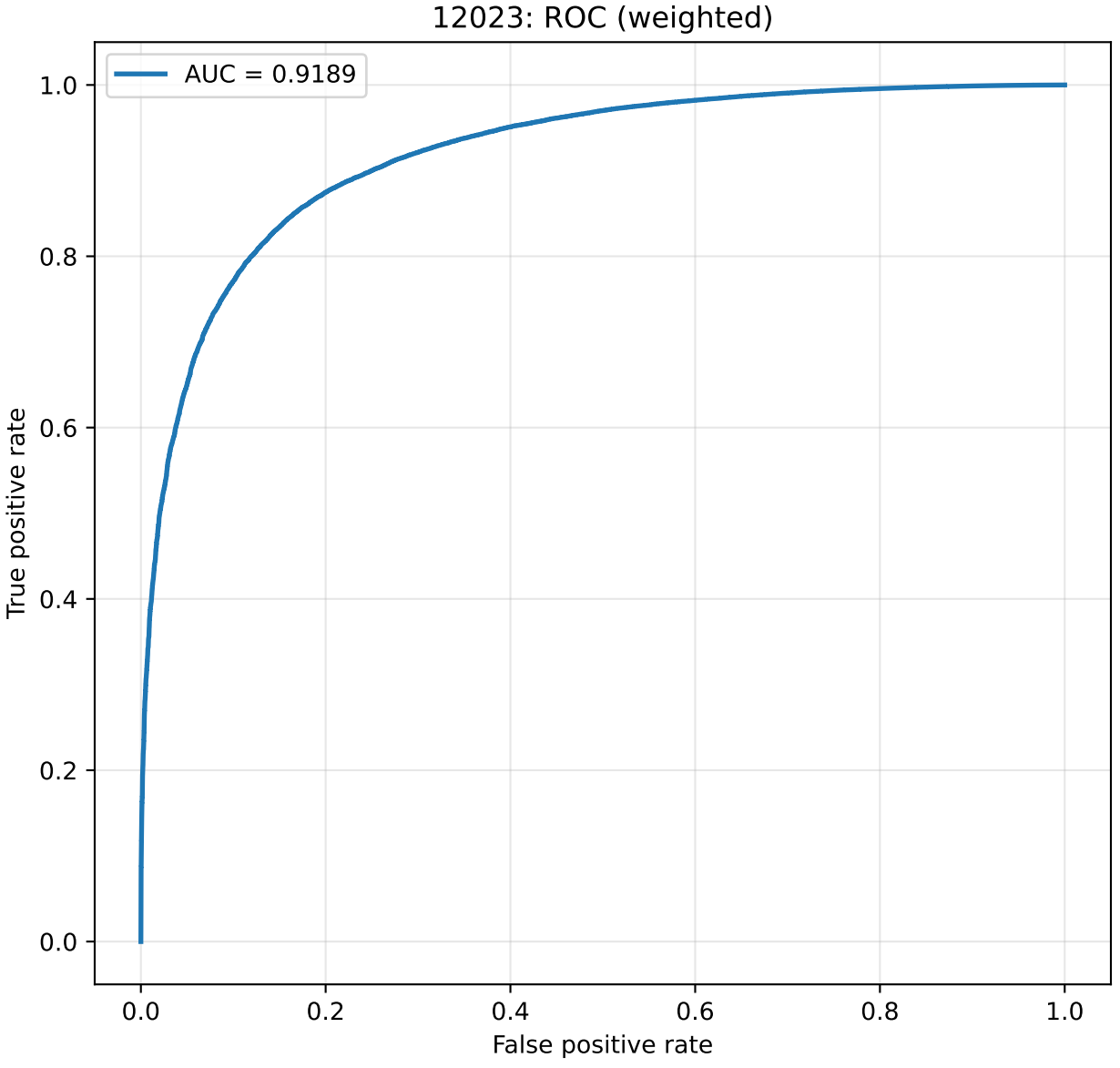
22022: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.33%	106211	0.001	0.20%	FALSE
1	0.31%	106211	0.001	0.20%	FALSE
2	0.33%	106211	0.001	0.20%	FALSE
3	0.32%	106211	0.001	0.20%	FALSE

22022: Top-10 feature importance (gain)
22022_featureimportance_DY.txt

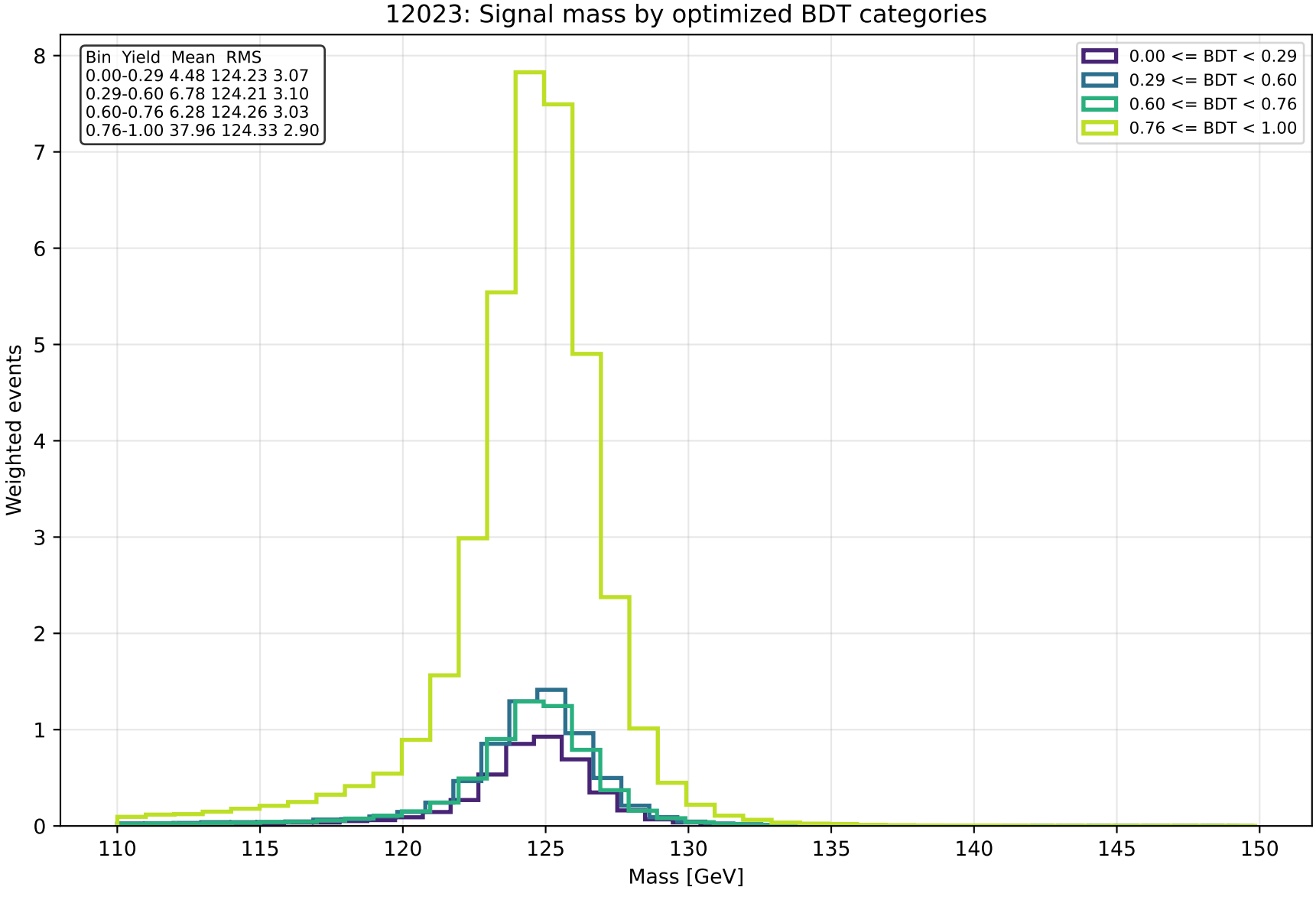
Feature	Gain %
pt_balance	22.93%
pt_mumu	9.72%
z	9.36%
dphi_mumu_jj	7.60%
deta_jj	7.10%
njets	6.37%
pt_j2	5.30%
pt_j1	4.85%
jet1_eta	3.36%
mu1_pt_over_m	2.82%

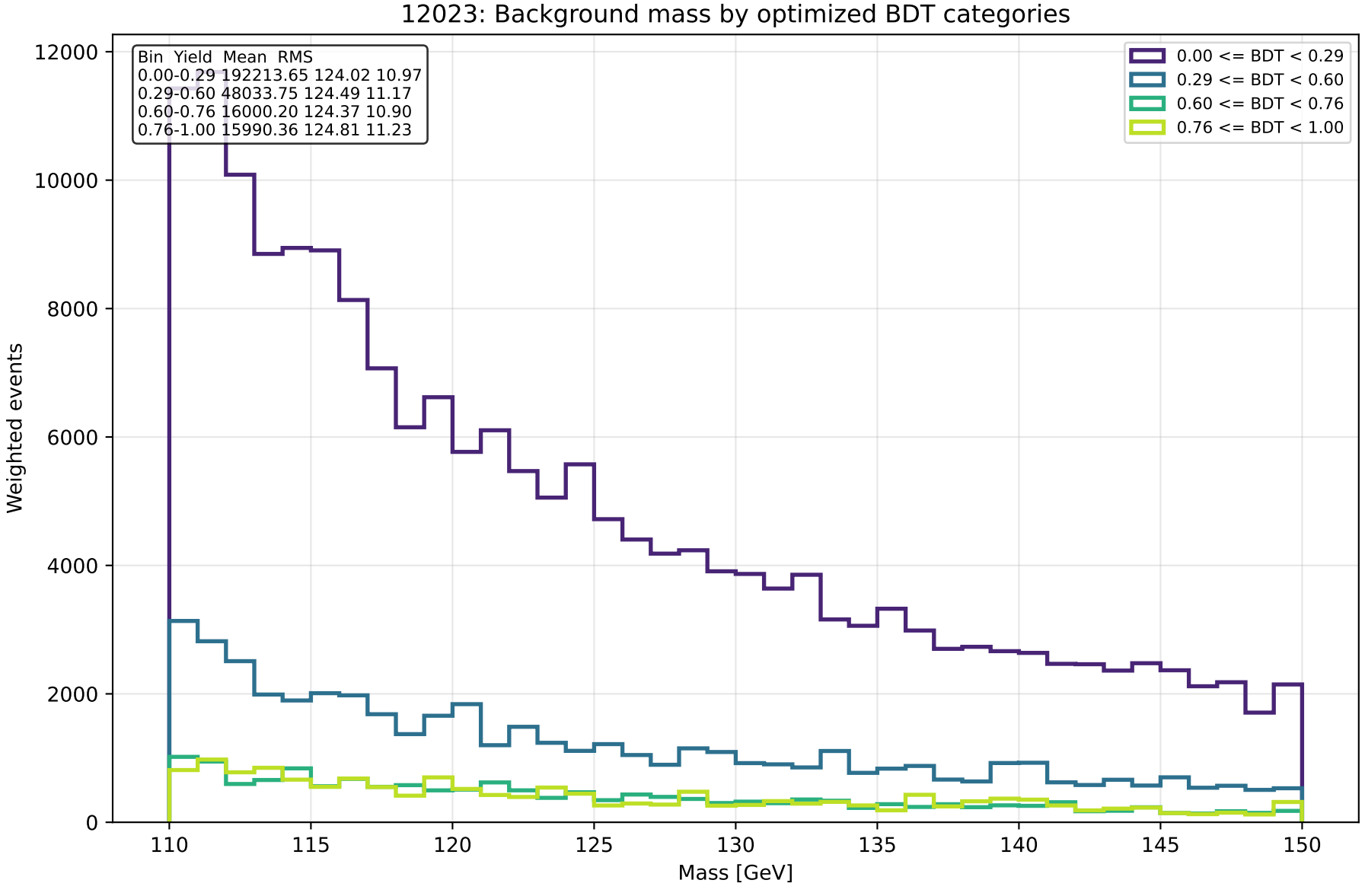

12023: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=272237.96)
Signal x4.9e+03 (sum=55.50)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

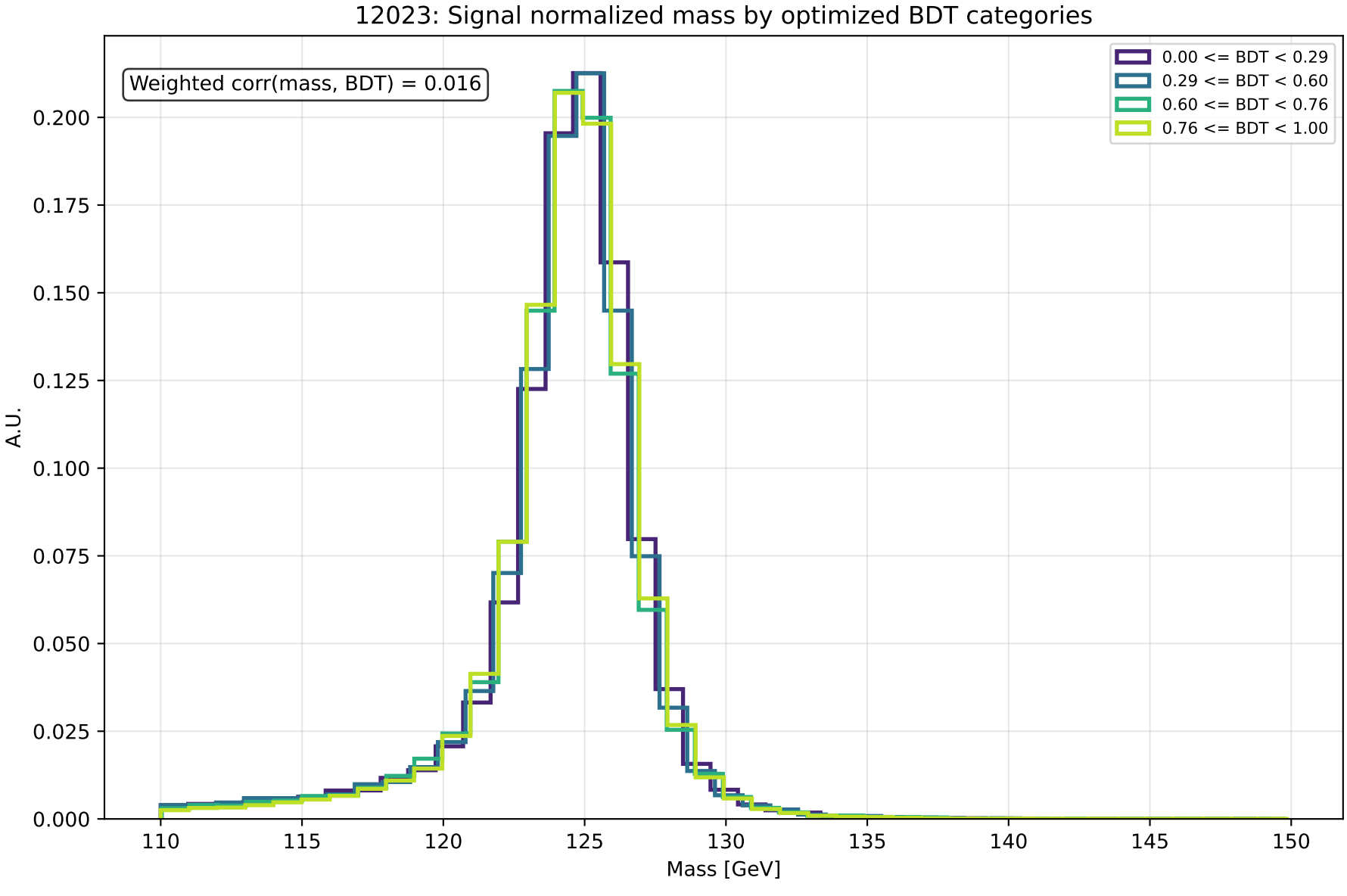

12023: ROC (weighted)
AUC = 0.9189
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

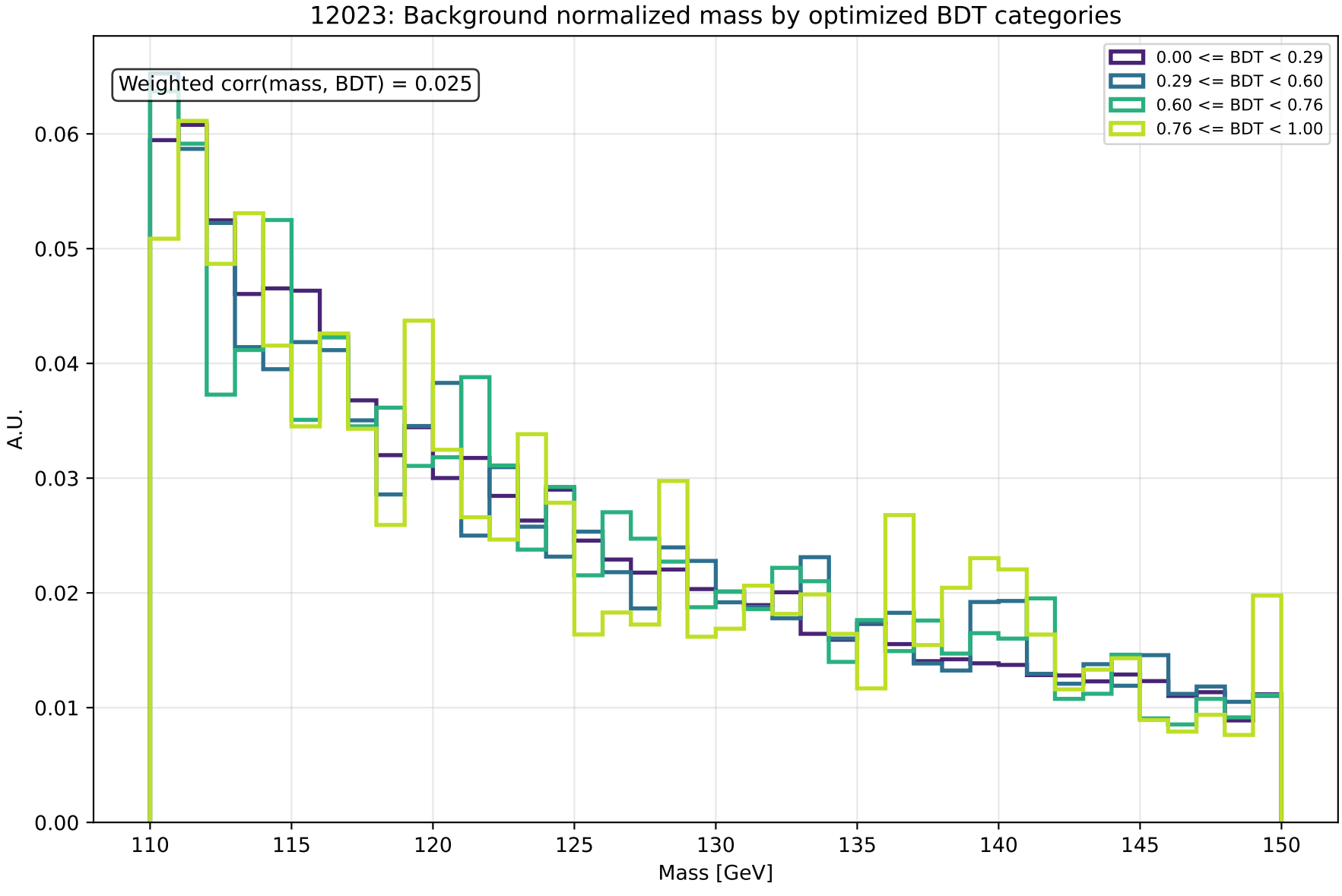
12023: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=0.000, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.287	4.481	192213.652	2.33e-05	0.010
2	0.287-0.599	6.783	48033.752	0.000141	0.031
3	0.599-0.764	6.276	16000.198	0.000392	0.050
4	0.764-1.000	37.960	15990.363	0.00237	0.300


12023: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.29 4.48 124.23 3.07
0.29-0.60 6.78 124.21 3.10
0.60-0.76 6.28 124.26 3.03
0.76-1.00 37.96 124.33 2.90
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
Mass [GeV]
0
1
2
3
4
5
6
7
8
110
115
120
125
130
135
140
145
150


12023: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.29 192213.65 124.02 10.97
0.29-0.60 48033.75 124.49 11.17
0.60-0.76 16000.20 124.37 10.90
0.76-1.00 15990.36 124.81 11.23
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
Mass [GeV]
0
2000
4000
6000
8000
10000
12000
110
115
120
125
130
135
140
145
150


12023: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.016
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

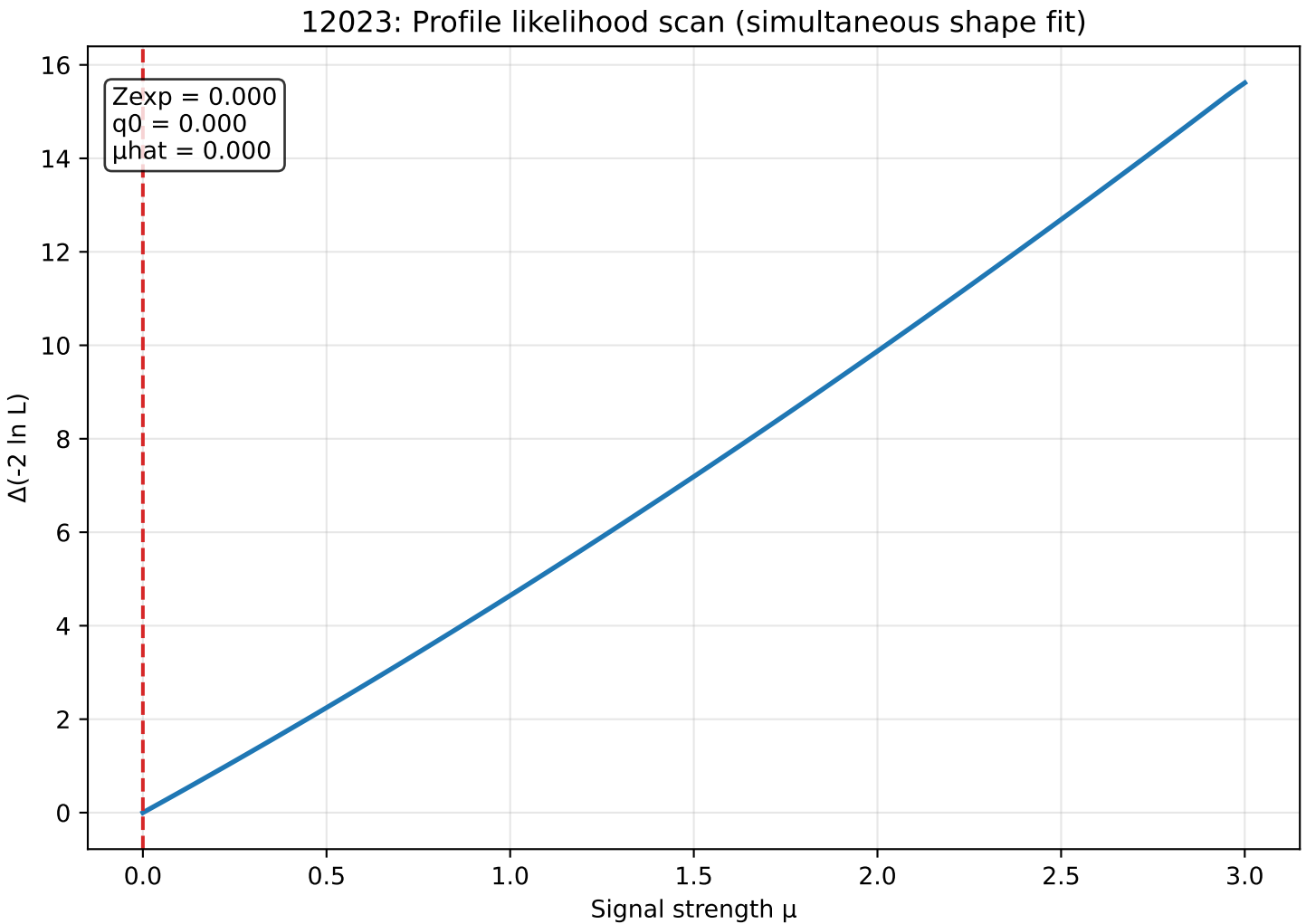

12023: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.025
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

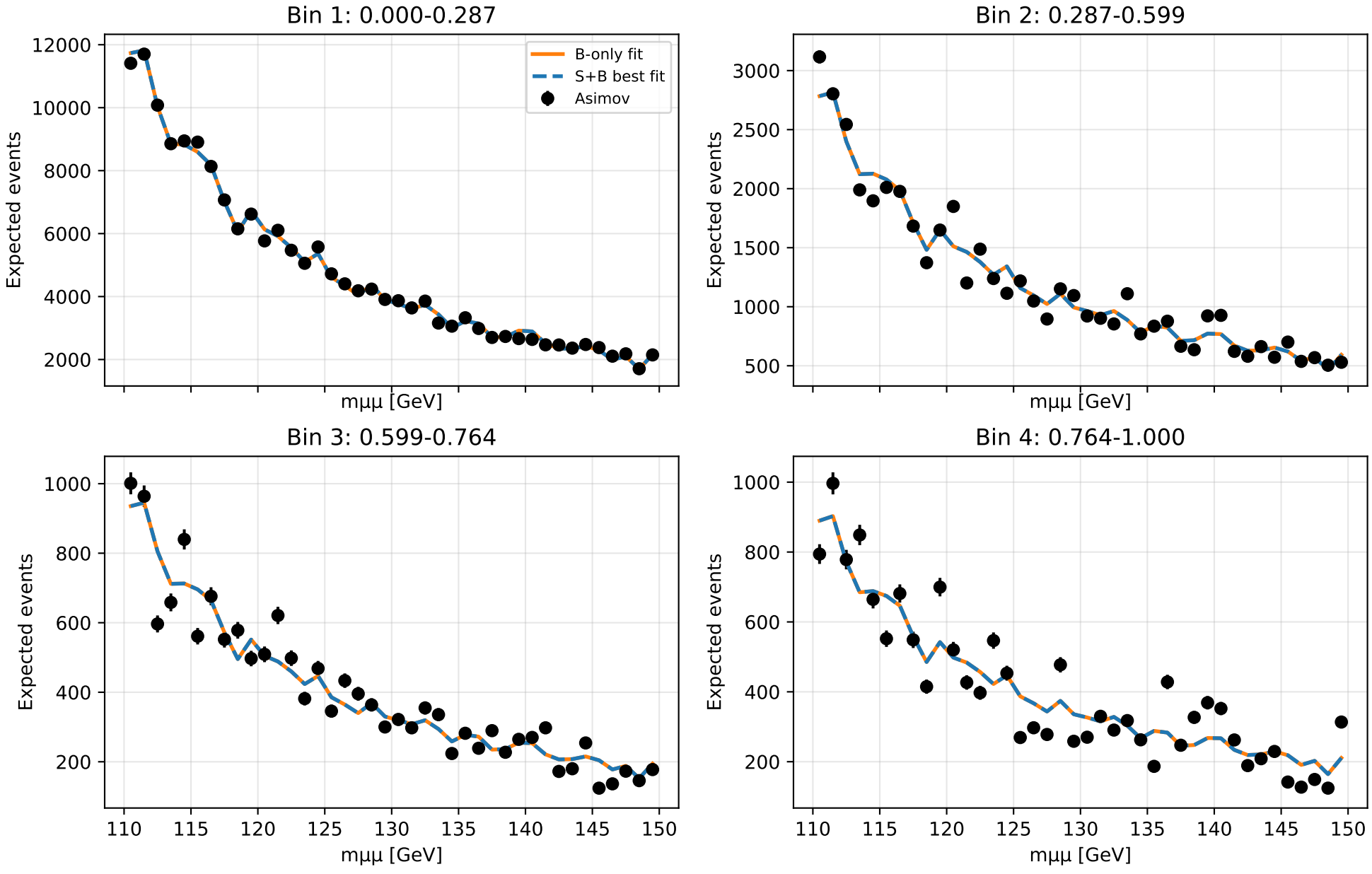
12023: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	0.0000
q0	0.0000
mu_hat	0.0000
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

12023: Shape-likelihood bin yields (Asimov)

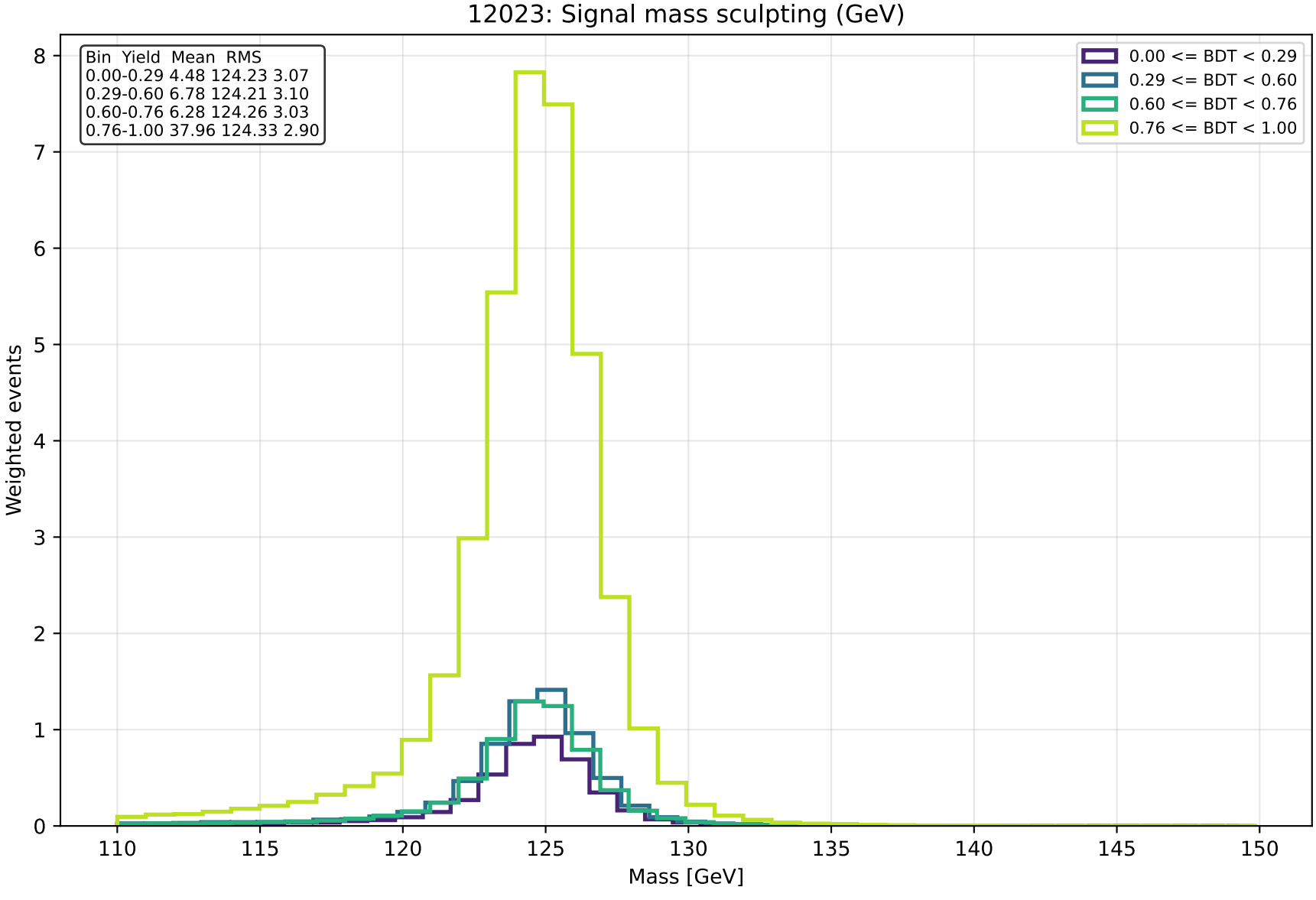
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.287	4.481	192213.652	192213.652	192213.652
2	0.287-0.599	6.783	48033.752	48033.752	48033.752
3	0.599-0.764	6.276	16000.198	16000.198	16000.198
4	0.764-1.000	37.960	15990.363	15990.363	15990.363

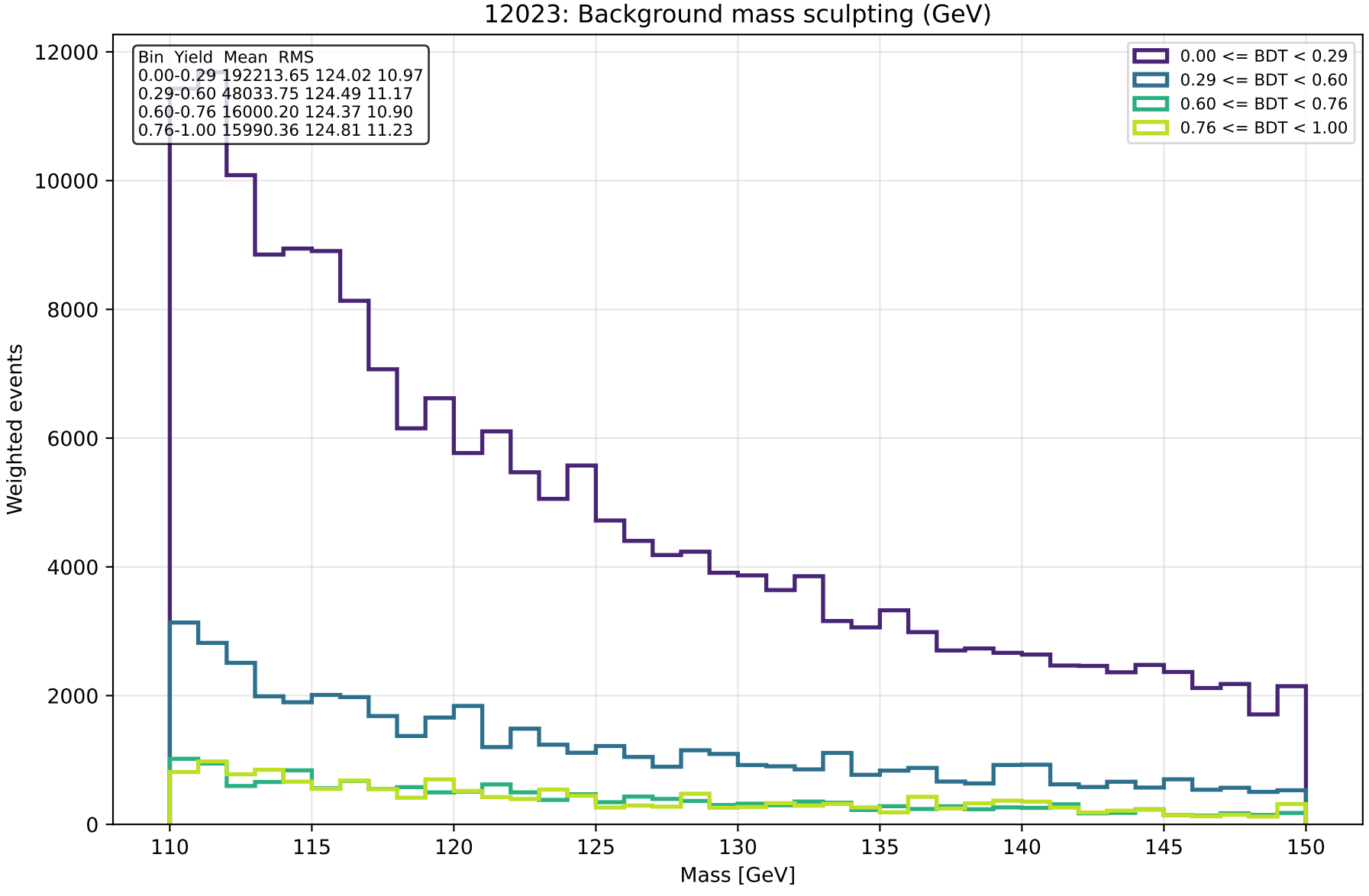

12023: Profile likelihood scan (simultaneous shape fit)
Zexp = 0.000
q0 = 0.000
μhat = 0.000
Δ(-2 ln L)
Signal strength μ
0
2
4
6
8
10
12
14
16
0.0
0.5
1.0
1.5
2.0
2.5
3.0

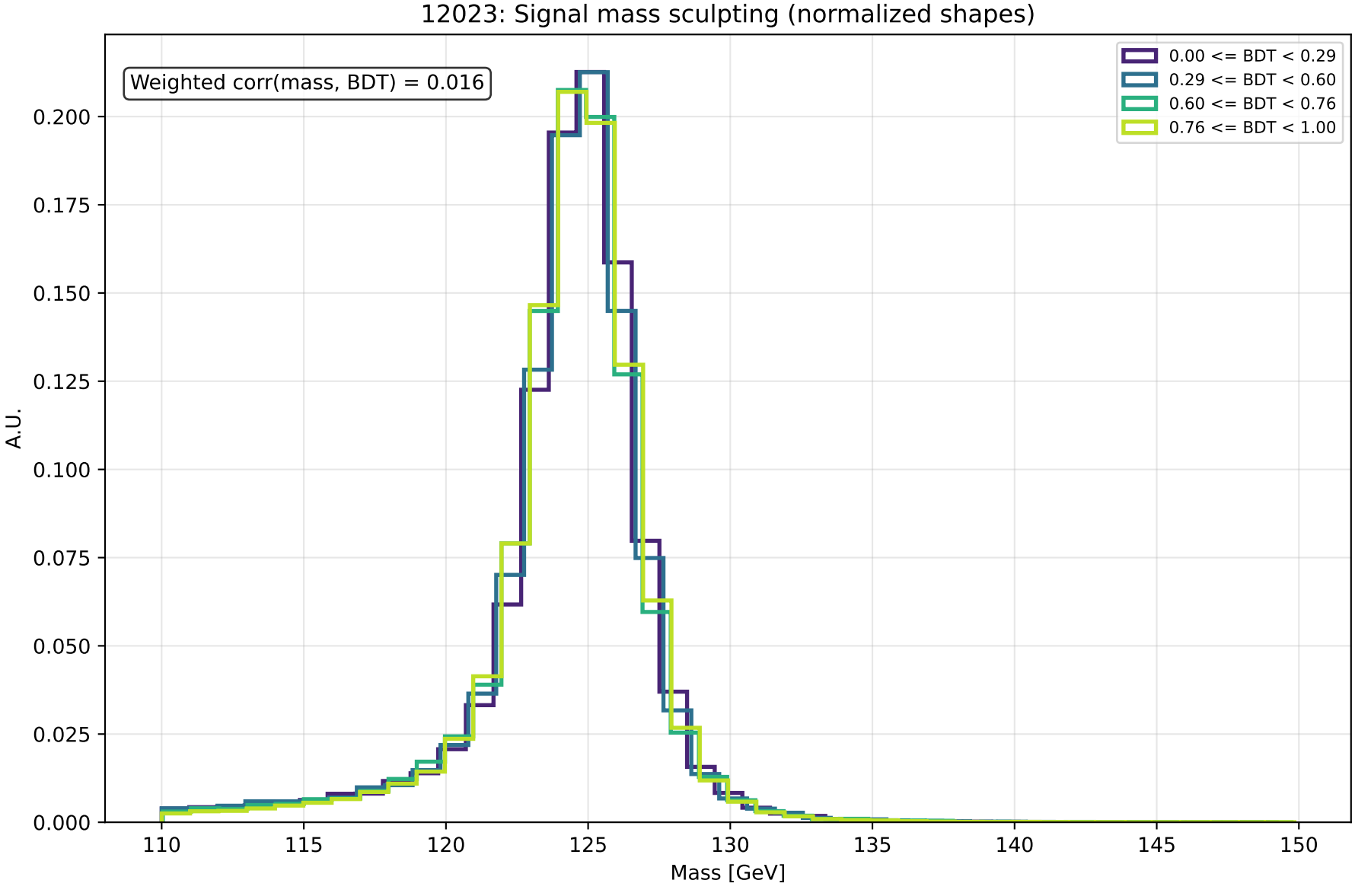

12023: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.287
Bin 2: 0.287-0.599
Bin 3: 0.599-0.764
Bin 4: 0.764-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

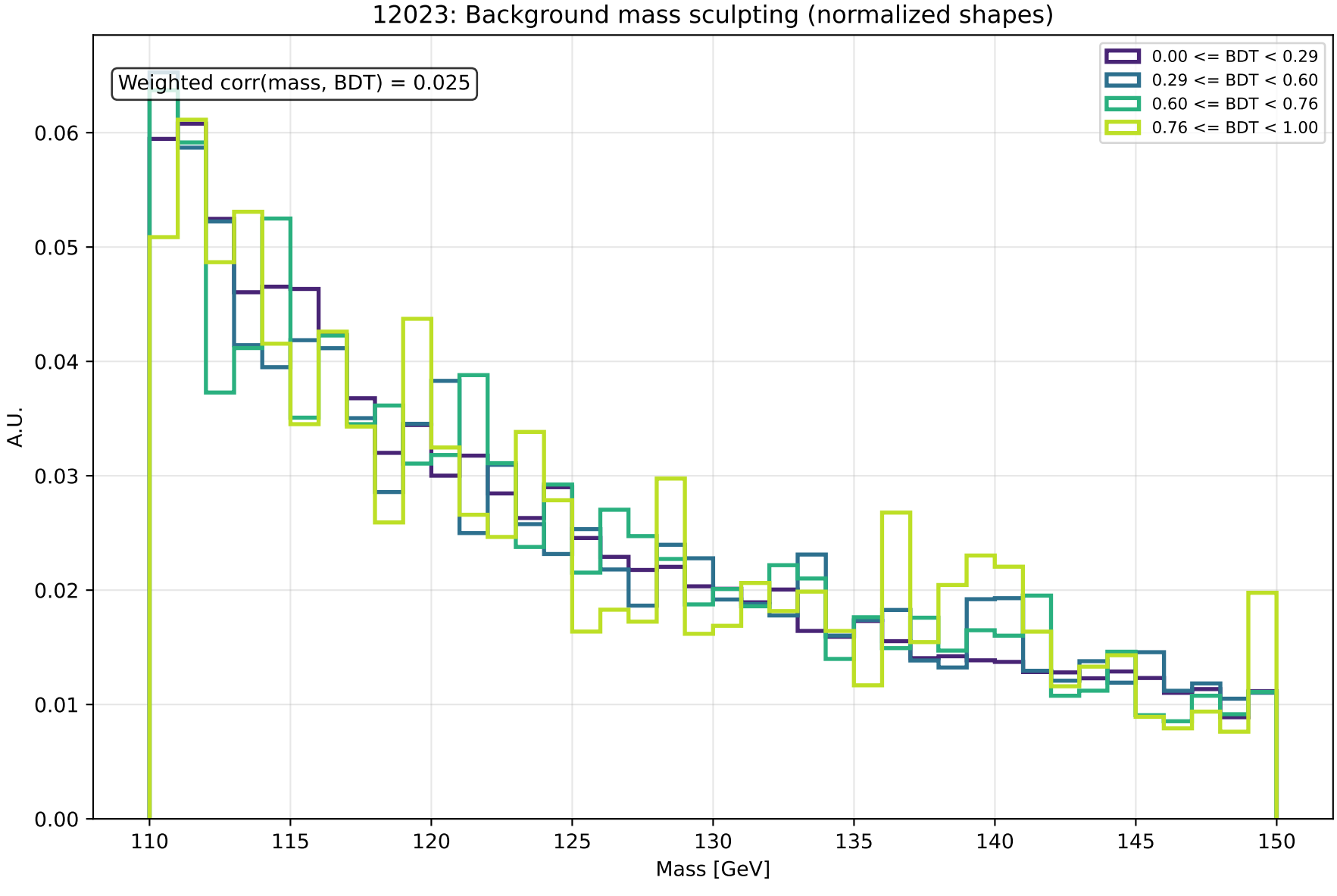
12023: Expected yield per optimized BDT bin (110-150 GeV)
S total=55.50, B total=272237.96

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.287	4.481	8.1%	100.0%	192213.652	0.000204
0.287-0.599	6.783	12.2%	91.9%	48033.752	0.000638
0.599-0.764	6.276	11.3%	79.7%	16000.198	0.00138
0.764-1.000	37.960	68.4%	68.4%	15990.363	0.00237

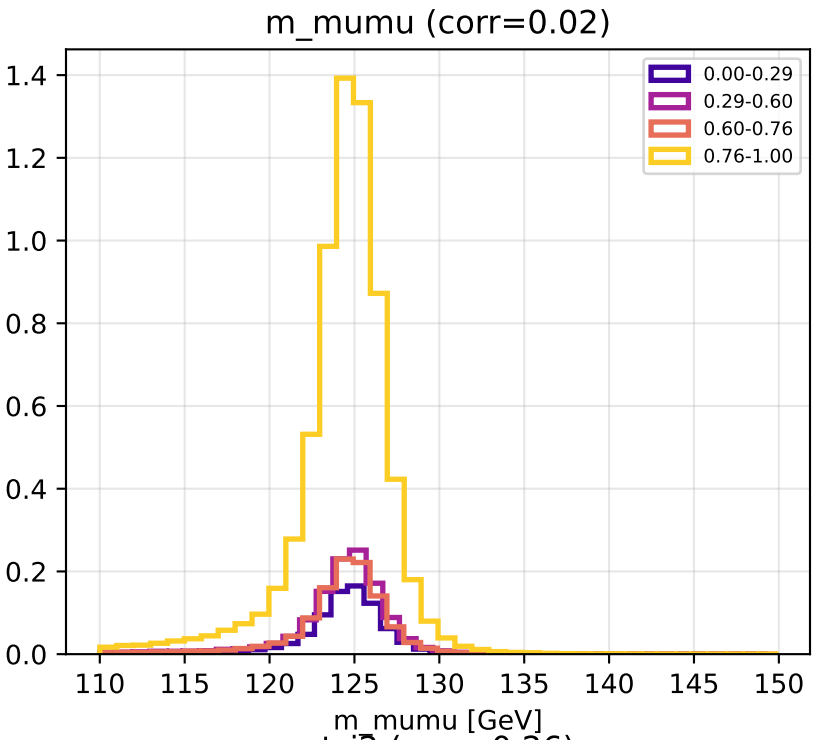

12023: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.29 4.48 124.23 3.07
0.29-0.60 6.78 124.21 3.10
0.60-0.76 6.28 124.26 3.03
0.76-1.00 37.96 124.33 2.90
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
Mass [GeV]
0
1
2
3
4
5
6
7
8
110
115
120
125
130
135
140
145
150

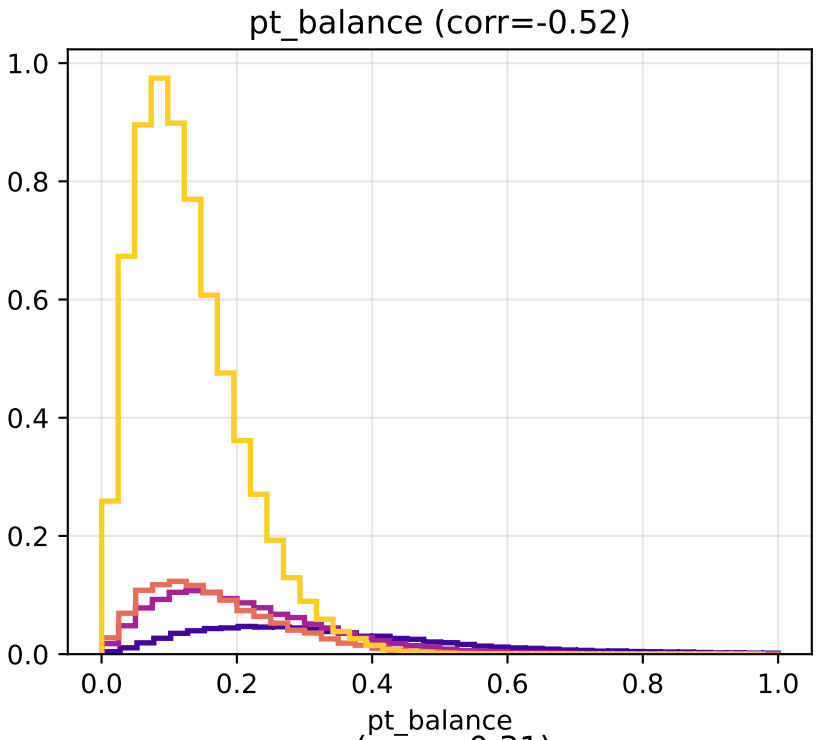

12023: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.29 192213.65 124.02 10.97
0.29-0.60 48033.75 124.49 11.17
0.60-0.76 16000.20 124.37 10.90
0.76-1.00 15990.36 124.81 11.23
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
Weighted events
0
2000
4000
6000
8000
10000
12000
110
115
120
125
130
135
140
145
150
Mass [GeV]

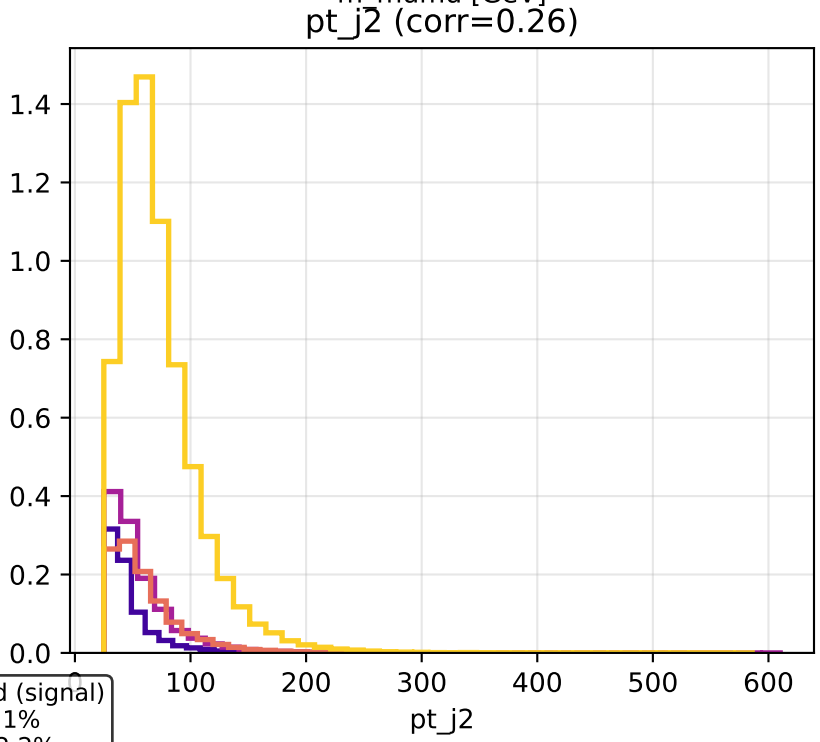

12023: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.016
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

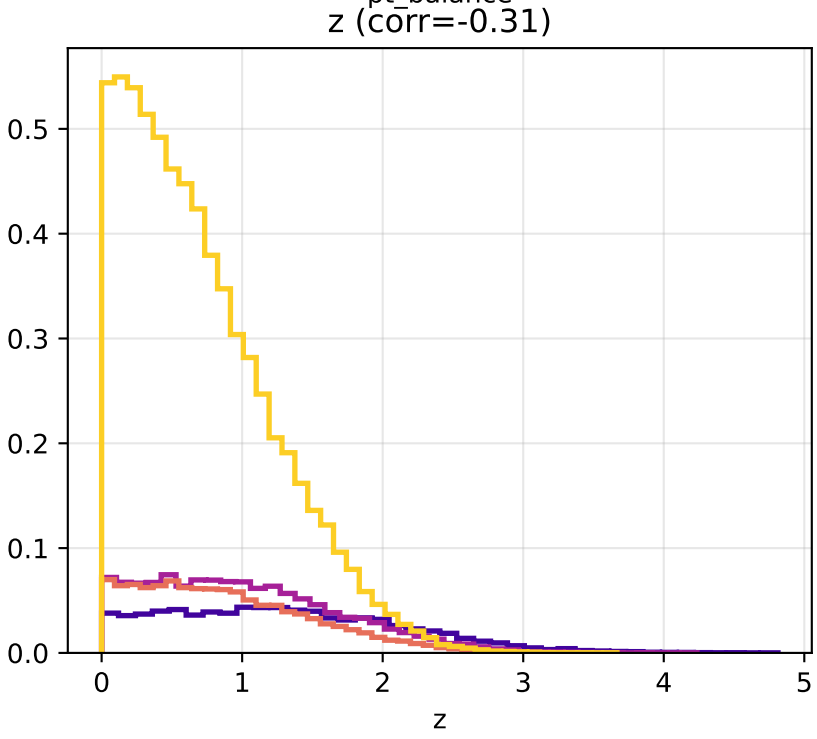

12023: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.025
0.00 <= BDT < 0.29
0.29 <= BDT < 0.60
0.60 <= BDT < 0.76
0.76 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

12023: Signal distributions by BDT bin

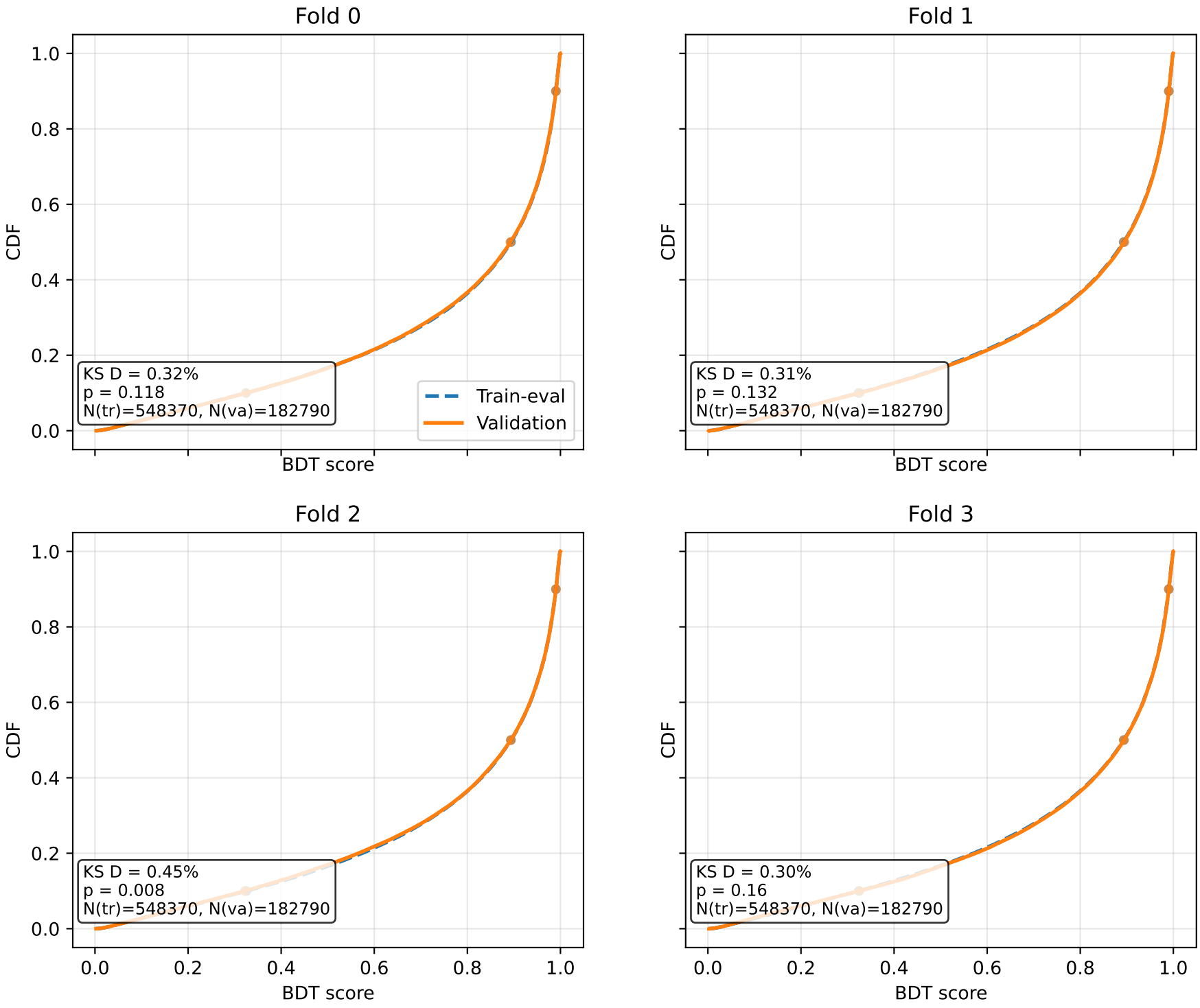








BDT bin yield (signal)
0.00-0.29: 8.1%
0.29-0.60: 12.2%
0.60-0.76: 11.3%
0.76-1.00: 68.4%

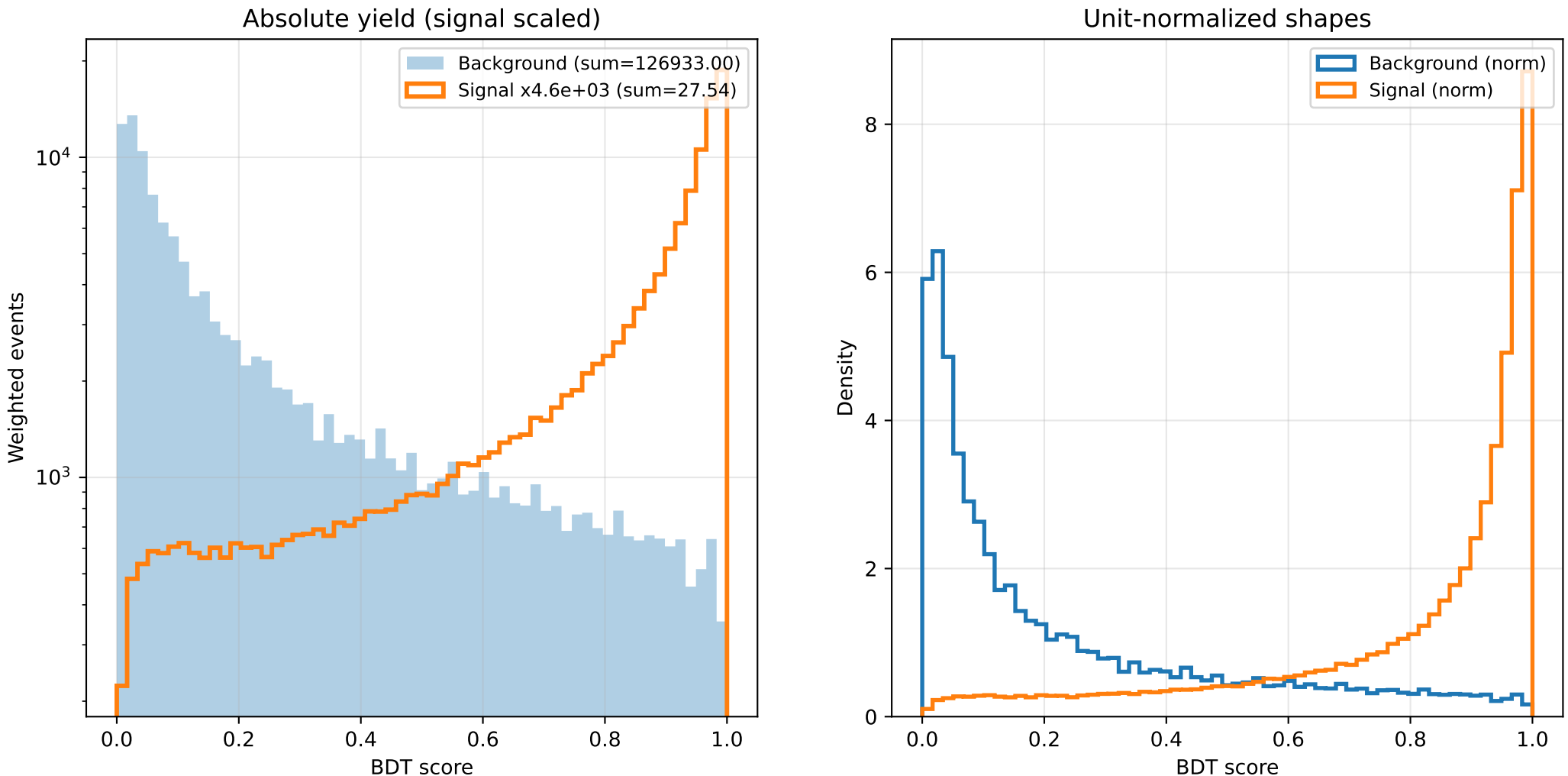

12023: CDF (train-eval vs validation)
Fold 0
KS D = 0.32%
p = 0.118
N(tr)=548370, N(va)=182790
Train-eval
Validation
Fold 1
KS D = 0.31%
p = 0.132
N(tr)=548370, N(va)=182790
Fold 2
KS D = 0.45%
p = 0.008
N(tr)=548370, N(va)=182790
Fold 3
KS D = 0.30%
p = 0.16
N(tr)=548370, N(va)=182790
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

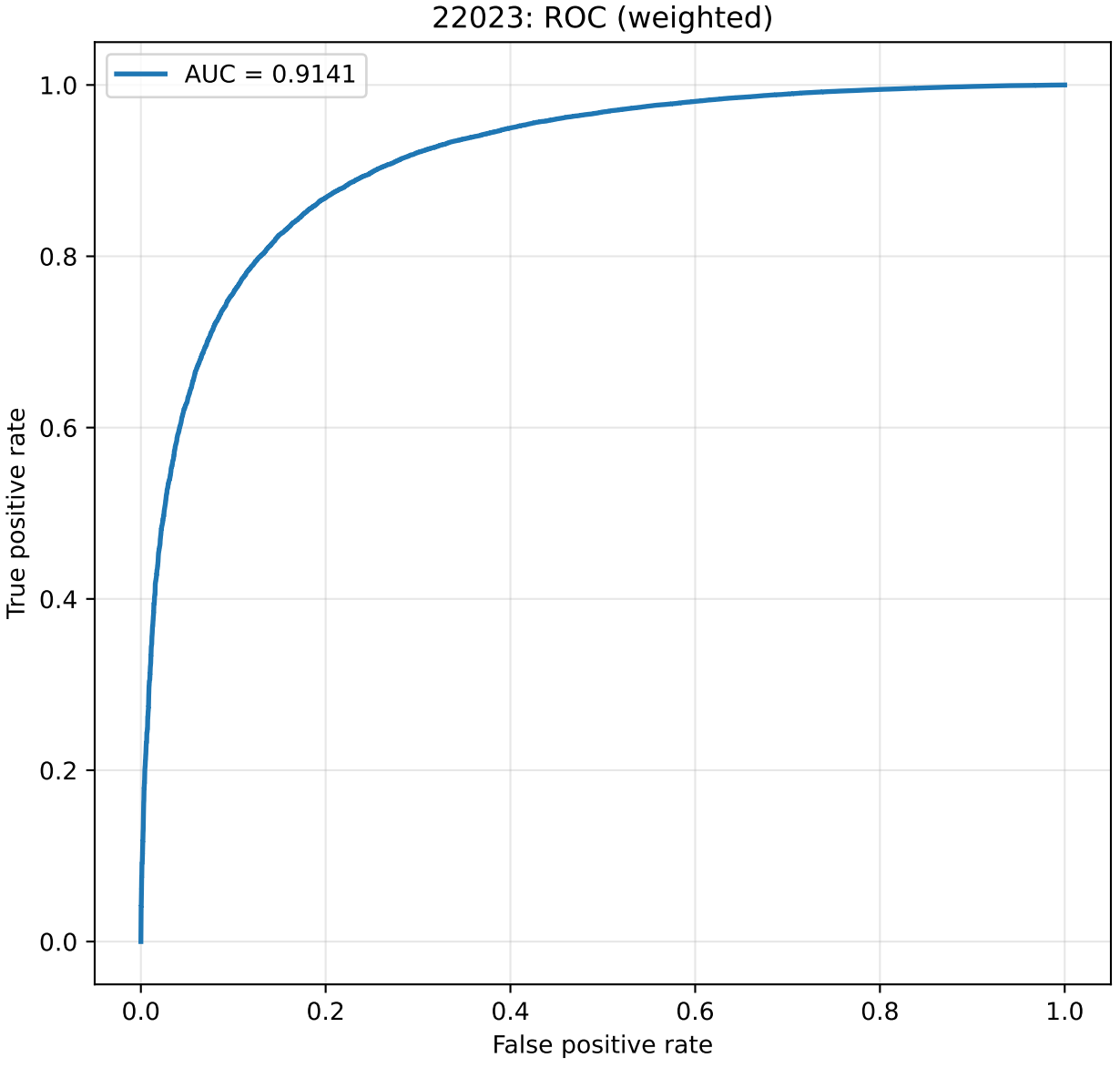
12023: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.32%	182790	0.001	0.20%	FALSE
1	0.31%	182790	0.001	0.20%	FALSE
2	0.45%	182790	0.001	0.20%	FALSE
3	0.30%	182790	0.001	0.20%	FALSE

12023: Top-10 feature importance (gain)
12023_featureimportance_DY.txt

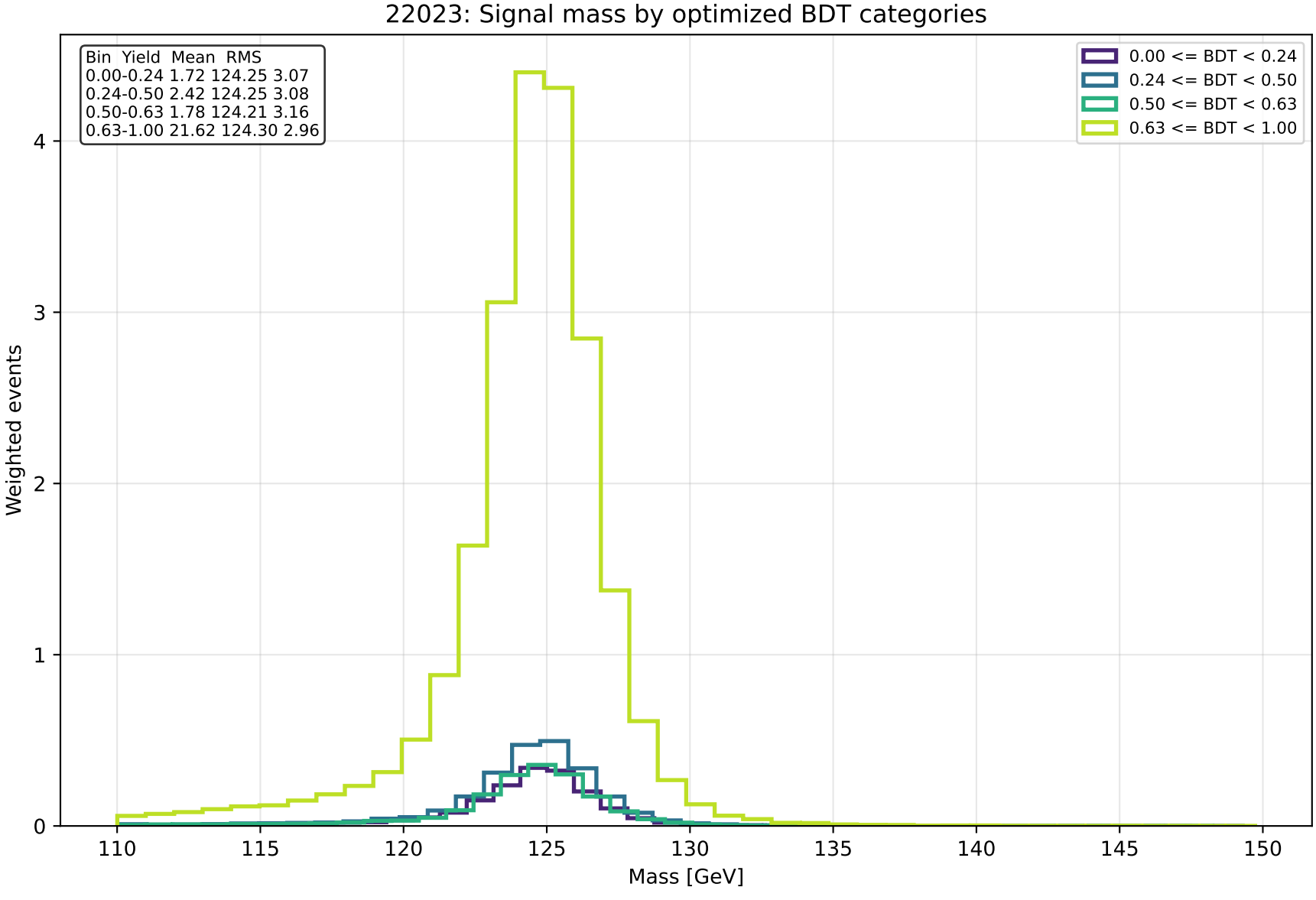
Feature	Gain %
pt_balance	15.97%
pt_j2	9.51%
z	8.51%
dphi_mumu_jj	8.17%
deta_jj	8.07%
pt_mumu	7.81%
pt_j1	7.06%
njets	6.03%
jet1_eta	4.01%
mu2_pt_over_m	2.94%

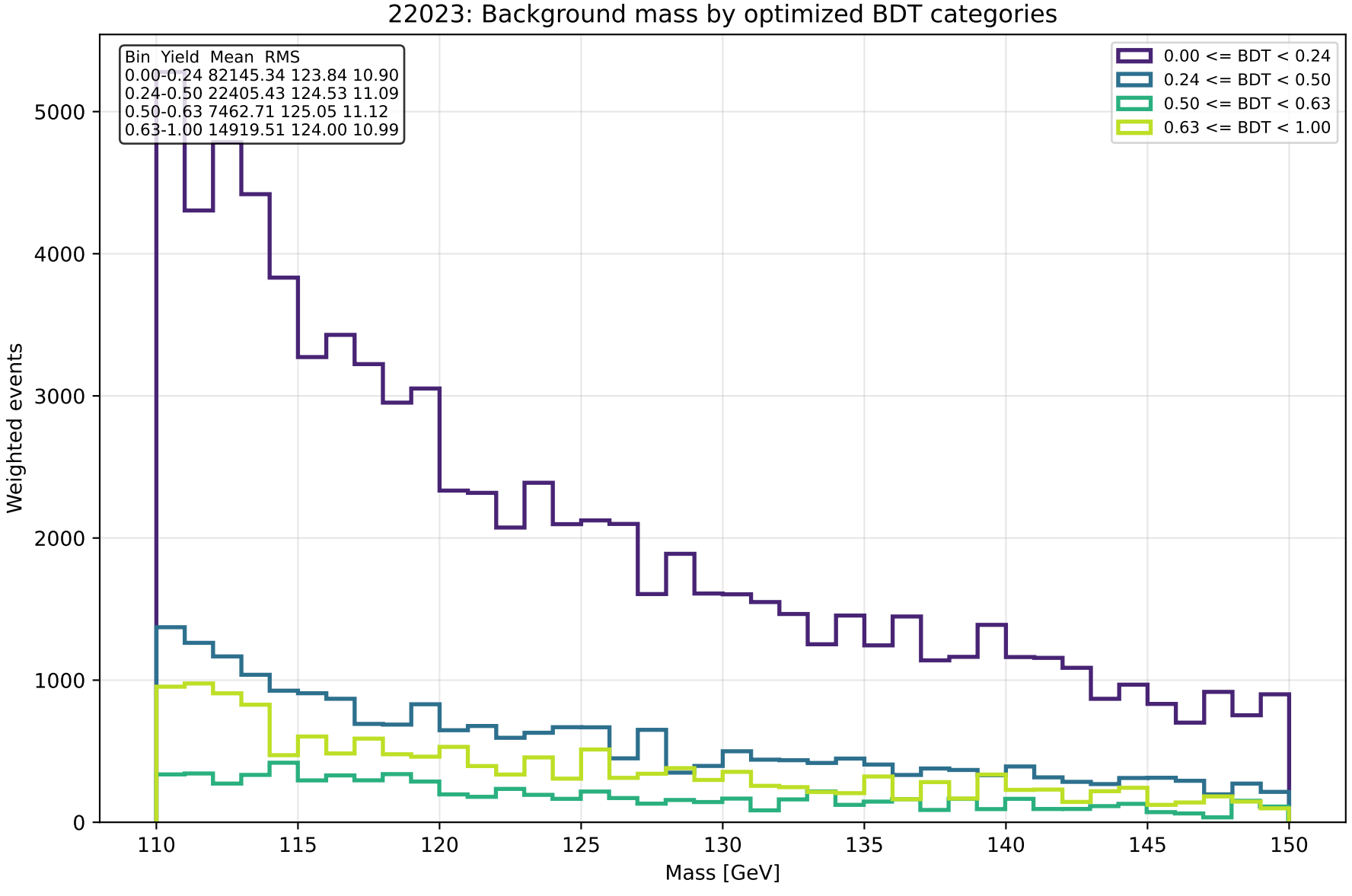

22023: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=126933.00)
Signal x4.6e+03 (sum=27.54)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
2
4
6
8
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

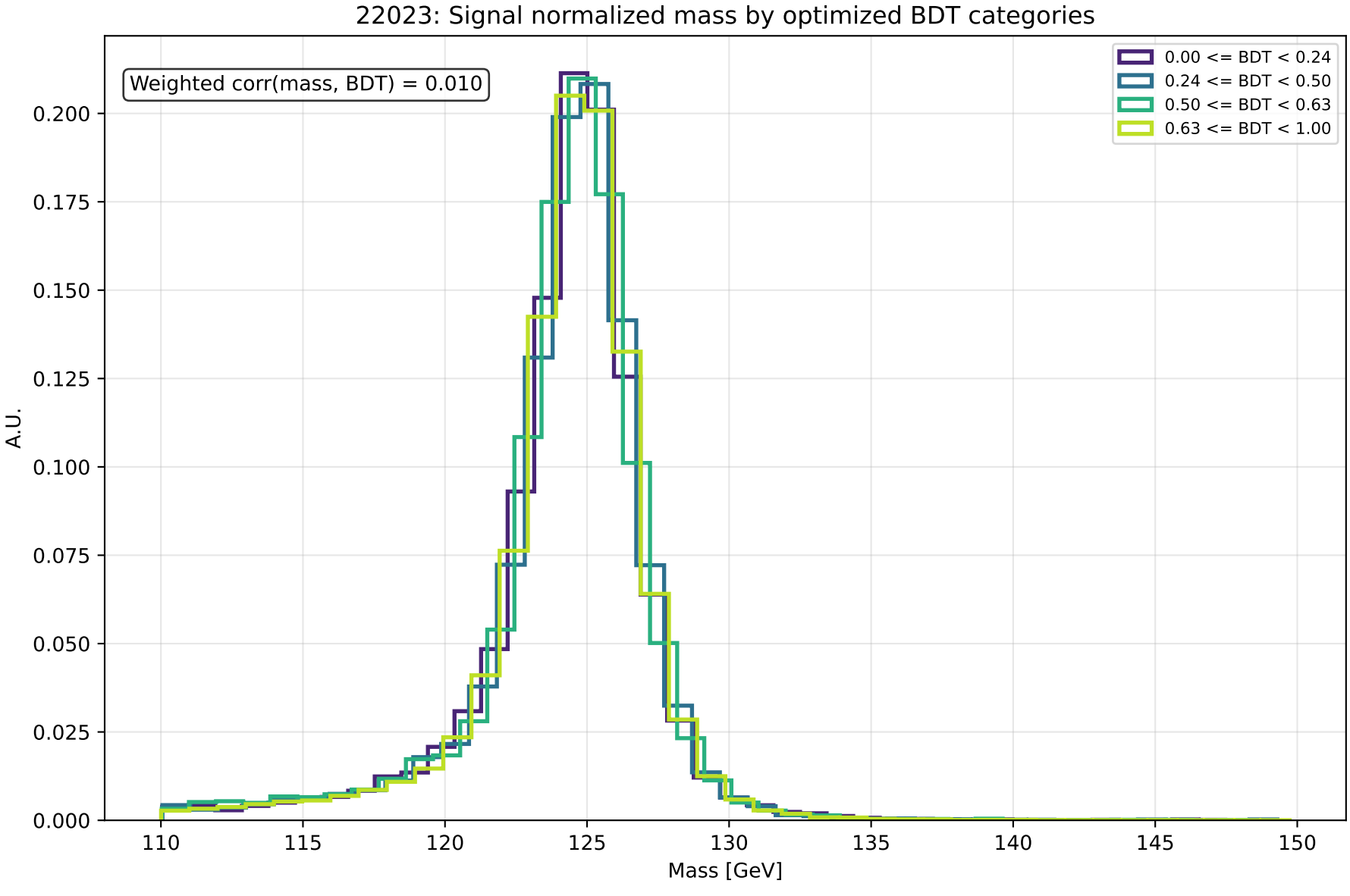

22023: ROC (weighted)
AUC = 0.9141
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

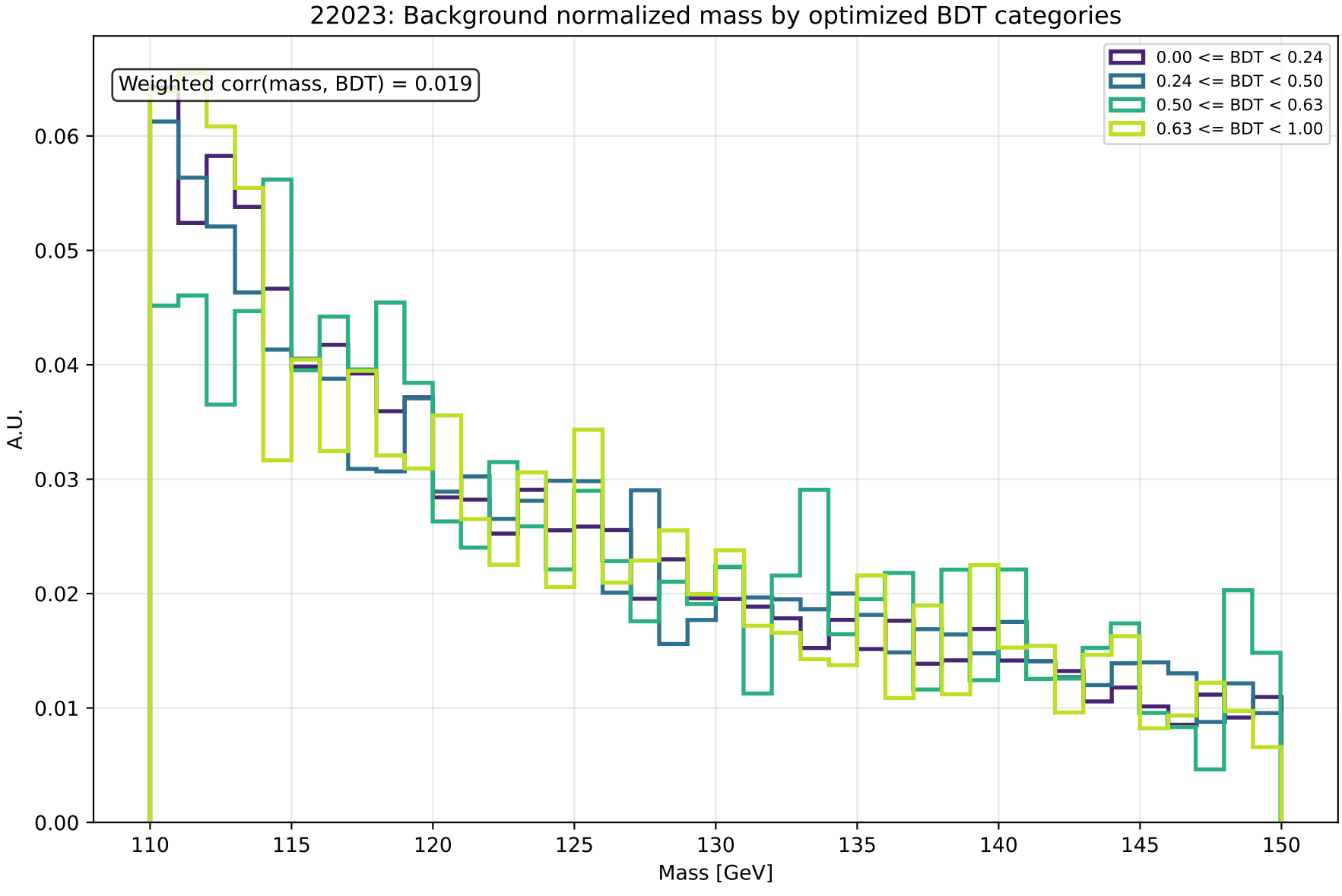
22023: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=0.214, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.241	1.716	82145.341	2.09e-05	0.006
2	0.241-0.503	2.425	22405.431	0.000108	0.016
3	0.503-0.633	1.780	7462.714	0.000239	0.021
4	0.633-1.000	21.616	14919.511	0.00145	0.177


22023: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.24 1.72 124.25 3.07
0.24-0.50 2.42 124.25 3.08
0.50-0.63 1.78 124.21 3.16
0.63-1.00 21.62 124.30 2.96
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
Weighted events
0
1
2
3
4
110
115
120
125
130
135
140
145
150
Mass [GeV]


22023: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.24 82145.34 123.84 10.90
0.24-0.50 22405.43 124.53 11.09
0.50-0.63 7462.71 125.05 11.12
0.63-1.00 14919.51 124.00 10.99
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
Weighted events
0
1000
2000
3000
4000
5000
110
115
120
125
130
135
140
145
150
Mass [GeV]


22023: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

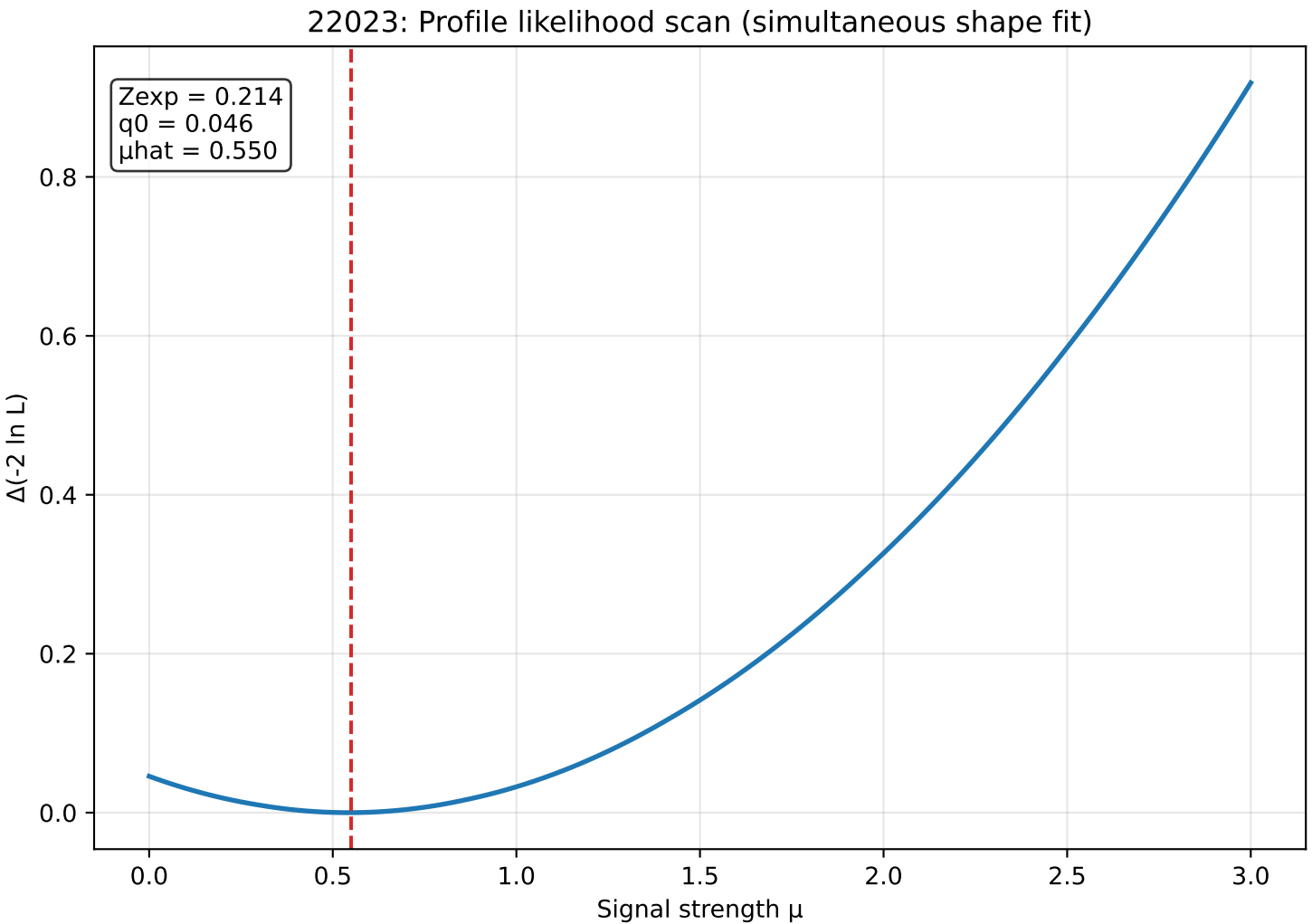

22023: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.019
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

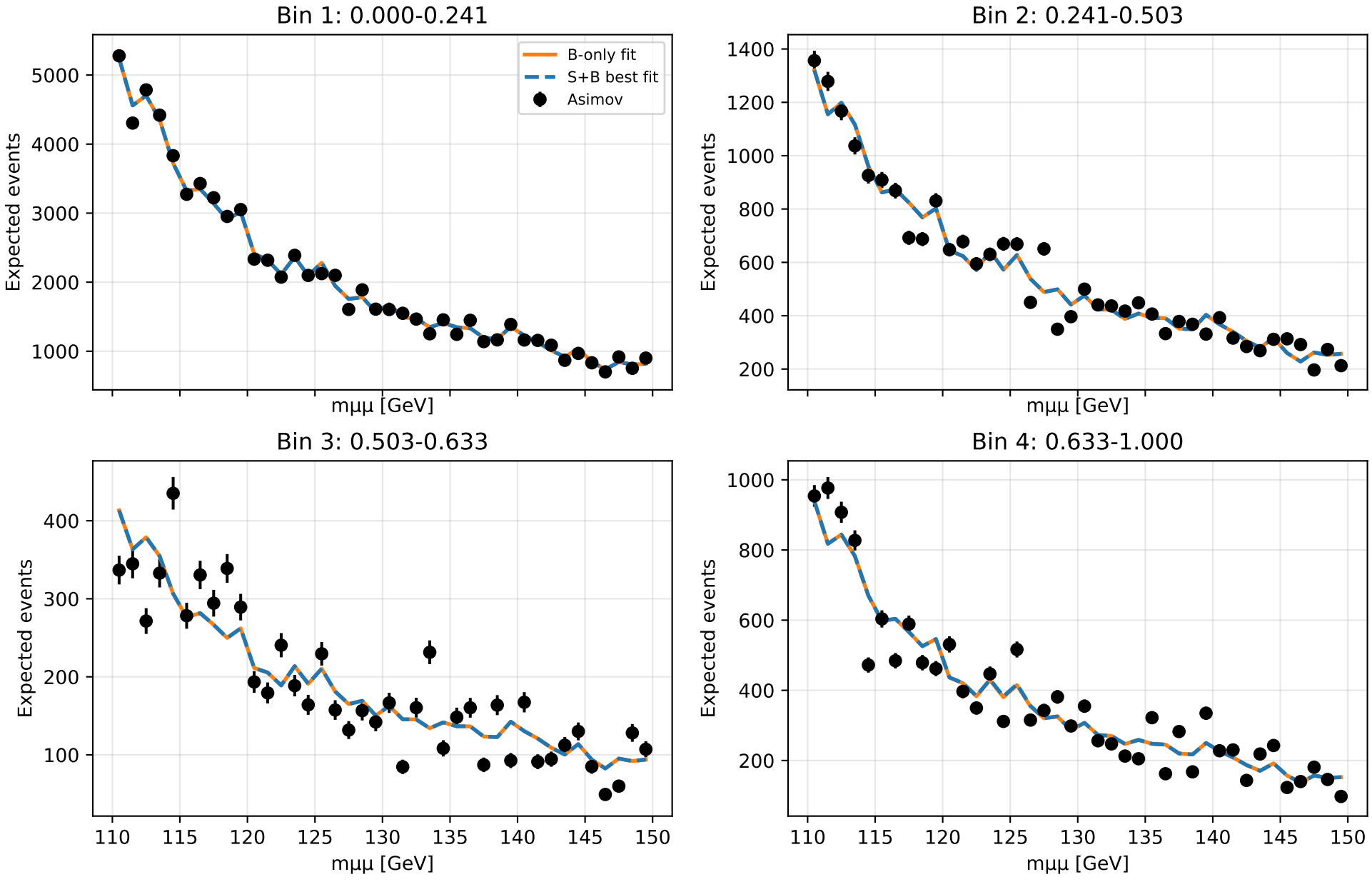
22023: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	0.2140
q0	0.0458
mu_hat	0.5500
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

22023: Shape-likelihood bin yields (Asimov)

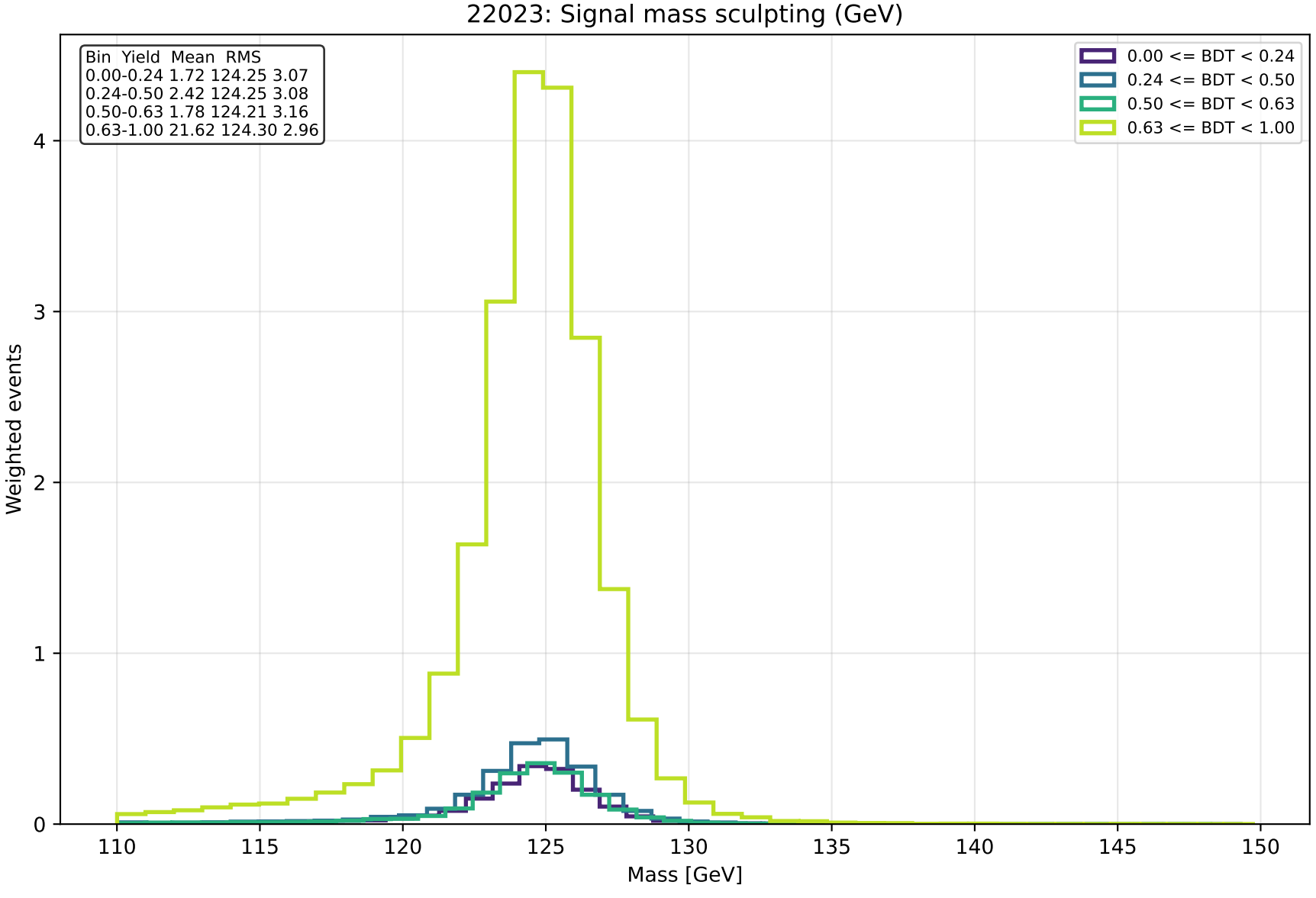
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.241	1.716	82145.341	82145.341	82146.284
2	0.241-0.503	2.425	22405.431	22405.431	22406.765
3	0.503-0.633	1.780	7462.714	7462.714	7463.693
4	0.633-1.000	21.616	14919.511	14919.511	14931.400

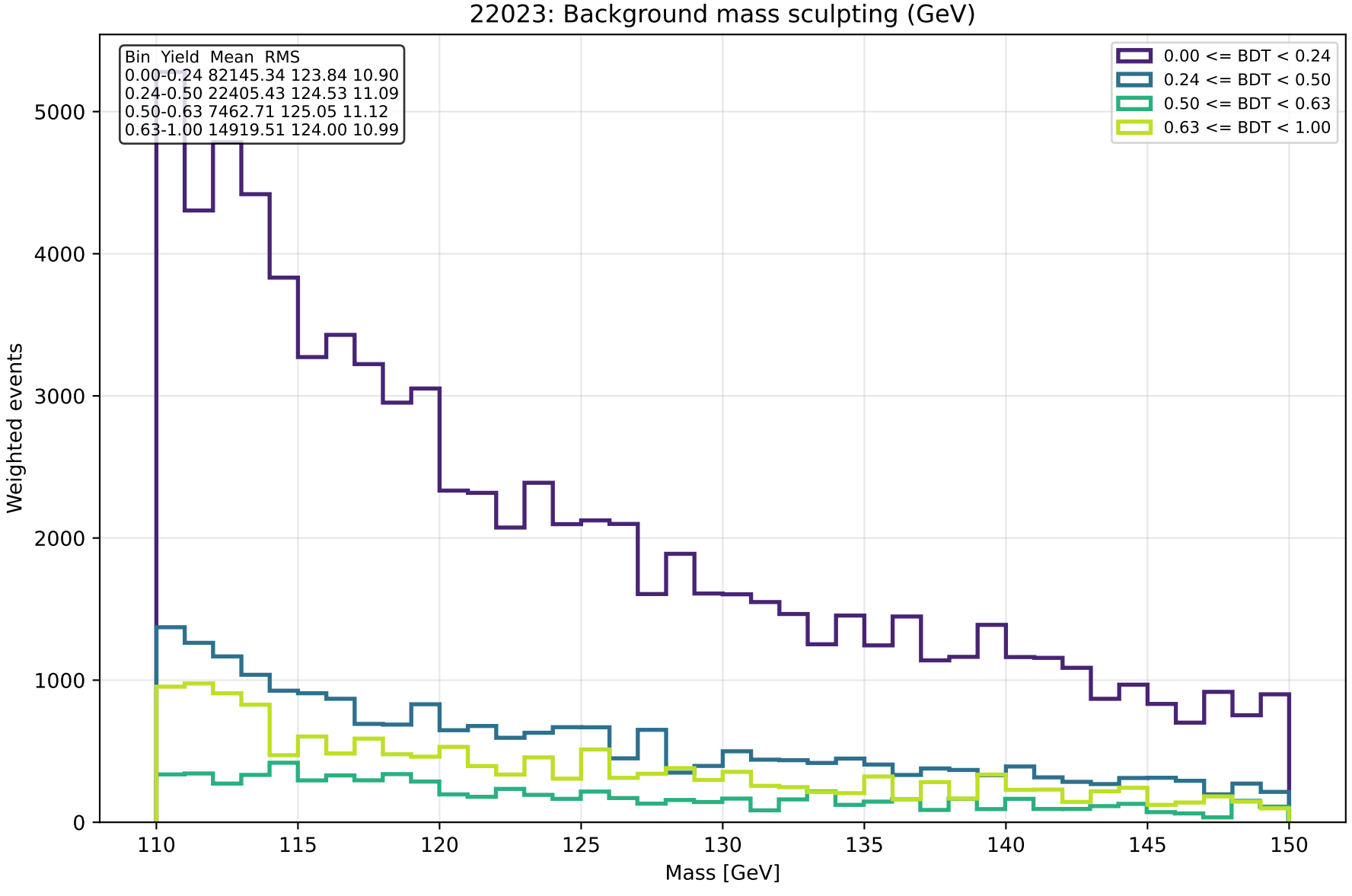

22023: Profile likelihood scan (simultaneous shape fit)
Zexp = 0.214
q0 = 0.046
μhat = 0.550
Δ(-2 ln L)
0.0
0.2
0.4
0.6
0.8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Signal strength μ

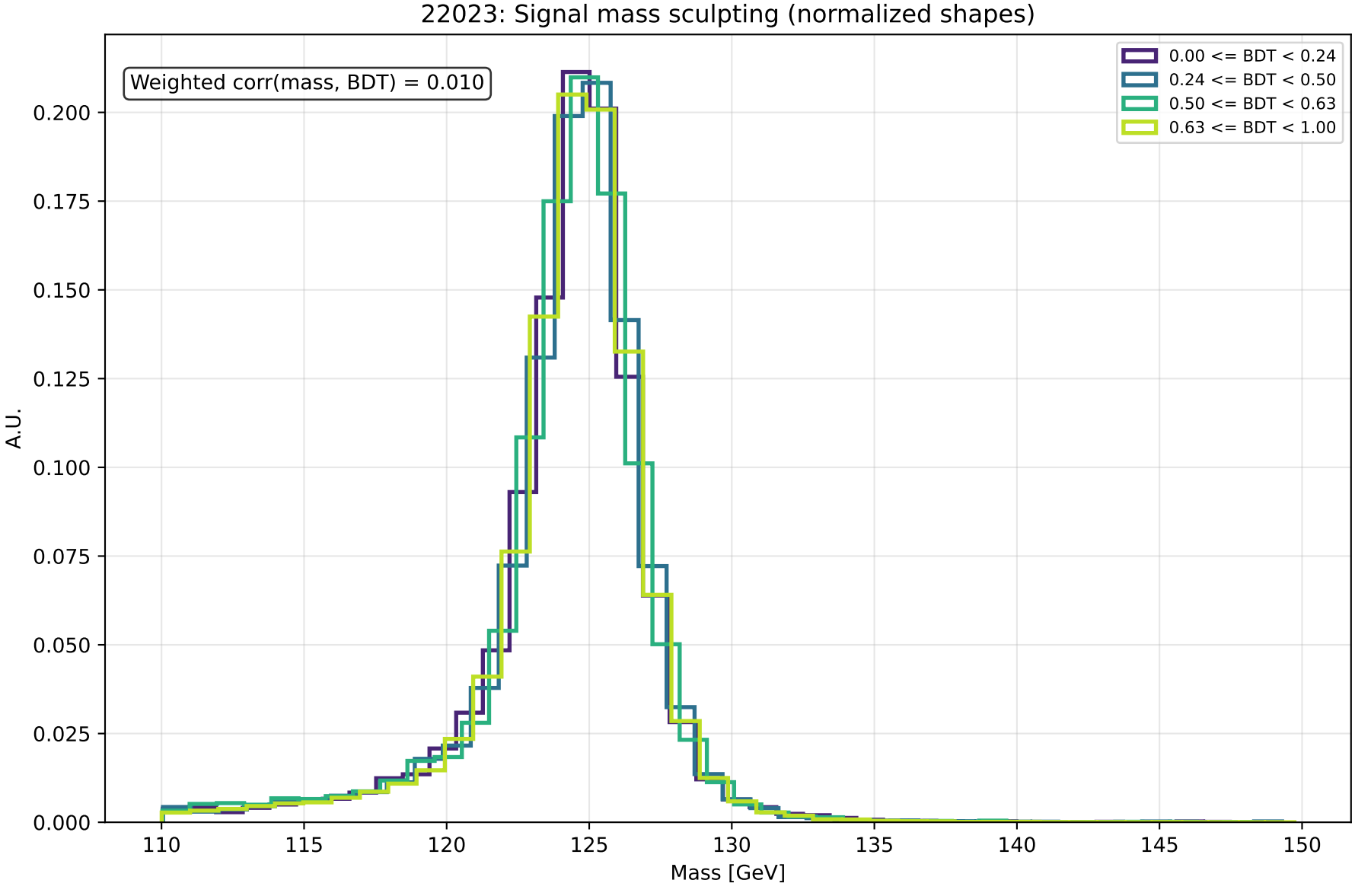

22023: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.241
Bin 2: 0.241-0.503
Bin 3: 0.503-0.633
Bin 4: 0.633-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

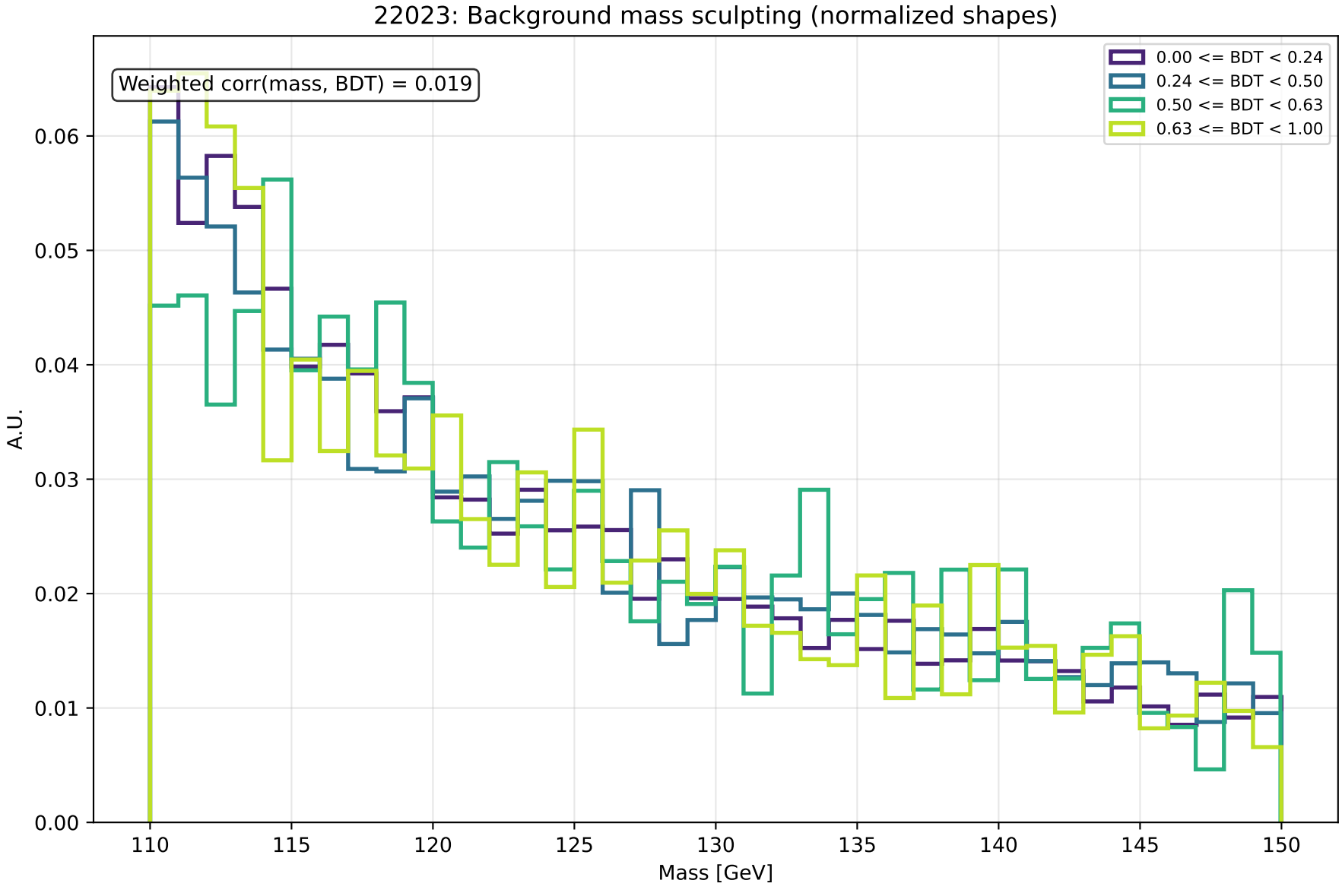
22023: Expected yield per optimized BDT bin (110-150 GeV)
S total=27.54, B total=126933.00

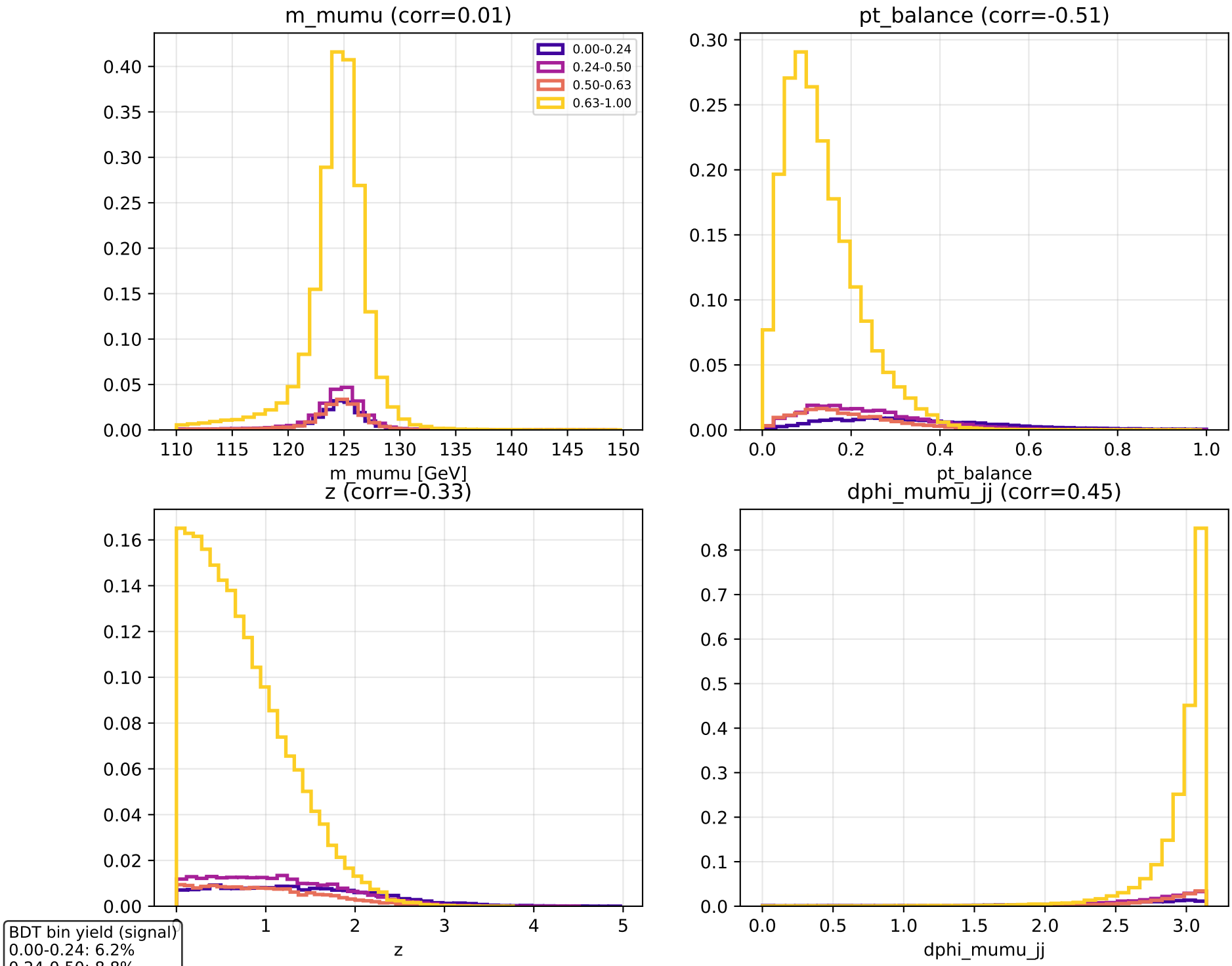
BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.241	1.716	6.2%	100.0%	82145.341	0.000217
0.241-0.503	2.425	8.8%	93.8%	22405.431	0.000577
0.503-0.633	1.780	6.5%	85.0%	7462.714	0.00105
0.633-1.000	21.616	78.5%	78.5%	14919.511	0.00145

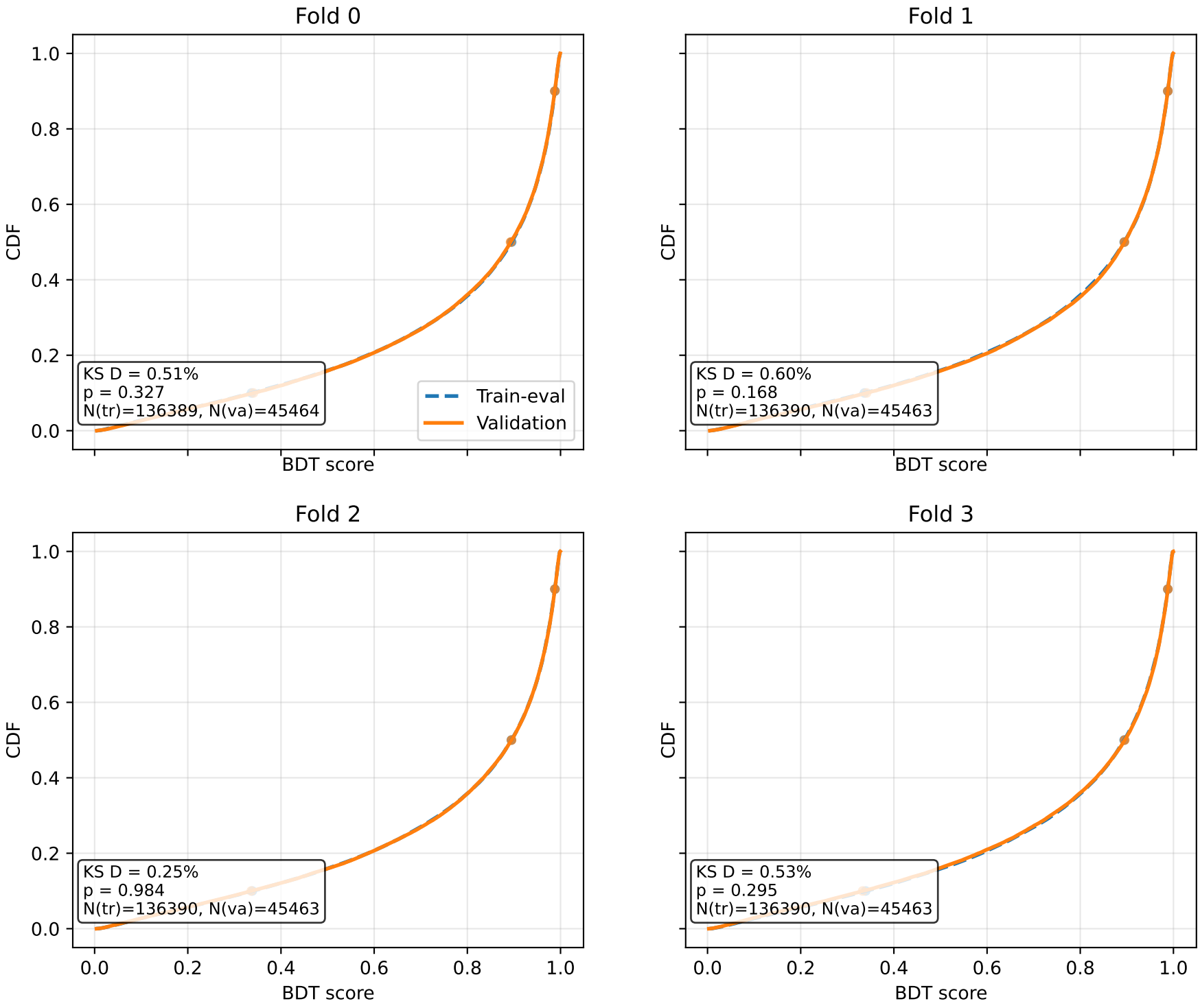

22023: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.24 1.72 124.25 3.07
0.24-0.50 2.42 124.25 3.08
0.50-0.63 1.78 124.21 3.16
0.63-1.00 21.62 124.30 2.96
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
Weighted events
0
1
2
3
4
110
115
120
125
130
135
140
145
150
Mass [GeV]


22023: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.24 82145.34 123.84 10.90
0.24-0.50 22405.43 124.53 11.09
0.50-0.63 7462.71 125.05 11.12
0.63-1.00 14919.51 124.00 10.99
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
Weighted events
0
1000
2000
3000
4000
5000
110
115
120
125
130
135
140
145
150
Mass [GeV]


22023: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]


22023: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.019
0.00 <= BDT < 0.24
0.24 <= BDT < 0.50
0.50 <= BDT < 0.63
0.63 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]


22023: Signal distributions by BDT bin
m_mumu (corr=0.01)
0.00-0.24
0.24-0.50
0.50-0.63
0.63-1.00
m_mumu [GeV]
pt_balance (corr=-0.51)
pt_balance
z (corr=-0.33)
z
dphi_mumu_jj (corr=0.45)
dphi_mumu_jj
BDT bin yield (signal)
0.00-0.24: 6.2%
0.24-0.50: 8.8%
0.50-0.63: 6.5%
0.63-1.00: 78.5%

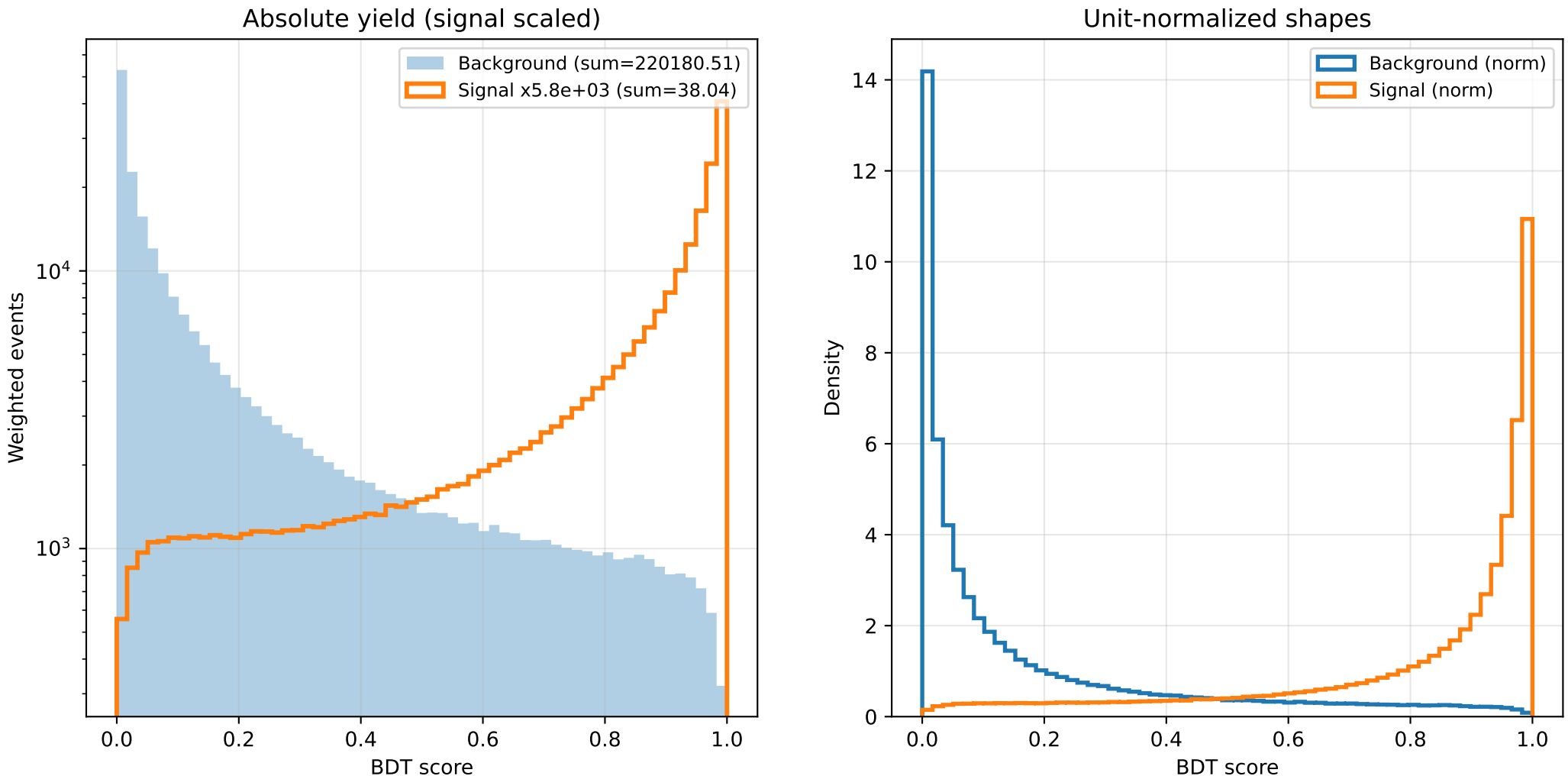

22023: CDF (train-eval vs validation)
Fold 0
KS D = 0.51%
p = 0.327
N(tr)=136389, N(va)=45464
Train-eval
Validation
Fold 1
KS D = 0.60%
p = 0.168
N(tr)=136390, N(va)=45463
Fold 2
KS D = 0.25%
p = 0.984
N(tr)=136390, N(va)=45463
Fold 3
KS D = 0.53%
p = 0.295
N(tr)=136390, N(va)=45463
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

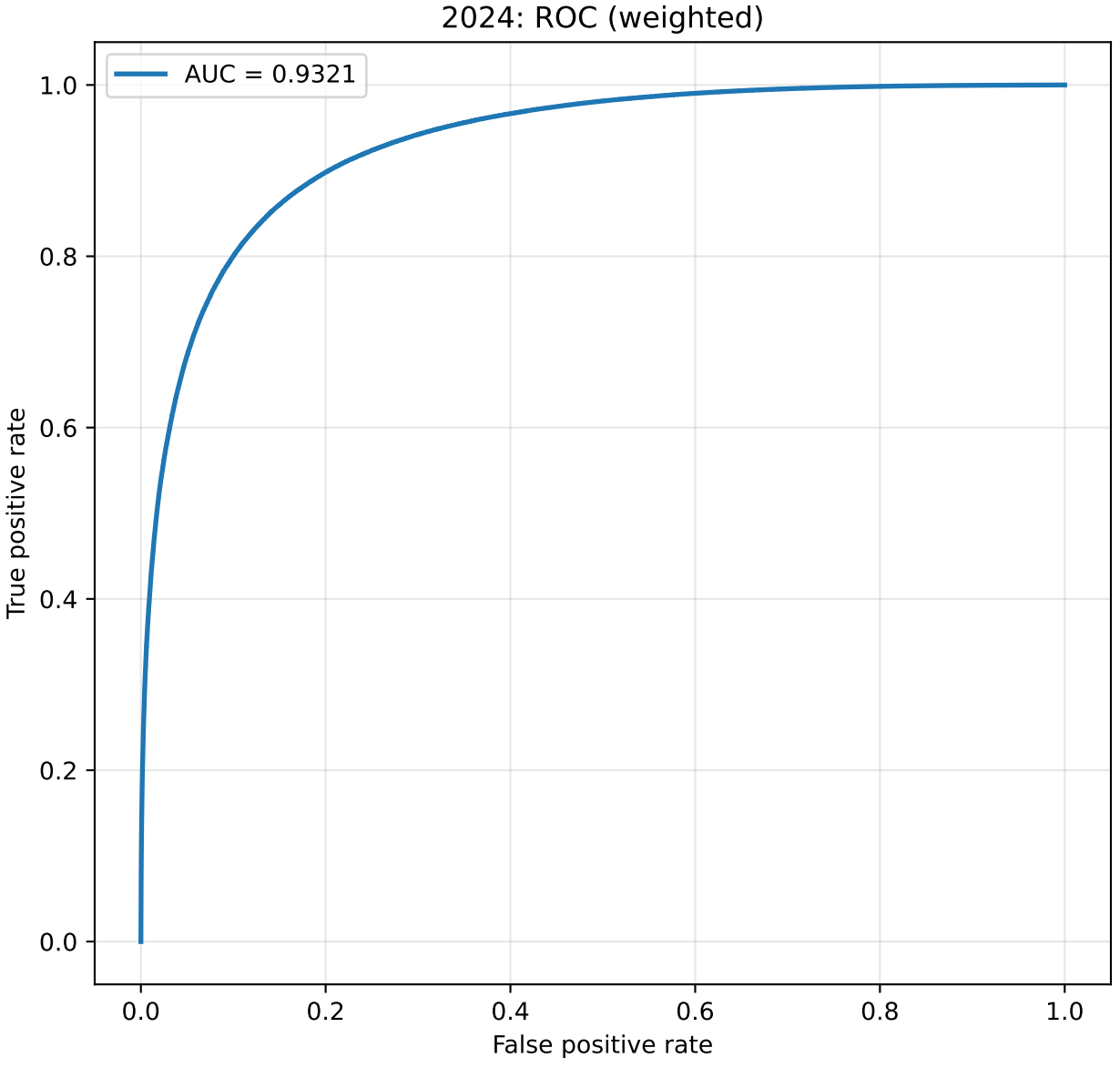
22023: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.51%	45464	0.001	0.20%	FALSE
1	0.60%	45463	0.001	0.20%	FALSE
2	0.25%	45463	0.001	0.20%	FALSE
3	0.53%	45463	0.001	0.20%	FALSE

22023: Top-10 feature importance (gain)
22023_featureimportance_DY.txt

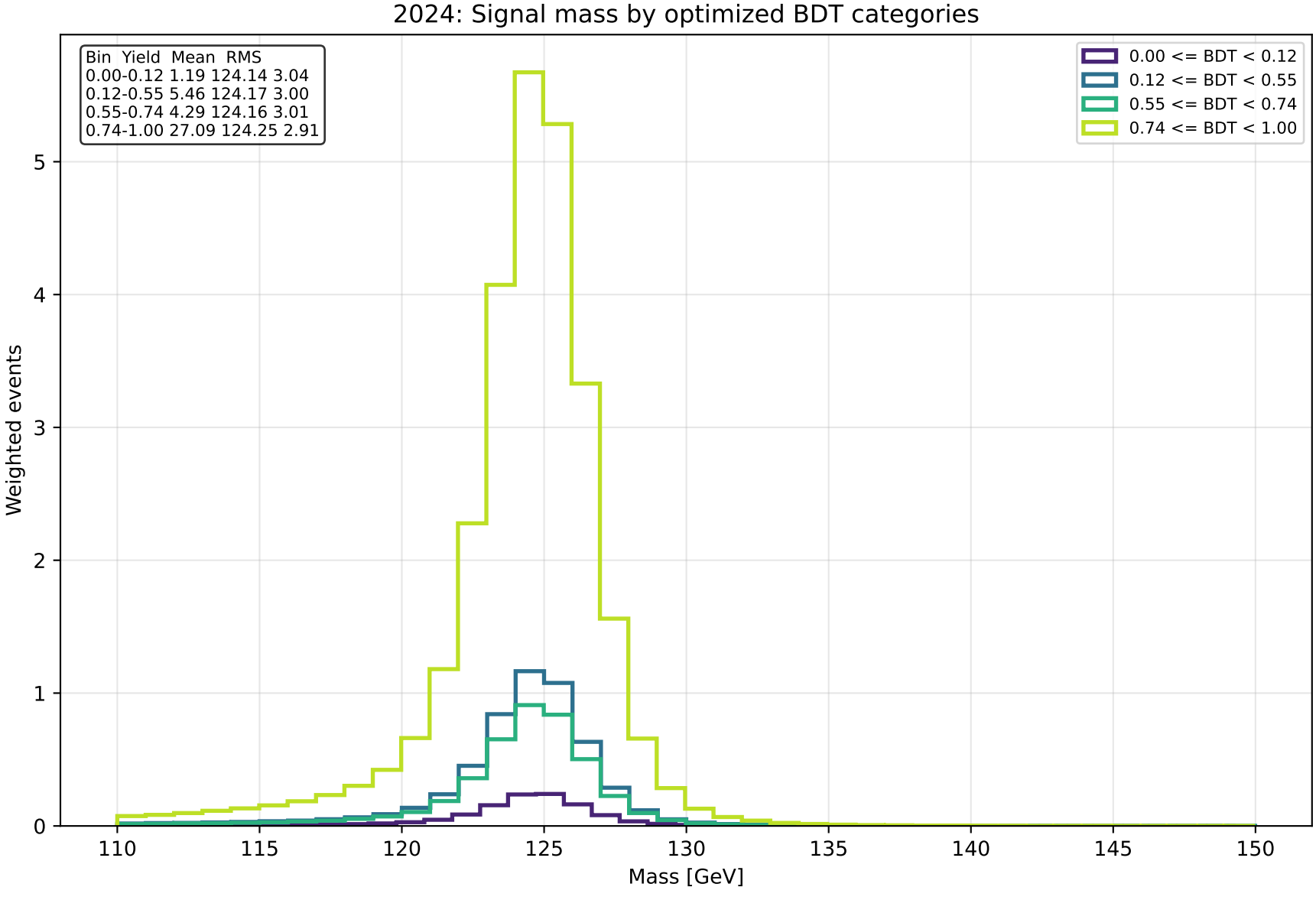
Feature	Gain %
pt_balance	14.21%
z	9.14%
dphi_mumu_jj	8.75%
pt_j2	7.31%
njets	7.09%
deta_jj	6.70%
pt_mumu	6.68%
pt_j1	6.29%
jet1_eta	4.03%
mu1_pt_over_m	3.55%

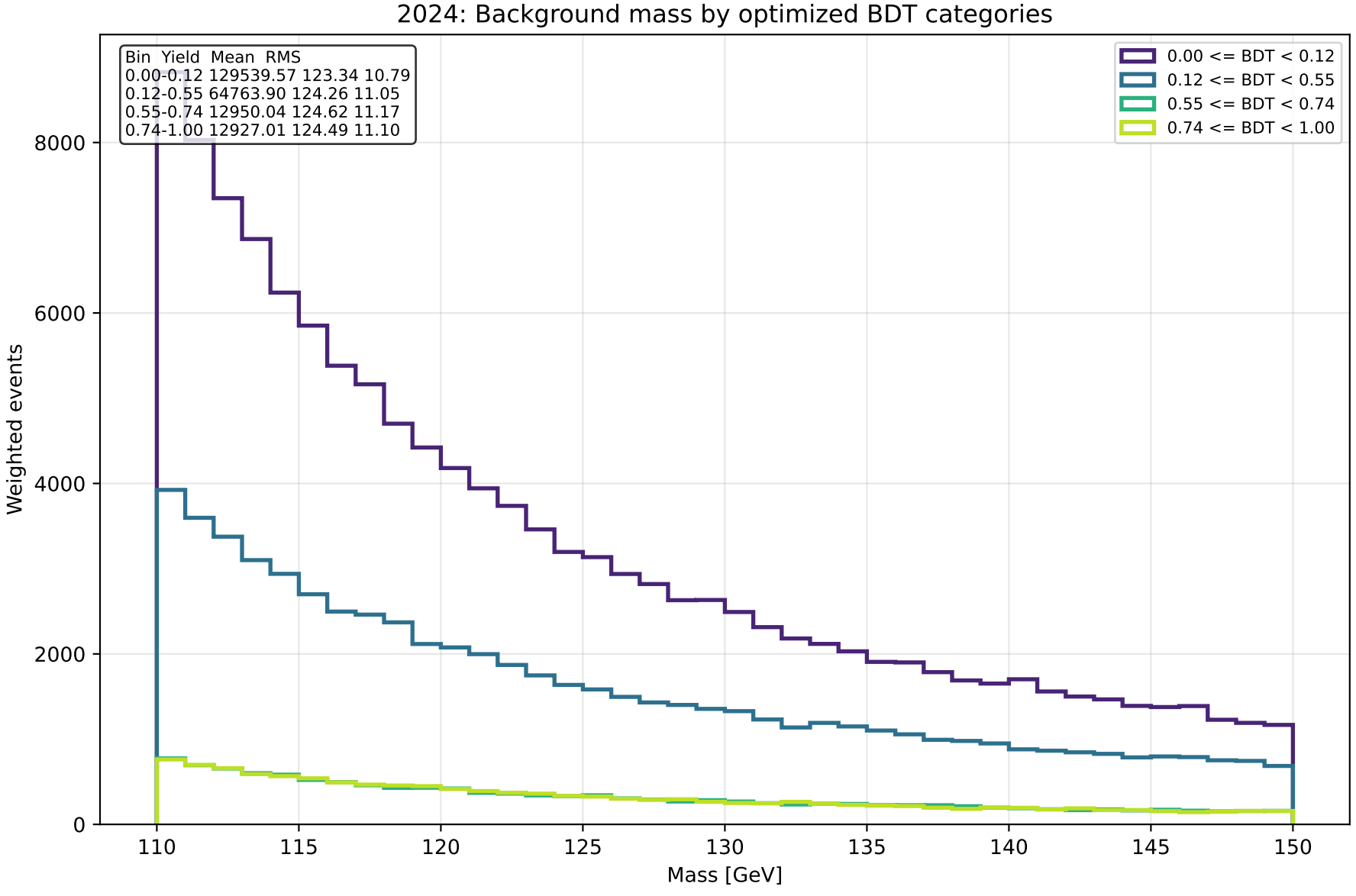

2024: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=220180.51)
Signal x5.8e+03 (sum=38.04)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
2
4
6
8
10
12
14
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

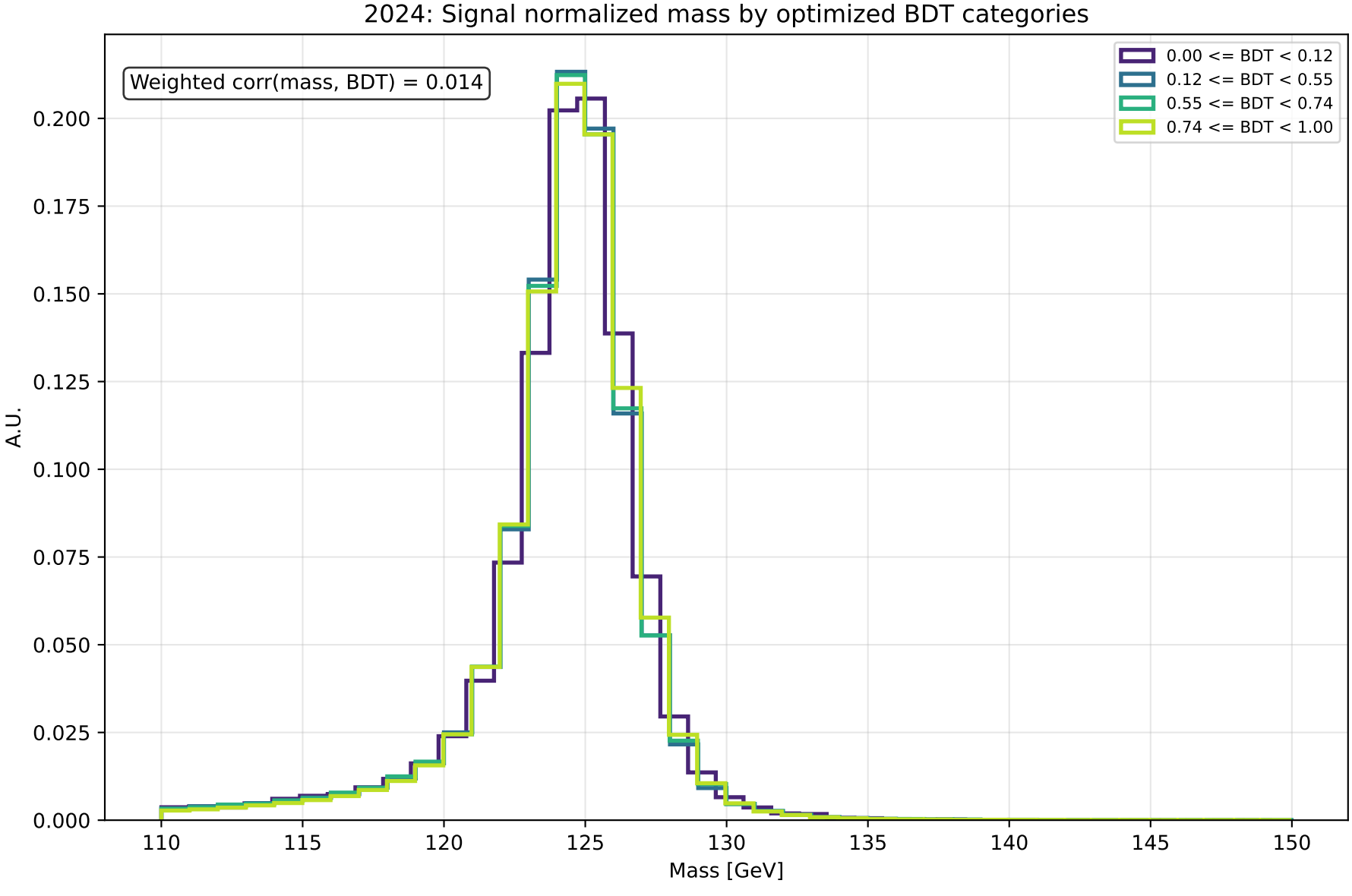

2024: ROC (weighted)
AUC = 0.9321
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

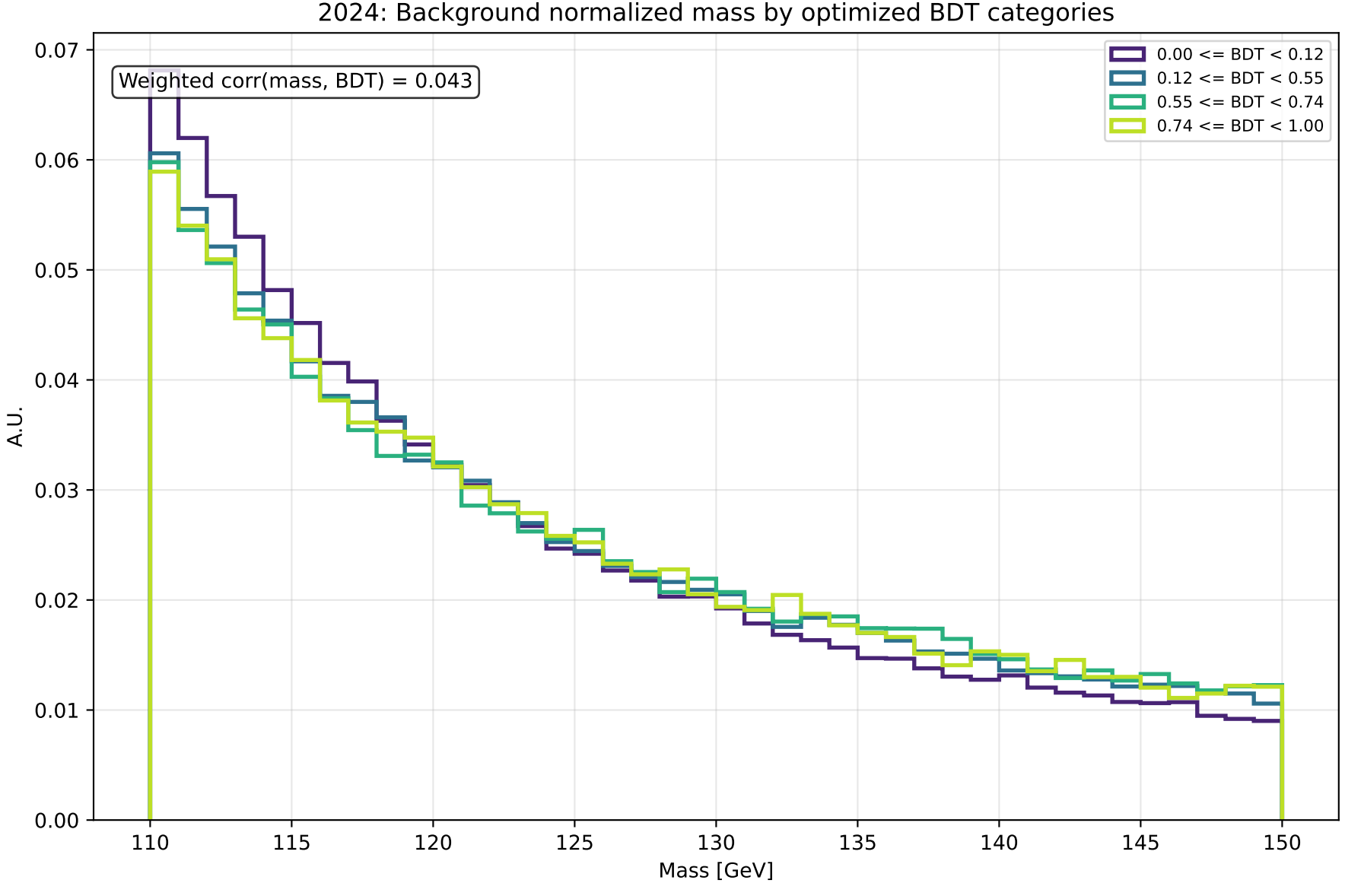
2024: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=1.581, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.122	1.194	129539.566	9.22e-06	0.003
2	0.122-0.546	5.465	64763.899	8.44e-05	0.021
3	0.546-0.739	4.288	12950.036	0.000331	0.038
4	0.739-1.000	27.094	12927.014	0.0021	0.238


2024: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.12 1.19 124.14 3.04
0.12-0.55 5.46 124.17 3.00
0.55-0.74 4.29 124.16 3.01
0.74-1.00 27.09 124.25 2.91
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
1
2
3
4
5
110
115
120
125
130
135
140
145
150
Mass [GeV]


2024: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.12 129539.57 123.34 10.79
0.12-0.55 64763.90 124.26 11.05
0.55-0.74 12950.04 124.62 11.17
0.74-1.00 12927.01 124.49 11.10
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
2000
4000
6000
8000
110
115
120
125
130
135
140
145
150
Mass [GeV]


2024: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.014
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

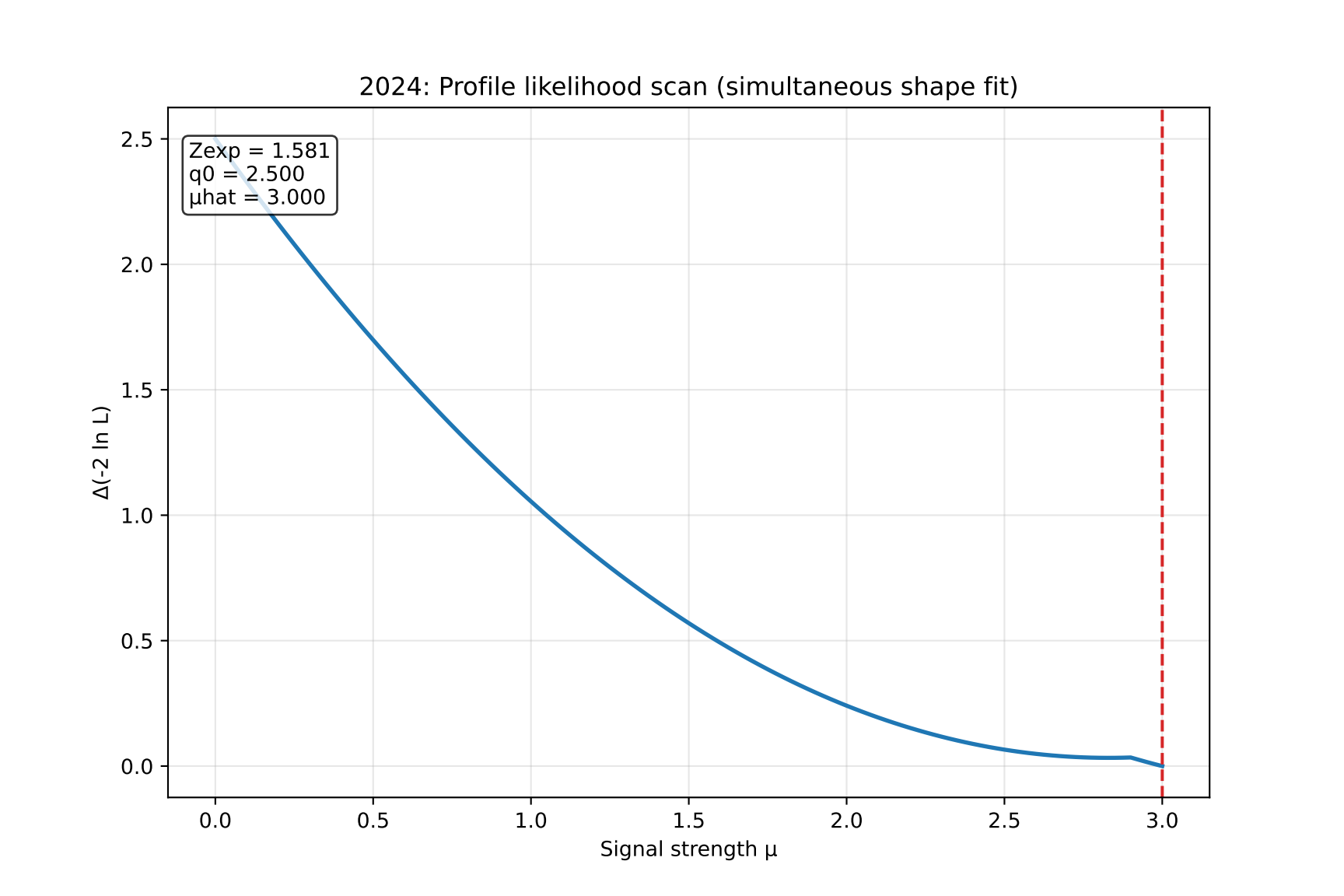

2024: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.043
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
110
115
120
125
130
135
140
145
150
Mass [GeV]

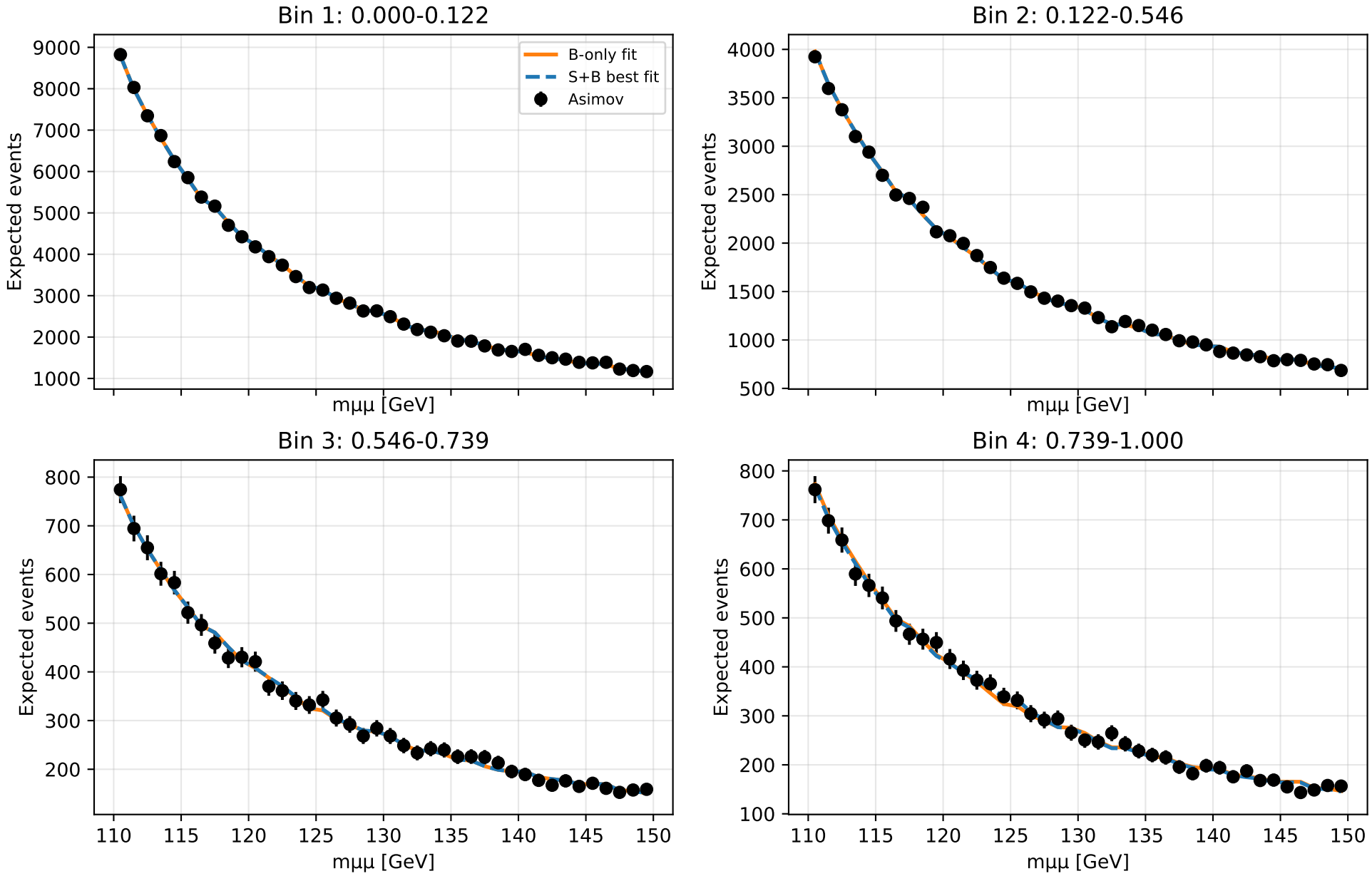
2024: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	1.5812
q0	2.5001
mu_hat	3.0000
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

2024: Shape-likelihood bin yields (Asimov)

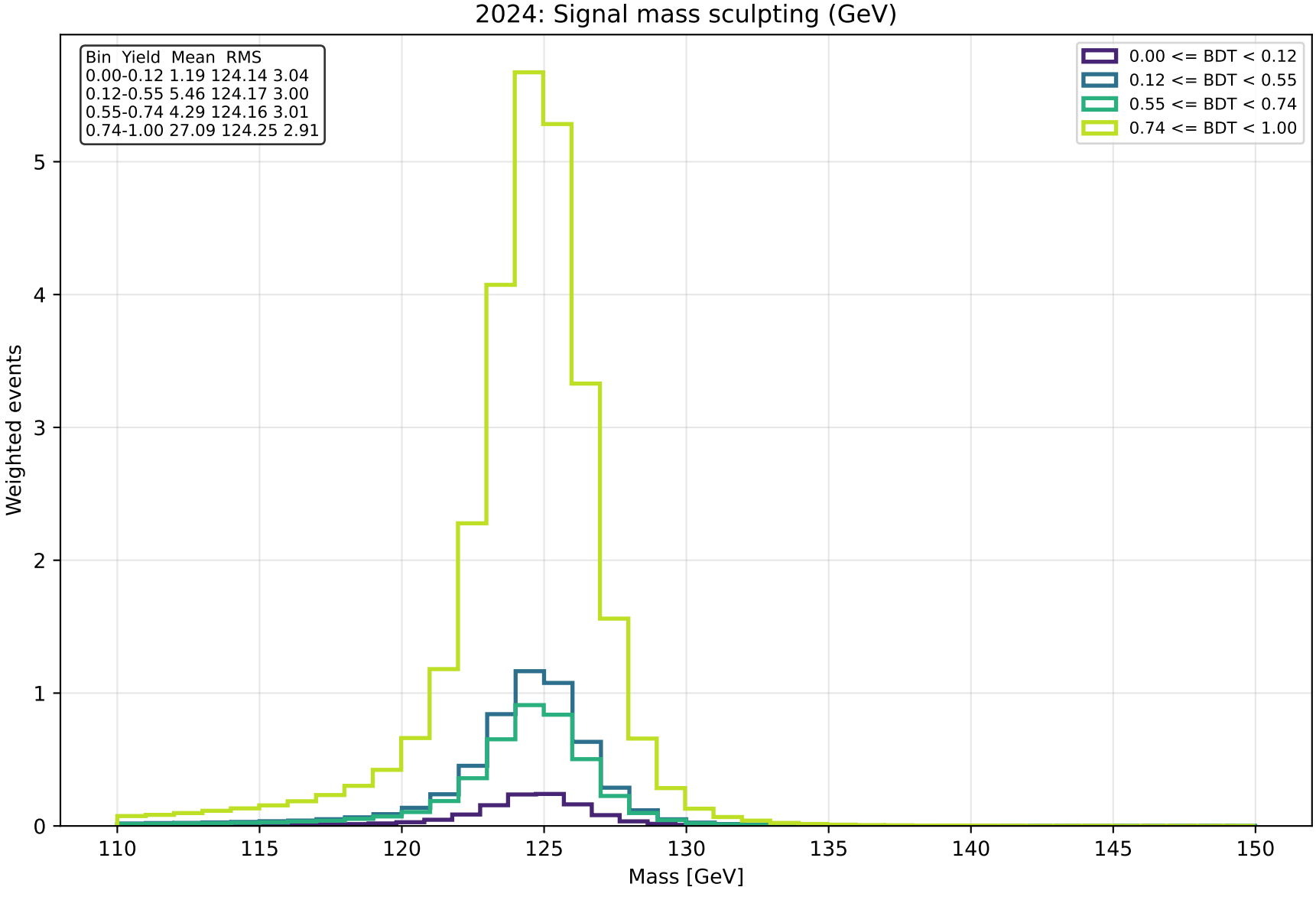
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.122	1.194	129539.566	129539.566	129543.156
2	0.122-0.546	5.465	64763.899	64763.899	64780.325
3	0.546-0.739	4.288	12950.036	12950.036	12962.926
4	0.739-1.000	27.094	12927.014	12927.014	12905.454

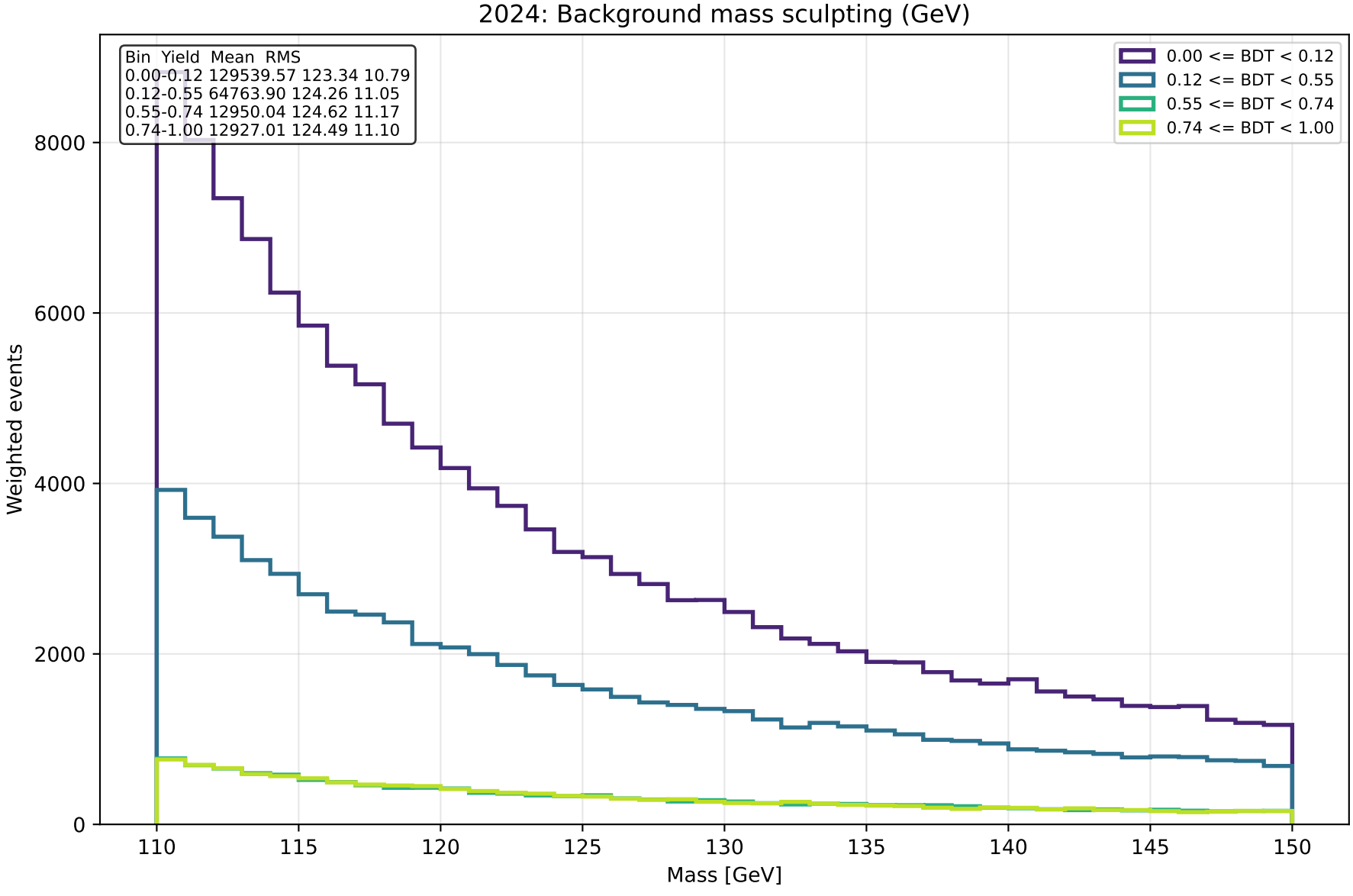

2024: Profile likelihood scan (simultaneous shape fit)
Zexp = 1.581
q0 = 2.500
μhat = 3.000
Δ(-2 ln L)
Signal strength μ
0.0
0.5
1.0
1.5
2.0
2.5
3.0

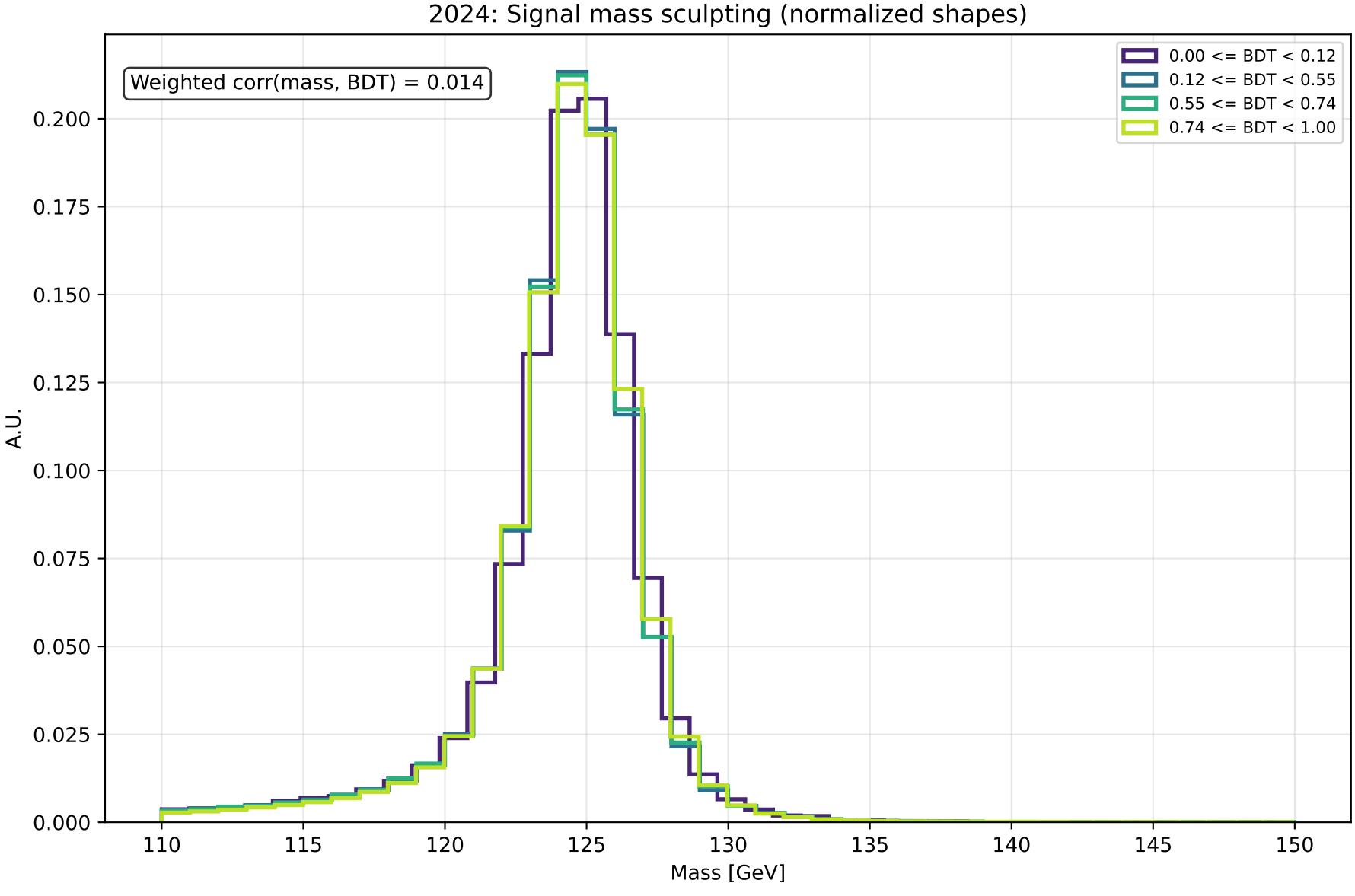

2024: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.122
Bin 2: 0.122-0.546
Bin 3: 0.546-0.739
Bin 4: 0.739-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

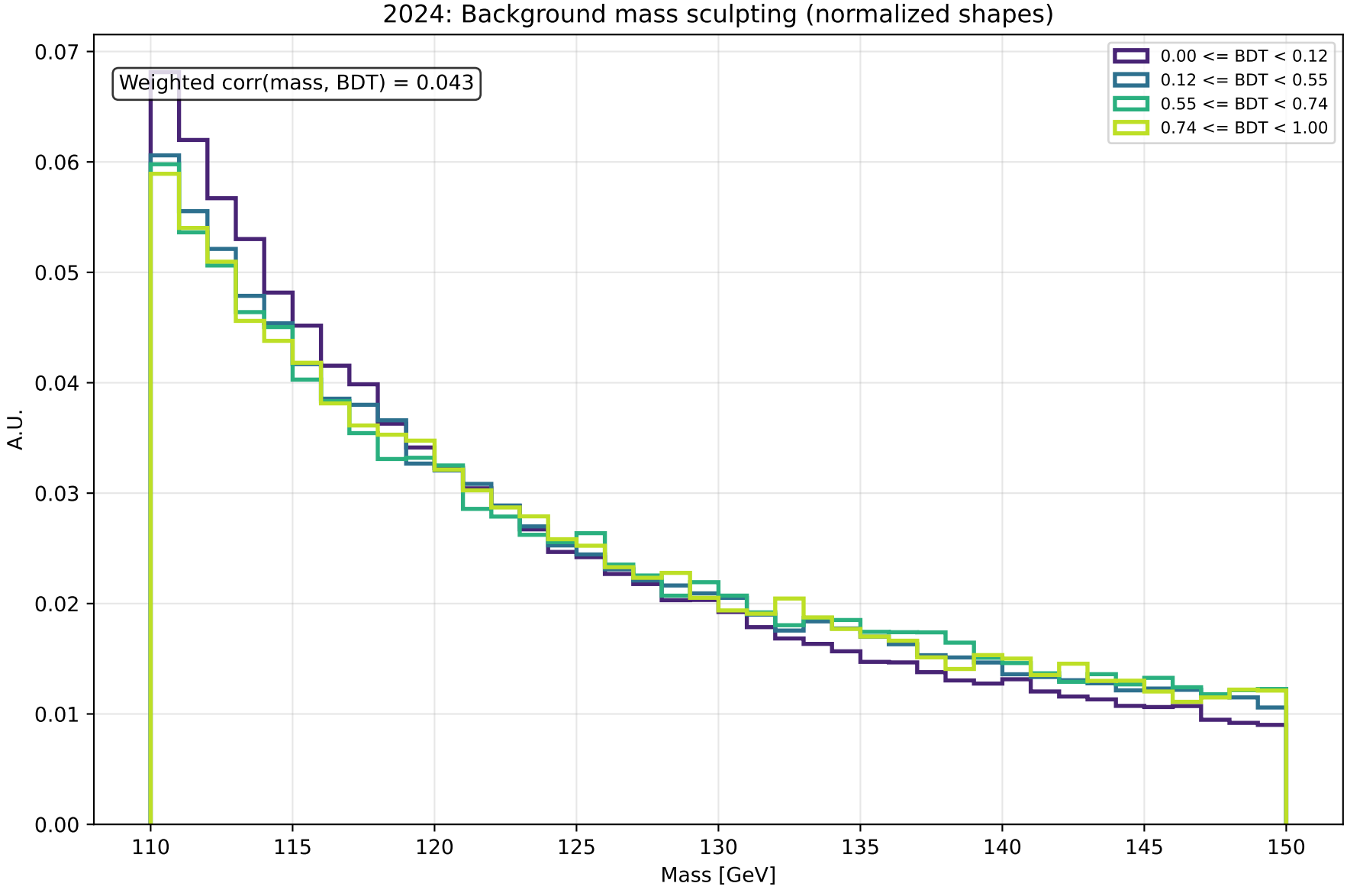
2024: Expected yield per optimized BDT bin (110-150 GeV)
S total=38.04, B total=220180.51

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.122	1.194	3.1%	100.0%	129539.566	0.000173
0.122-0.546	5.465	14.4%	96.9%	64763.899	0.000407
0.546-0.739	4.288	11.3%	82.5%	12950.036	0.00121
0.739-1.000	27.094	71.2%	71.2%	12927.014	0.0021

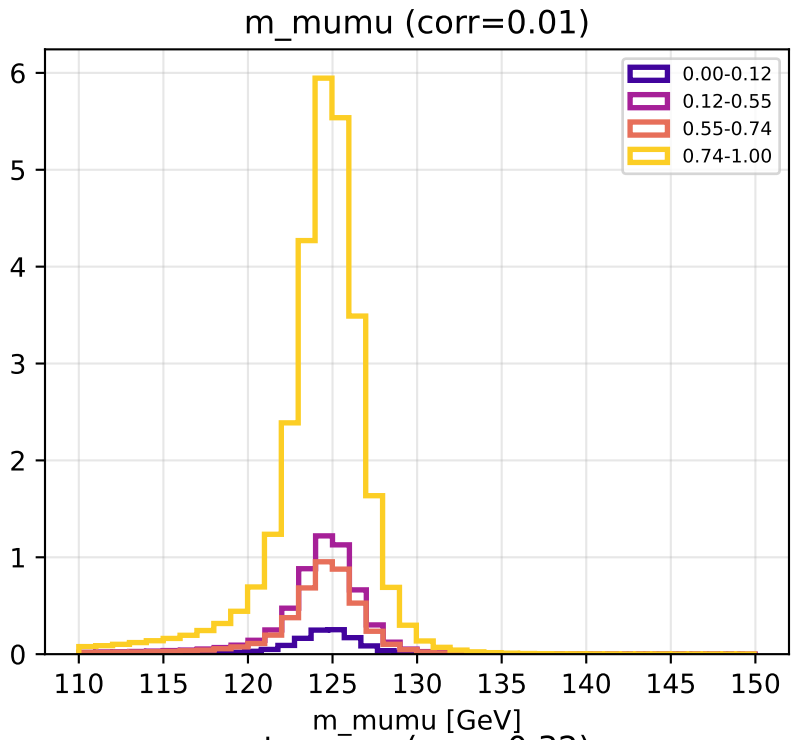

2024: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.12 1.19 124.14 3.04
0.12-0.55 5.46 124.17 3.00
0.55-0.74 4.29 124.16 3.01
0.74-1.00 27.09 124.25 2.91
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
1
2
3
4
5
110
115
120
125
130
135
140
145
150
Mass [GeV]

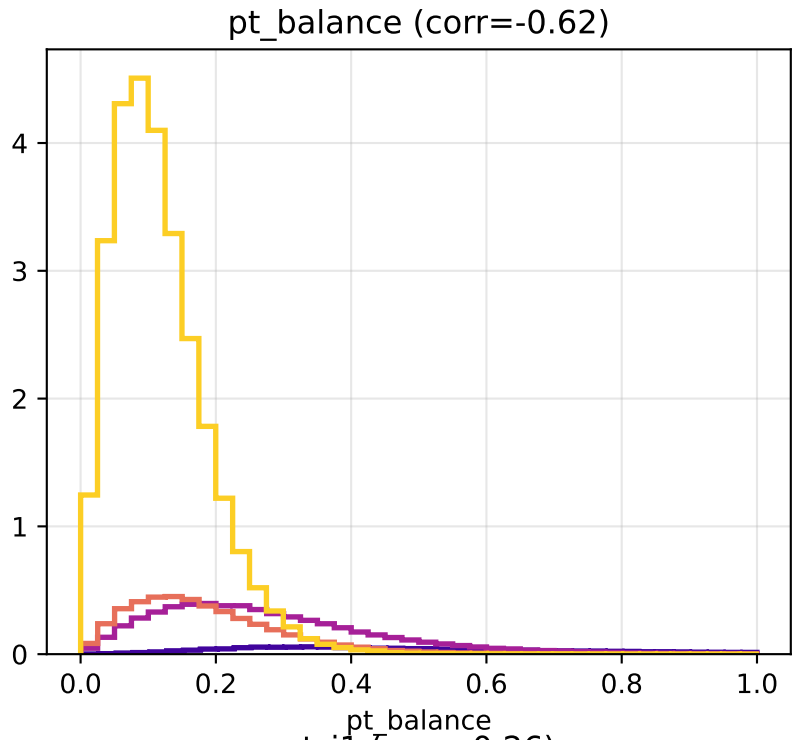

2024: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.12 129539.57 123.34 10.79
0.12-0.55 64763.90 124.26 11.05
0.55-0.74 12950.04 124.62 11.17
0.74-1.00 12927.01 124.49 11.10
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
Weighted events
0
2000
4000
6000
8000
110
115
120
125
130
135
140
145
150
Mass [GeV]

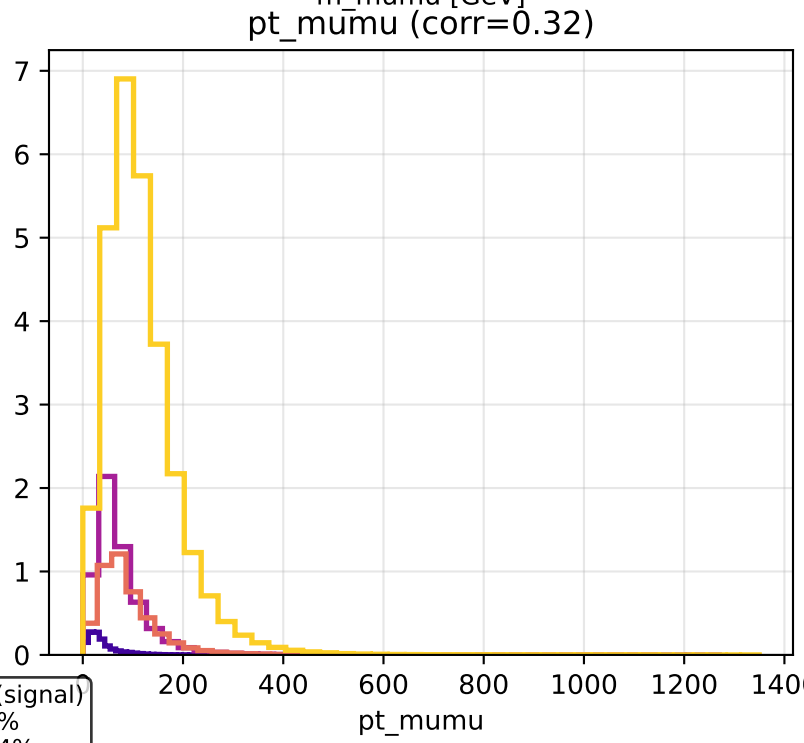

2024: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.014
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
110
115
120
125
130
135
140
145
150
Mass [GeV]

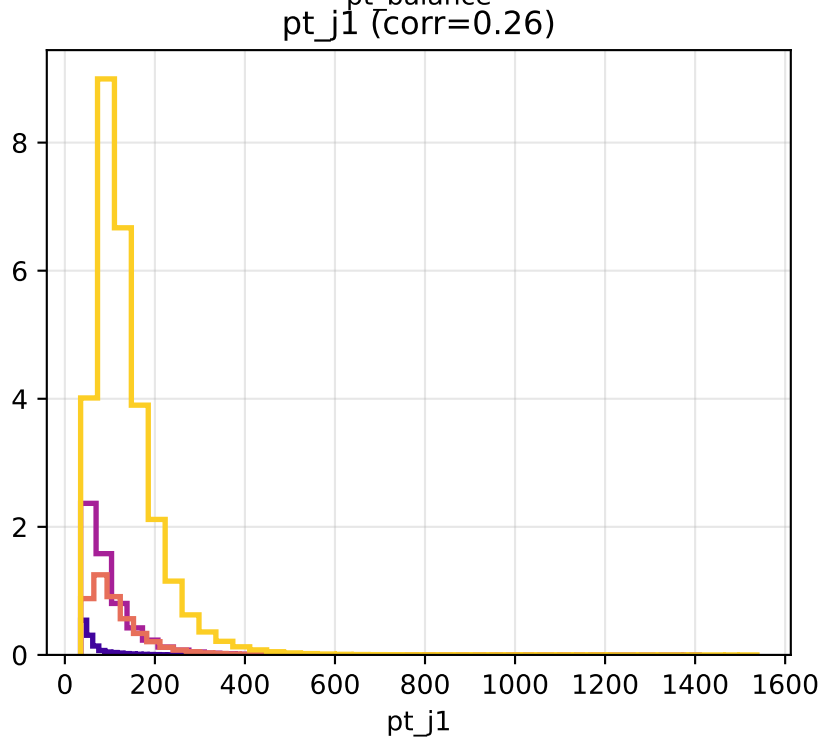

2024: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.043
0.00 <= BDT < 0.12
0.12 <= BDT < 0.55
0.55 <= BDT < 0.74
0.74 <= BDT < 1.00
A.U.
Mass [GeV]
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
110
115
120
125
130
135
140
145
150

2024: Signal distributions by BDT bin

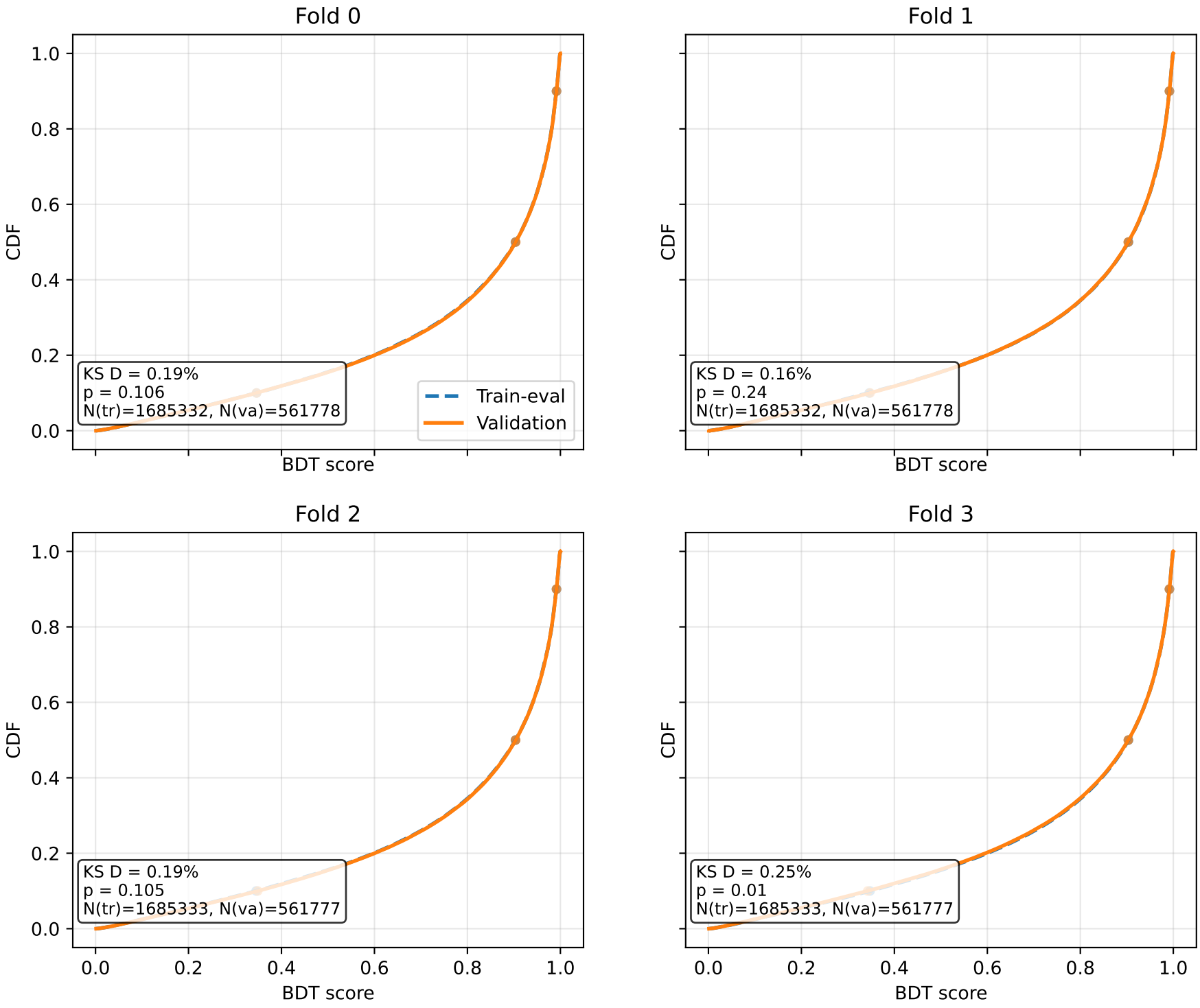








BDT bin yield (signal)
0.00-0.12: 3.1%
0.12-0.55: 14.4%
0.55-0.74: 11.3%
0.74-1.00: 71.2%

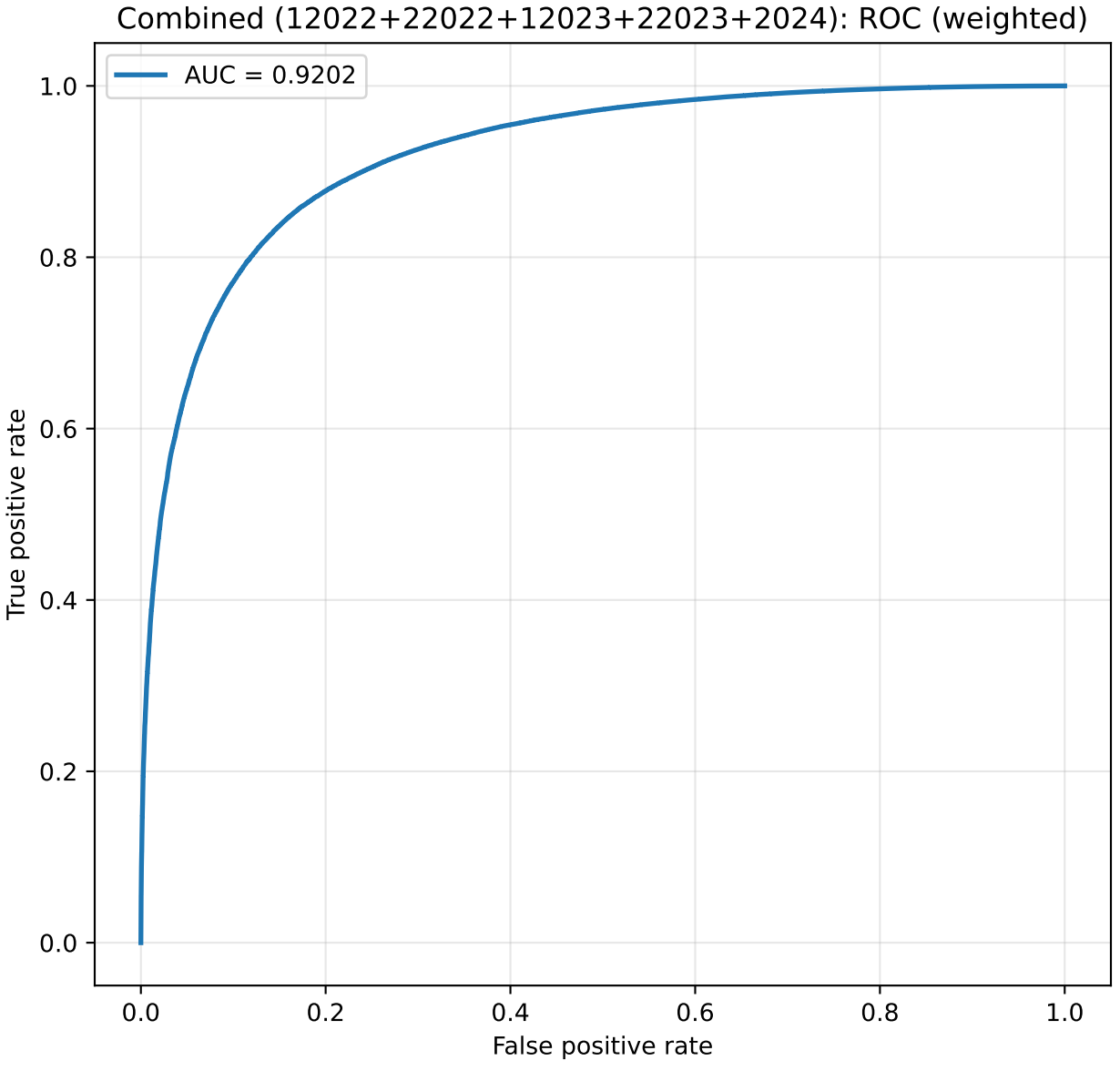

2024: CDF (train-eval vs validation)
Fold 0
KS D = 0.19%
p = 0.106
N(tr)=1685332, N(va)=561778
Train-eval
Validation
Fold 1
KS D = 0.16%
p = 0.24
N(tr)=1685332, N(va)=561778
Fold 2
KS D = 0.19%
p = 0.105
N(tr)=1685333, N(va)=561777
Fold 3
KS D = 0.25%
p = 0.01
N(tr)=1685333, N(va)=561777
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

2024: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.19%	561778	0.001	0.20%	TRUE
1	0.16%	561778	0.001	0.20%	TRUE
2	0.19%	561777	0.001	0.20%	TRUE
3	0.25%	561777	0.001	0.20%	FALSE

2024: Top-10 feature importance (gain)
2024_featureimportance_background.txt

Feature	Gain %
pt_balance	33.48%
pt_mumu	11.43%
pt_j1	11.10%
z	9.63%
dphi_mumu_jj	7.52%
deta_jj	7.02%
pt_j2	6.56%
njets	3.13%
jet1_eta	2.27%
mu1_pt_over_m	1.77%


Combined (12022+22022+12023+22023+2024): ROC (weighted)
AUC = 0.9202
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

Combined (12022+22022+12023+22023+2024): Shape-likelihood summary

Quantity	Value
Combined Eras	12022+22022+12023+22023+2024
Expected Z	0.0000
q0	0.0000
mu_hat	0.0000
N_BDT bins	6
N_mass bins	40
poly order	1

Combined (12022+22022+12023+22023+2024): BDT bin yields

Bin	BDT range	S	B	S/B
1	0.000-0.127	5.706	464907.811	1.23e-05
2	0.127-0.275	7.807	154954.154	5.04e-05
3	0.275-0.461	11.236	103286.880	0.000109
4	0.461-0.600	11.151	51645.200	0.000216
5	0.600-0.771	21.247	51630.281	0.000412
6	0.771-1.000	120.429	51538.669	0.00234

Cross-era stability of optimized BDT category boundaries
(smaller boundary std = more stable across eras)

N bins	#eras	<Zexp(profile)>	Zexp(profile) mi	Zexp(profile) ma	mean std(edge)	max std(edge)
4	5	0.448	0.000	1.581	0.0798	0.0951
5	5	0.400	0.000	1.577	0.0647	0.0804
6	5	0.362	0.000	1.567	0.0356	0.0433